Quantum Entanglement

Leonard Susskind

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Transcript-derived notes curated by LazyingArt LLC with Video2Book

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Chapter 1

Special Relativity: Frames, Simultaneity, and the Lorentz Transformation

These notes follow Leonard Susskind's lecture, with curation for this edition by LazyingArt LLC. The lecture opens by setting the agenda for the quarter: first special relativity done carefully as kinematics, then relativistic dynamics, and only later a more conceptual pass through general relativity. We therefore begin where the lecture itself begins in earnest, with the meaning of a frame of reference, with clocks and meter sticks, and with the insistence that the right way to start a relativity problem is to draw spacetime.

1.1 A Short Framing Note

Relativity, in the sense used here, is the statement that the laws of physics should not depend on the uniform motion of the observer. That idea is older than Einstein. Susskind likes to make the point with a train moving so smoothly that nothing inside it rattles or shakes. In such a train, the laws of juggling are the same as they are in a train standing at the station. If we throw a ball upward, we do not have to compensate for the train's motion in order to catch it again. The physics inside the smoothly moving train is the same physics.

What Einstein changes is not the relativity principle itself, but the structure of space and time needed to preserve it once light is brought into the story. Newton imagined a universal time, almost a God's time, common to all clocks everywhere. Einstein keeps the relativity principle and abandons the universality of simultaneity. The lecture proceeds exactly in that order: first Galileo, then the crisis created by light, then the reconstruction of space and time.

1.2 Reference Frames as Lattices of Meter Sticks and Clocks

Definition 1.1. A reference frame is an imaginary lattice of meter sticks filling space, with a clock at each lattice point. An event is a point in spacetime labeled by a spatial location in that lattice and by the reading of the appropriate clock.

Figure 1.1: Rest-frame time axis and the emerging spacetime lattice.

The lecture immediately turns this operational definition into a picture. Time is drawn vertically, space horizontally, and for the moment we suppress the y and z directions and work only with one spatial coordinate x. Every point on the board then represents a place together with a time.

The horizontal axis is the set of events with $t = 0$. The vertical line through the origin is the worldline of the point $x = 0$. The next vertical line is the worldline of the point $x = 1$, then $x = 2$, and so on. As the clocks tick upward in time, the meter-stick lattice produces a family of constant-position worldlines. When we add horizontal lines of constant time, the whole picture becomes a spacetime lattice.

The board at this point is still sparse, and that is important. Susskind is not yet racing toward formulas. He is teaching us how to think about a frame: as a grid of locations and synchronized clocks, made visible in spacetime before any transformation law is written down.

1.3 Galilean Relativity from the Spacetime Picture

The lecture now brings back the train picture and makes it geometrical. Let the stationary observer use coordinates (x, t), and let the moving observer ride in a train whose origin moves with constant velocity v. In the stationary frame the moving origin has the worldline

$$x = vt. \tag{1.1}$$

The next point of the moving lattice traces a parallel line shifted by a fixed amount, then the next one, and so on. In a spacetime plot the moving family of constant-position lines is tilted.

That tilt is the whole content of Galilean kinematics. An event with stationary coordinates (x, t) has, for the moving observer, the same time but a different spatial coordinate, because the moving

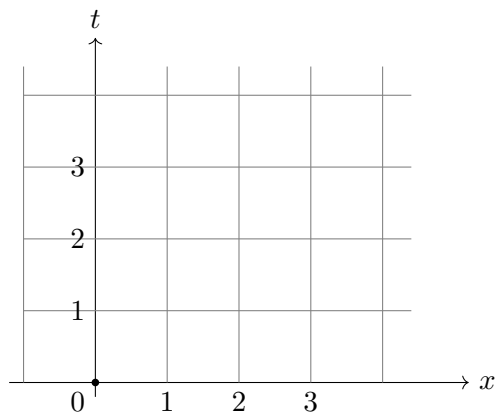


Figure 1.2: A clean reconstruction of the stationary spacetime lattice: constant-position worldlines are vertical and constant-time slices are horizontal.

origin has shifted by the amount vt. Reading directly from the diagram, one arrives at

$$t' = t, \tag{1.2}$$
$$x' = x - vt. \tag{1.3}$$

Susskind is careful about the order here. He does not begin from algebra and then draw a picture afterward. He draws the picture and then reads the equations off the board.

We can invert the relation as well. Since $t = t'$, the inverse form is

$$t = t', \tag{1.4}$$
$$x = x' + vt'. \tag{1.5}$$

The lower inverse equation is cropped in the board frame, and Susskind briefly phrases it as $x = x' + vt$. But because $t = t'$ in Galilean relativity, the intended inverse equation is the canonical one just written.

This is also the point in the lecture where the train analogy and the mathematics meet cleanly. The first pair of equations tells the stationary observer how to convert his description into the moving observer's description; the second pair tells us how to go back. Galilean relativity is, in that sense, a theory of transformations between frames.

1.4 What Galileo Gets Right

Susskind now pauses and asks the right question: what is invariant? He does not want the transformation law to float free of physics.

Take two events at the same time t, with positions x_1 and x_2. In the moving frame,

$$x_1' = x_1 - vt, \tag{1.6}$$
$$x_2' = x_2 - vt. \tag{1.7}$$

Subtracting the two,

$$x_2' - x_1' = (x_2 - vt) - (x_1 - vt) = x_2 - x_1. \tag{1.8}$$

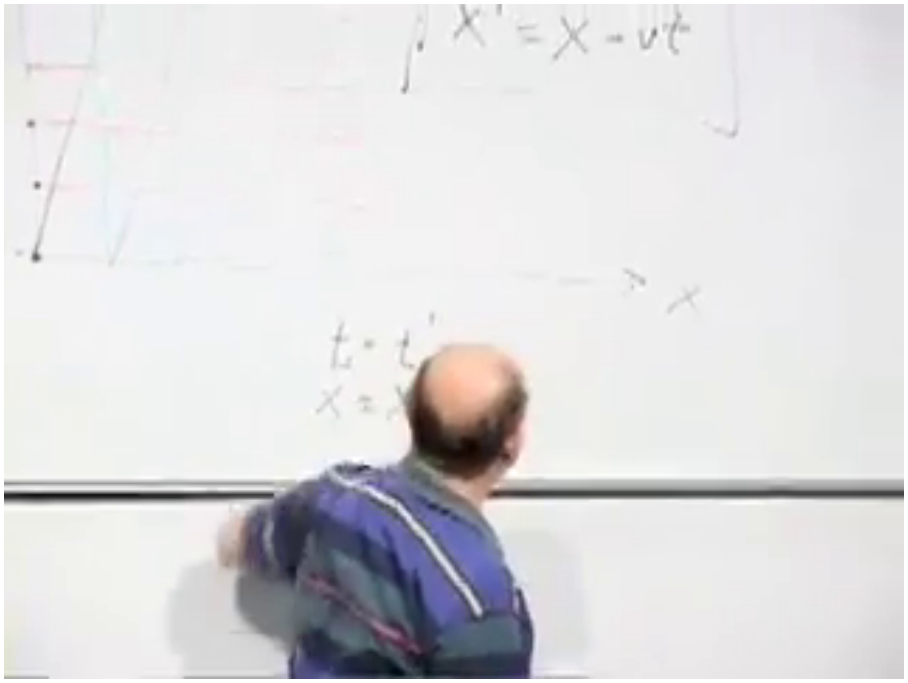


Figure 1.3: Galilean coordinate shift beside the spacetime sketch.

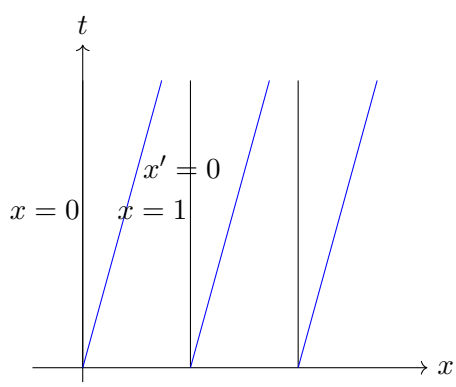


Figure 1.4: Stationary worldlines and the tilted moving family from which the Galilean shift is read.

So the spatial separation of two simultaneous events is invariant under Galilean transformations. The individual coordinates are not invariant, but their difference is.

At this stage the lecture also raises a natural worry. How do we know that time really remains universal when a clock is carried around? Galileo himself did not know in any strict modern sense. It was, as Susskind says, frankly a guess, though a very good one. For ordinary clocks moved around at ordinary speeds, the agreement is fantastically precise. Only at sufficiently high speeds does the guess fail, and that failure is exactly what special relativity is about.

Now let a particle follow a trajectory $x(t)$. The moving observer sees

$$x'(t) = x(t) - vt. \tag{1.9}$$

So position is not invariant. Different observers attach different spatial coordinates to the same particle. Differentiating once,

$$\frac{dx'}{dt} = \frac{dx}{dt} - v, \tag{1.10}$$

and therefore velocity is not invariant either. This is obvious from ordinary experience: if I move along beside your arrow, I judge its speed differently from you.

Differentiate once more:

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2}, \tag{1.11}$$

because the relative velocity v between inertial frames is constant. Acceleration is invariant.

This is the Newtonian payoff. If the force between two objects depends only on the distance between them, and if that distance is invariant, then the force is also invariant. If acceleration is invariant as well, then

$$F = ma \tag{1.12}$$

has precisely the right structure to serve as a law of motion that all inertial observers can share, provided mass itself is invariant. In this sense Newton's laws are invariant under Galilean coordinate transformations.

The lecture's point is not merely historical courtesy. Galileo and Newton were not making a naive mistake. For mechanics at low velocities, the whole scheme is extraordinarily successful.

1.5 Maxwell's Challenge and Einstein's Return to First Principles

The next move is the crisis. Galilean relativity works beautifully for mechanics, but electromagnetism changes the stakes. Maxwell's equations predict a definite speed for light, and if Maxwell's equations are taken seriously as laws of nature, then the speed of light ought to be the same for every inertial observer.

Galilean kinematics says otherwise. If a light ray in the stationary frame satisfies

$$x = ct, \tag{1.13}$$

then the moving observer would assign

$$x' = x - vt = (c - v)t. \tag{1.14}$$

So under a Galilean transformation the moving observer sees light propagate at speed $c - v$, not at c. That is exactly what one would expect for sound waves or water waves. Indeed Susskind stresses that this was not a crazy expectation. It was the natural expectation.

The historical temptation, before Einstein, was to conclude that there must be a preferred frame, a kind of God's frame, in which Maxwell's equations are exactly right and in which light really travels at speed c. Einstein's move is more radical and more economical. He keeps the relativity principle and forces space and time to adapt. In the lecture this comes in the form of two postulates:

1. The laws of physics are identical in every inertial frame.
2. The speed of light is the same in every inertial frame.

But these postulates do not yet tell us how to assign coordinates. So Einstein goes back to first principles and asks what it means for clocks at different places to be synchronized. The answer is operational. Put an observer at the midpoint between two endpoint clocks. Emit flashes from the endpoints when the endpoint clocks read the same time. If the midpoint observer receives the flashes simultaneously, the clocks are synchronized.

This is the point where the lecture stops being Newtonian in spirit. If a second observer moves past that synchronized apparatus, the two flashes no longer arrive at the moving midpoint at the same instant. The moving observer must therefore conclude that the original emissions were not simultaneous. Simultaneity is not absolute.

1.5.1 Question & Answer

Question. Why not synchronize clocks with sound instead of light?

Answer. One can synchronize clocks with sound in a frame in which the air is at rest. But then one immediately discovers that moving observers measure different sound speeds. Sound propagates in a medium, and motion relative to that medium matters. In an open-air train moving through still air, sound would be a bad messenger of relativity. Light is chosen for the opposite reason: Einstein's postulate is that light has the same speed for every inertial observer, so light is exactly the signal with which the notion of simultaneity must be rebuilt.

The lecture now asks its central quantitative question: if the stationary observer calls one pair of distant events simultaneous, which pair does the moving observer call simultaneous instead?

1.6 The Geometry of Relativity of Simultaneity

Susskind now slows down and makes the question geometric. He sets $c = 1$, so that space and time are measured in compatible units and light rays run at 45°. A right-moving light ray has the form

$$x = t + d, \tag{1.15}$$

and a left-moving light ray has the form

$$x = -t + d'. \tag{1.16}$$

In particular, the light ray from the origin satisfies $x = t$.

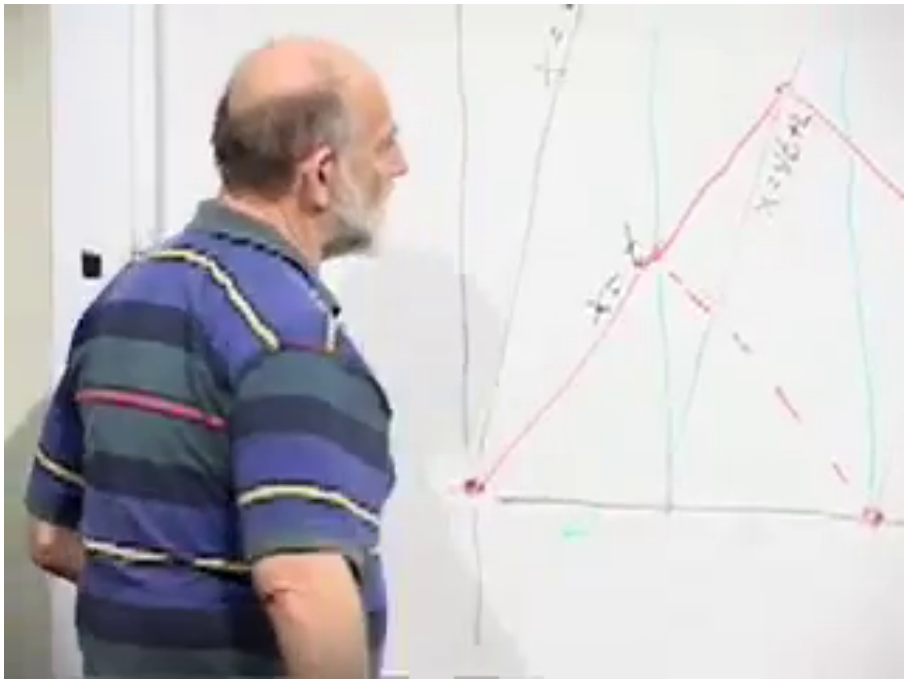


Figure 1.5: Light-ray line $x = t$ in the simultaneity construction.

He then lays out the rest-frame synchronization experiment with endpoints separated by equal distances L. The midpoint observer sits between them. Once the moving frame is added, the question becomes concrete: which event on the right-moving family is simultaneous, in the moving sense, with the event at the origin?

On the board only the label $x = t$ is clearly legible. The shifted moving worldlines are reconstructed from the transcript and from the geometry as

$$x = vt, \qquad x = vt + L, \qquad x = vt + 2L. \tag{1.17}$$

One transcript line later drops the factor of t, but the construction itself makes the intended form unambiguous.

The lecture's rhythm at this point is exact: it first argues qualitatively that the two frames will disagree, and then asks, "Can we do it quantitatively?" The answer is yes.

Worked derivation. Let A be the point where the forward light ray $x = t$ meets the shifted worldline $x = vt + L$. Then A is determined by

$$x = t, \tag{1.18}$$

$$x = vt + L. \tag{1.19}$$

Substituting $x = t$ into the second equation gives

$$t = vt + L, \tag{1.20}$$

so

$$(1 - v)t = L, \tag{1.21}$$

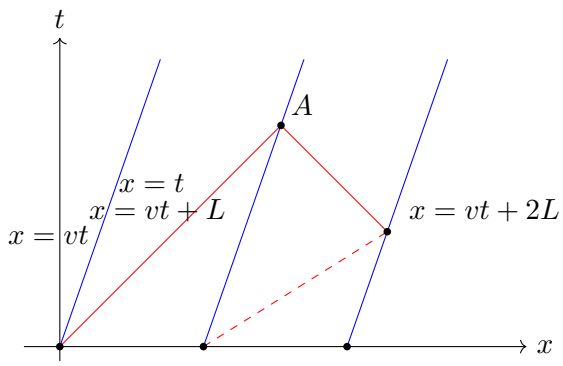


Figure 1.6: Reconstruction of the simultaneity argument. The screenshot preserves the board evidence; the redraw makes the equations explicit.

and therefore

$$t_A = \frac{L}{1-v}, \qquad x_A = \frac{L}{1-v}. \tag{1.22}$$

Now follow a backward-going light ray from A. A left-moving light ray has the form

$$x = -t + b, \tag{1.23}$$

or equivalently

$$x + t = b. \tag{1.24}$$

To determine b, use the fact that the ray passes through A. At A, we have $x = t = L/(1-v)$, so

$$b = x_A + t_A = \frac{2L}{1-v}. \tag{1.25}$$

Thus the backward light ray is

$$x + t = \frac{2L}{1-v}. \tag{1.26}$$

The event we want is where this backward light ray meets the next shifted worldline $x = vt + 2L$. So we solve

$$x + t = \frac{2L}{1-v}, \tag{1.27}$$

$$x = vt + 2L. \tag{1.28}$$

Substituting the second equation into the first yields

$$vt + 2L + t = \frac{2L}{1-v}. \tag{1.29}$$

Rearranging,

$$(1+v)t = 2L\left(\frac{1}{1-v} - 1\right) = 2L\frac{v}{1-v}, \tag{1.30}$$

and therefore

$$t = \frac{2Lv}{1-v^2}. \tag{1.31}$$

Putting this back into $x = vt + 2L$, we obtain

$$x = \frac{2L}{1 - v^2}. \tag{1.32}$$

What matters is not these coordinates by themselves, but their ratio:

$$\frac{t}{x} = v. \tag{1.33}$$

So the moving observer's simultaneity line through the origin is

$$t = vx. \tag{1.34}$$

This is the lecture's geometric payoff. The moving observer's time axis is

$$x = vt, \tag{1.35}$$

because that is the worldline of the moving origin. The moving observer's line of simultaneity is

$$t = vx, \tag{1.36}$$

because those are the points assigned the same moving time. The two are tilted symmetrically with respect to the light rays.

Restoring dimensions requires only one correction:

$$t = \frac{v}{c^2}x. \tag{1.37}$$

This formula makes clear why simultaneity appears absolute in ordinary life. The denominator c^2 is enormous. Only when velocities become appreciable compared with the speed of light does the tilt become visible.

1.7 From Tilted Axes to the Lorentz Transformation

The lecture now performs an instructive cleanup. It erases the clutter of the construction and keeps only what the construction has taught us: the unprimed axes, the primed axes, and the light rays. In units with $c = 1$, the picture has what Susskind calls a lovely symmetry.

The lecture later shifts on the board to upper-case X, T, but the substance is unchanged. We keep lower-case x, t throughout.

The geometry already tells us the zero-sets of the primed coordinates. Since the moving origin is the line $x = vt$, the quantity x' must vanish there. That is the surviving content of Galileo's old formula $x' = x - vt$: even if the exact coefficient changes, the zero-set must remain the same. This motivates

$$x' = \frac{x - vt}{F(v)}. \tag{1.38}$$

Likewise, because the moving simultaneity axis is the line $t = vx$, the quantity t' must vanish there. So we write

$$t' = \frac{t - vx}{G(v)}. \tag{1.39}$$

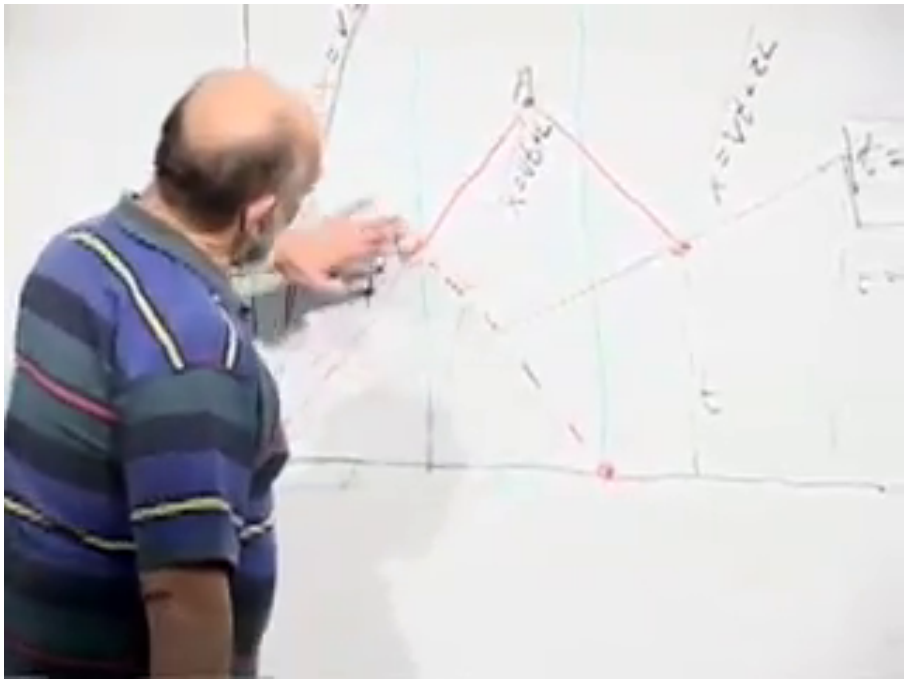


Figure 1.7: Symmetric $c = 1$ spacetime diagram.

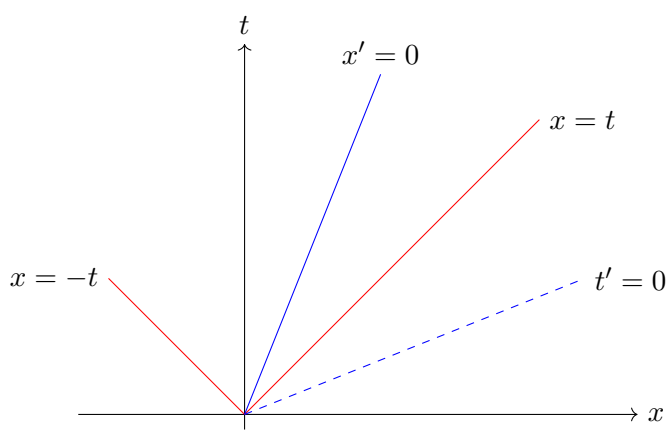


Figure 1.8: Cleaned $c = 1$ diagram showing the primed time axis, the primed simultaneity axis, and the two light rays.

Here $F(v)$ and $G(v)$ are unknown denominator functions. This matters: when the lecture later concludes that $F(v) = \sqrt{1-v^2}$, it is the denominator that equals that square root.

Now use the invariance of light speed. A light ray satisfying $x = t$ in the unprimed frame must satisfy $x' = t'$ in the primed frame. Substituting $x = t$ gives

$$x' = \frac{(1-v)t}{F(v)}, \tag{1.40}$$

$$t' = \frac{(1-v)t}{G(v)}. \tag{1.41}$$

These must agree for every such light ray, so

$$F(v) = G(v). \tag{1.42}$$

Hence

$$x' = \frac{x - vt}{F(v)}, \qquad t' = \frac{t - vx}{F(v)}. \tag{1.43}$$

We still do not know $F(v)$. To determine it, multiply through:

$$F(v)x' = x - vt, \tag{1.44}$$

$$F(v)t' = t - vx. \tag{1.45}$$

To solve for x, multiply the second equation by v and add it to the first. The vt terms cancel:

$$(1-v^2)x = F(v)(x' + vt'), \tag{1.46}$$

so

$$x = \frac{F(v)(x' + vt')}{1-v^2}. \tag{1.47}$$

By the same algebra,

$$t = \frac{F(v)(t' + vx')}{1-v^2}. \tag{1.48}$$

Now comes what Susskind calls the other half of the key. The forward and inverse transformations should have the same form, except that the sign of the relative velocity reverses. This reciprocity condition demands

$$\frac{F(v)}{1-v^2} = \frac{1}{F(v)}. \tag{1.49}$$

Multiplying through by $F(v)$, we obtain

$$F(v)^2 = 1 - v^2, \tag{1.50}$$

and therefore

$$F(v) = \sqrt{1-v^2}. \tag{1.51}$$

The Lorentz transformation in units with $c = 1$ is now fixed:

$$x' = \frac{x - vt}{\sqrt{1-v^2}}, \tag{1.52}$$

$$t' = \frac{t - vx}{\sqrt{1-v^2}}. \tag{1.53}$$

By reciprocity, the inverse relations are

$$x = \frac{x' + vt'}{\sqrt{1 - v^2}}, \tag{1.54}$$

$$t = \frac{t' + vx'}{\sqrt{1 - v^2}}. \tag{1.55}$$

To restore dimensions, we replace v^2 by v^2/c^2, and in the time equation we divide the mixed term by c^2:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \tag{1.56}$$

$$t' = \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}. \tag{1.57}$$

If one later introduces

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \tag{1.58}$$

these become the familiar textbook formulas. But the lecture reaches them first in denominator form, and it is worth keeping that sequence.

A direct check. The lecture closes by checking the defining property of the transformation. If

$$x = ct, \tag{1.59}$$

then

$$x' = \frac{(c - v)t}{\sqrt{1 - v^2/c^2}}, \tag{1.60}$$

$$t' = \frac{(1 - v/c)t}{\sqrt{1 - v^2/c^2}}. \tag{1.61}$$

Hence

$$\frac{x'}{t'} = \frac{c - v}{1 - v/c} = c. \tag{1.62}$$

So a right-moving light ray still travels with speed c in the primed frame. The same computation shows that a left-moving light ray satisfying $x = -ct$ becomes $x' = -ct'$.

The lecture also insists on the low-velocity limit. When $v \ll c$, the factors v^2/c^2 and $(v/c^2)x$ are negligible in ordinary circumstances, and the Lorentz transformation reduces to

$$t' \approx t, \qquad x' \approx x - vt. \tag{1.63}$$

Galileo is recovered as an approximation. But the new thing, the genuinely bizarre thing in Susskind's phrasing, is already visible: the surfaces $t =$ constant and $t' =$ constant are different surfaces. Simultaneity is frame-dependent.

1.8 Summary

This lecture unfolds in a deliberate order. We begin with a reference frame understood operationally as a lattice of meter sticks and synchronized clocks. We draw spacetime before we write equations. From that geometry we read the Galilean transformation and then test it: simultaneous distance is invariant, position and velocity are not, acceleration is, and so Newtonian mechanics acquires the right invariant form.

Only after Galileo has been allowed to look strong does the real problem appear. Maxwell and light do not fit Galilean kinematics. Einstein's response is not to abandon relativity, but to return to first principles and rebuild synchronization with light signals. From that reconstruction comes the relativity of simultaneity.

The concrete light-ray construction then gives the moving time axis $x = vt$ and the moving simultaneity axis $t = vx$ in units with $c = 1$. From there, plus the invariance of light speed and the reciprocity of the two frames, the Lorentz transformation follows. The lecture stops before length contraction and time dilation are derived. Its real achievement is more basic: it has shown us that time is not simply time, and that what counts as synchronous in one frame is not what counts as synchronous in another.

Chapter 2

Proper Time, Time Dilation, and Length Contraction

This lecture, though delivered inside the entanglement track, is really a special-relativity lecture. These notes follow Leonard Susskind's order of attack as preserved in the LazyingArt LLC course curation: start with Galileo, correct the kinematics so that light has the same speed for everybody, then use the resulting spacetime geometry to understand proper time, time dilation, the twin paradox, signal exchange, and finally length contraction. The argument is deliberately picture-first. We are not handed isolated formulas; we are shown why each one appears when it does.

2.1 Review: Galilean Relativity and Its Spacetime Grid

Susskind begins by fixing a practical convention. Sometimes the speed of light will be written explicitly as c, and sometimes it will be set equal to 1. We will follow that convention. Most of the algebra is cleaner in units with $c = 1$, and whenever dimensions matter we can restore the missing factors of c by dimensional analysis.

The pre-Einsteinian starting point is Galilean relativity. For motion along a single spatial direction,

$$x' = x - vt, \qquad t' = t. \tag{2.1}$$

This is Newton's world. The moving origin drifts through space, but time is universal. If we solve for the unprimed coordinates, we find

$$x = x' + vt = x' + vt', \qquad t = t'. \tag{2.2}$$

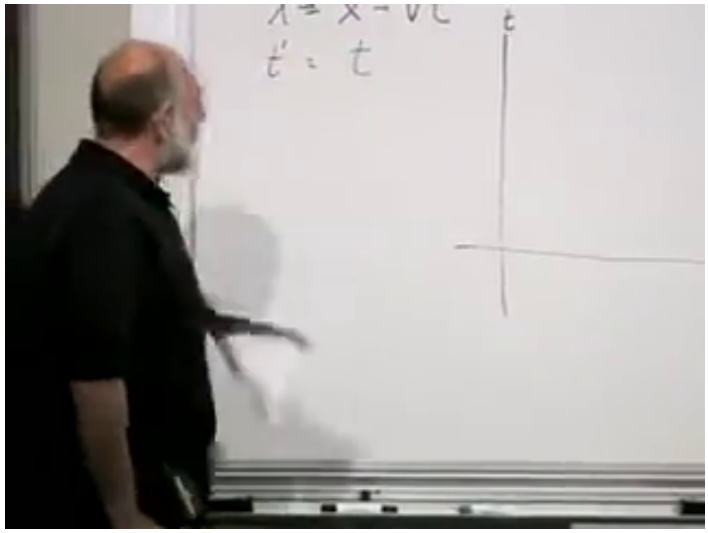
The structure is reciprocal: going from one observer to the other only reverses the sign of the velocity.

Susskind phrases the story in train-and-station language. He imagines himself on the moving train and the audience at rest on the platform, while also reminding us that the whole point of relativity is that either side may regard itself as the one at rest. But before Einstein is allowed to enter, one wants to see exactly what Galileo would have drawn.

The first geometric fact is that the moving observer's spatial origin is the locus $x' = 0$. In the unprimed coordinates this becomes

$$x = vt \iff x' = 0. \tag{2.3}$$

That line is the t'-axis, drawn on the same (x, t) graph as the original axes. The line tilts because the moving origin drifts, but the constant-time slices do not. Since $t' = t$, a surface of constant t' is also a surface of constant t. In Galilean relativity all observers share the same simultaneity.



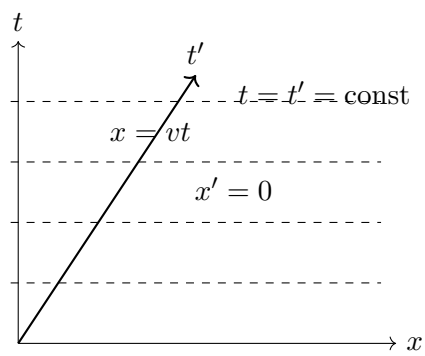


Figure 2.1: The board at the start of the Galilean review, followed by a clean reconstruction of the spacetime picture described immediately afterward in the lecture. The moving origin traces the slanted line $x = vt$, but the constant-time slices remain horizontal because $t' = t$.

This is the baseline. Einstein's correction will be intelligible only if we keep this old grid in view.

2.2 From Galileo to Lorentz

The reason the Galilean formulas fail is equally simple and equally profound: they are inconsistent with the statement that the speed of light is the same in every inertial frame. The corrected formulas are the Lorentz transformations. In units with $c = 1$,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \qquad t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{2.4}$$

Restoring dimensions gives

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}, \qquad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}. \tag{2.5}$$

Susskind makes two immediate points. First, for everyday velocities the correction is negligible:

$$\sqrt{1-\frac{v^2}{c^2}} \approx 1 \qquad \text{when} \qquad v \ll c. \tag{2.6}$$

So Newtonian physics survives as the low-velocity approximation. Special relativity is not a repudiation of Galileo; it is the refinement that becomes necessary when v approaches c.

Second, the really new feature is not the denominator in x' but the fact that t' is no longer equal to t. Newton believed in a universal time shared by everyone. Einstein's formula says that once the speed of light is universal, time itself must mix with space.

There is also a directional asymmetry built into the formula. If the relative motion is along the x-axis, then the transverse coordinates are unchanged:

$$y' = y, \qquad z' = z. \tag{2.7}$$

Only the coordinate along the direction of motion gets entangled with time.

Susskind also pauses on the denominator. When v approaches c, the factor $\sqrt{1-v^2/c^2}$ tends to zero. The algebra is warning us that one inertial frame cannot move at exactly the speed of light relative to another. Long before one reaches that point, however, measurable deviations from Galilean intuition begin to appear.

2.2.1 Question & Answer

Question. Why do y and z stay unchanged while x mixes with t? And what is the real invariant content of the transverse meter-stick argument?

Answer. The lecture's answer is operational and symmetrical. Imagine two meter sticks oriented transversely to the motion, one carried on the train and one at rest on the platform. Consider an observer halfway between the two frames, a bicyclist in Susskind's language, who sees them approaching with equal and opposite velocities. In that halfway frame the setup is completely symmetric. There is no reason for one transverse rod to pass the other looking shorter. So the transverse directions cannot contract.

But the deeper point is about events. When two endpoints meet, that meeting is a spacetime coincidence. One may mark it by a flash of light, but the flash is only evidence; the coincidence itself is the event. If two events occur at the same spacetime point in one frame, they occur at the same spacetime point in every frame. Thus the transverse rod argument is not merely a visual story. It rests on the invariance of events.

2.3 Light-Cone Coordinates and the Geometry of Lorentz Transformations

At this point Susskind deliberately changes the way he looks at the formulas. He says, in effect: let me show you another way to write the Lorentz transformation. First he gets rid of the c's, because they are a nuisance for the algebra, and then he adds and subtracts the two equations.

Adding the transformed coordinates gives

$$t' + x' = \frac{(t - vx) + (x - vt)}{\sqrt{1 - v^2}} \tag{2.8}$$

$$= \frac{(t + x)(1 - v)}{\sqrt{1 - v^2}} \tag{2.9}$$

$$= (t + x)\sqrt{\frac{1 - v}{1 + v}}. \tag{2.10}$$

Subtracting gives

$$t' - x' = \frac{(t - vx) - (x - vt)}{\sqrt{1 - v^2}} \tag{2.11}$$

$$= \frac{(t - x)(1 + v)}{\sqrt{1 - v^2}} \tag{2.12}$$

$$= (t - x)\sqrt{\frac{1 + v}{1 - v}}. \tag{2.13}$$

This is not merely algebraic housekeeping. The combinations

$$t + x, \qquad t - x \tag{2.14}$$

are coordinates along the two 45° null directions in a spacetime diagram with $c = 1$. In other words, they are adapted to the light cone itself. Under a Lorentz transformation, each of these null coordinates is simply rescaled, and the two scale factors are inverses of one another.

That is the geometric content Susskind wants us to see. A Lorentz transformation is a squeeze: one lightlike direction is stretched while the other is compressed by the reciprocal amount. The crucial thing is what does *not* change. The 45° directions remain 45°. Light rays stay light rays. That is precisely the geometric meaning of the principle that the speed of light is frame-independent.

Once one sees the Lorentz transformation this way, the next question presents itself almost automatically. If the transformation squeezes spacetime but leaves the light cone intact, what quantity does it preserve?

2.4 Proper Time as the Invariant Interval

Before he writes down the formula, Susskind does something wise. He explains what kind of object proper time is supposed to be. Imagine a collection of stationary observers, each carrying a synchronized clock. Their network defines a coordinate time. Now imagine one observer moving through that network with a wristwatch. The wristwatch does not carry the whole stationary grid along with it; it carries only its own ticking. Proper time will be the time read by that moving clock.

With that operational picture in the background, Susskind turns to the geometric analogue of Euclidean distance. For ordinary rotations in the plane, the invariant quantity is

$$s^2 = x^2 + y^2 = x'^2 + y'^2. \tag{2.15}$$

Under a Lorentz transformation, the corresponding invariant is different by a minus sign:

$$t'^2 - x'^2 = t^2 - x^2. \tag{2.16}$$

Proposition 2.1. *Under the Lorentz transformation in $1+1$ dimensions, the quantity $t^2 - x^2$ is invariant.*

Proof. Starting from

$$x' = \frac{x - vt}{\sqrt{1-v^2}}, \qquad t' = \frac{t - vx}{\sqrt{1-v^2}}, \tag{2.17}$$

we compute

$$\begin{aligned} t'^2 - x'^2 &= \frac{(t-vx)^2 - (x-vt)^2}{1-v^2} && (2.18)\\ &= \frac{t^2 - 2tvx + v^2x^2 - x^2 + 2xvt - v^2t^2}{1-v^2} && (2.19)\\ &= \frac{(1-v^2)t^2 - (1-v^2)x^2}{1-v^2} && (2.20)\\ &= t^2 - x^2. && (2.21) \end{aligned}$$

The cross terms cancel, and the common factor $1 - v^2$ drops out. □

In three spatial dimensions the corresponding statement is

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2. \tag{2.22}$$

The lecture mostly works along one axis, but this fuller expression makes clear that the invariant does not care which spatial direction we happened to choose for the relative motion.

Definition 2.2. If one event is the origin and a second event lies in the time-like region, its proper time from the origin is

$$\tau = \sqrt{t^2 - x^2} \tag{2.23}$$

in units with $c = 1$. More generally, one applies the same invariant to coordinate differences between two events:

$$\Delta\tau^2 = \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2. \tag{2.24}$$

Why the square root? Because the analogy is with ordinary distance. In Euclidean geometry the distance between two points is the square root of the invariant quadratic form. Here the invariant quadratic form has a minus sign in the spatial directions, but the logic is the same.

Now the sign difference begins to matter. If $t^2 > x^2$, the quantity under the square root is positive and the proper time is real. If $t^2 = x^2$, the proper time is zero. If $t^2 < x^2$, the separation is space-like, and in the lecture Susskind says bluntly that the proper time is imaginary. The polished statement is that space-like separated points are not related by ordinary elapsed clock time, but his warning is useful: Minkowski geometry is not Euclidean geometry with a new accent; the minus sign changes everything.

The operational interpretation now falls into place. For a clock at rest in the unprimed frame, $x = 0$, so

$$\tau = \sqrt{t^2} = t. \tag{2.25}$$

Thus proper time agrees with ordinary time for an observer at rest.

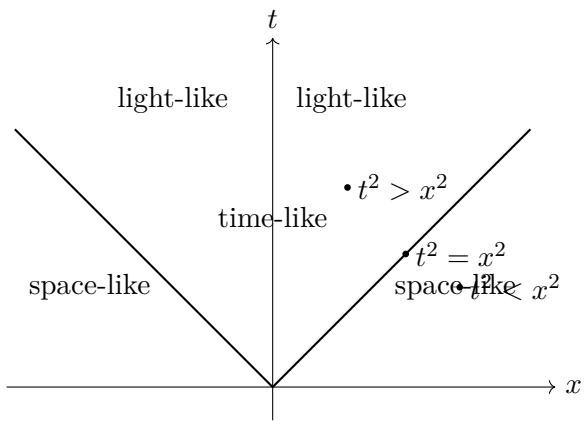


Figure 2.2: The light cone in $1+1$ dimensions. Inside the cone the interval is time-like and the proper time is real; on the cone the proper time is zero; outside the cone the separation is space-like.

For a moving clock, we may always transform to the frame in which the clock is momentarily at rest. Along that worldline $x' = 0$, and so

$$\tau = \sqrt{t'^2 - x'^2} = t'. \tag{2.26}$$

This is the heart of the lecture's definition: moving clocks tick off proper time. The invariant interval is not just a formula that everyone agrees on; it is the quantity that the traveling wristwatch actually reads.

2.5 Time Dilation and Relativity of Simultaneity

Now Susskind uses proper time as a tool. Consider the simplest possible moving worldline: the origin of the primed frame itself. In the unprimed coordinates this worldline is

$$x = vt \iff x' = 0. \tag{2.27}$$

So the slanted line in the spacetime diagram is not merely a particle trajectory. It is the moving observer's time axis.

To find the time read by the moving clock, use the invariant interval:

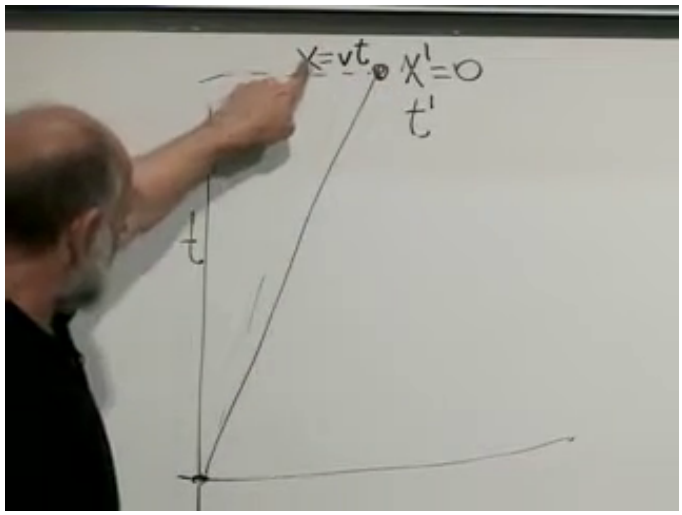
$$t'^2 - x'^2 = t^2 - x^2. \tag{2.28}$$

On the moving origin, $x' = 0$, while in the stationary frame $x = vt$. Therefore

$$t'^2 = t^2 - x^2 \tag{2.29}$$
$$= t^2 - v^2 t^2 \tag{2.30}$$
$$= (1 - v^2)t^2. \tag{2.31}$$

Taking the positive square root, we obtain

$$t' = t\sqrt{1 - v^2}. \tag{2.32}$$



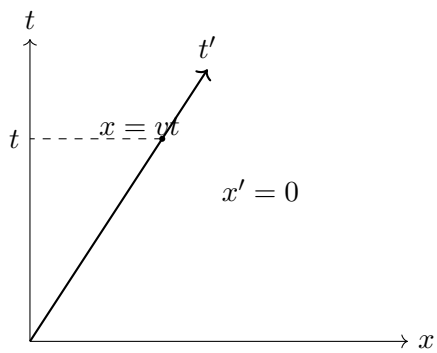


Figure 2.3: The key board frame for the moving-origin construction, paired with a clean reconstruction. The slanted line is identified on the board with both $x = vt$ and $x' = 0$; in the primed frame it is the t'-axis.

With the speed of light restored,

$$t' = t\sqrt{1 - \frac{v^2}{c^2}}. \tag{2.33}$$

This is the famous time-dilation formula. Between the same two spacetime events, the moving clock records less elapsed time than the synchronized stationary grid. In the stationary observer's description, the moving clock runs slow.

The lecture is slightly ragged in speech at this point, but the invariant interval leaves no ambiguity. Once one has written $t'^2 = (1 - v^2)t^2$, the result is fixed.

2.5.1 Question & Answer

Question. How can each observer say that the other clock runs slow? That sounds contradictory.

Answer. The contradiction is only apparent, because the two observers are not comparing the same pairs of events. The stationary observer uses surfaces $t = \text{const}$. The moving observer uses

surfaces $t' = \text{const}$. From

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}, \tag{2.34}$$

we see that holding t' fixed means

$$t - vx = \text{const}. \tag{2.35}$$

So the moving observer's simultaneity slices are tilted relative to the stationary observer's.

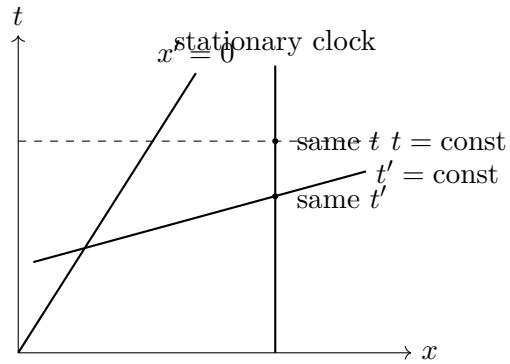


Figure 2.4: Relativity of simultaneity. The stationary observer compares clock readings on a horizontal slice $t = \text{const}$, while the moving observer compares them on a tilted slice $t' = \text{const}$. The two statements about "the other clock" therefore refer to different event pairs.

This is the lecture's resolution of the reciprocity puzzle. Moving clocks appear slow to stationary observers, and stationary clocks appear slow to moving observers, because the very notion of "at the same time" is frame-dependent.

2.6 Twin Paradox, Ideal Clocks, and Accelerated Motion

Having established that clocks measure proper time, Susskind turns to the twin paradox. Two twins are born at the same event. One stays at home on a vertical worldline. The other is shot outward at speed v, turns around, and comes back. When they meet again, they compare everything that can function as a clock: wristwatches, heartbeats, beard growth, metabolic aging. The question is not which clock was more poetic. The question is which worldline accumulated more proper time.

Let the stay-at-home twin measure a time T on the outward leg and another T on the return leg. Then

$$\tau_{\text{home}} = 2T. \tag{2.36}$$

For the traveling twin, the turning point has coordinates

$$x = vT, \qquad t = T. \tag{2.37}$$

Therefore one leg contributes

$$\tau_{\text{leg}}^2 = T^2 - x^2 \tag{2.38}$$

$$= T^2 - v^2T^2 \tag{2.39}$$

$$= (1 - v^2)T^2, \tag{2.40}$$

so

$$\tau_{\text{leg}} = T\sqrt{1 - v^2}. \tag{2.41}$$

The return leg contributes the same amount, hence

$$\tau_{\text{travel}} = 2T\sqrt{1 - v^2}. \tag{2.42}$$

Since $1 - v^2 < 1$, the traveling twin returns younger.

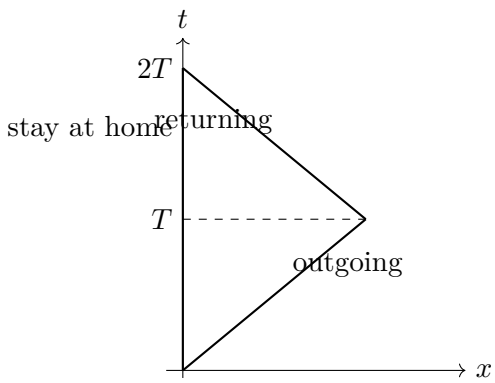


Figure 2.5: Twin paradox as a comparison of proper times along two worldlines: one straight and one broken. The clocks disagree because the paths through spacetime are different, not because the geometry is inconsistent.

Susskind is careful here. He does not leave the result as a mere slogan. The traveling twin is younger for the same reason that a broken path in geometry has a different length from a straight one. Minkowski geometry is unfamiliar, but it is still geometry.

2.6.1 Question & Answer

Question. Does the presence of acceleration destroy the argument? After all, the turnaround is not an inertial segment.

Answer. No one should pretend the acceleration is absent. It is present, and it is one of the things that breaks the symmetry between the twins. But that does not spoil the proper-time calculation.

First, the sharp corner in the spacetime diagram is only an idealization. We may smooth it into a finite turnaround without changing the conclusion.

Second, the lecture introduces an additional physical assumption: the assumption of an ideal clock. A sufficiently small, well-built clock is taken to be insensitive to modest accelerations, so that it continues to tick off proper time. Then a curved worldline can be approximated by many short inertial segments, and one adds the proper times segment by segment:

$$\tau_{\text{path}} \approx \sum_i \sqrt{\Delta t_i^2 - \Delta x_i^2}. \tag{2.43}$$

This is the spacetime analogue of computing the length of an ordinary curve by dividing it into many small straight pieces.

Susskind is explicit that this ideal-clock behavior is a physical assumption, not something proved from the Lorentz algebra alone. Small, faithful clocks may be engineered to withstand the relevant accelerations; big sloppy clocks may not. The geometry tells us what an ideal clock records.

The lecture then pushes the same issue into a more operational form. Instead of comparing final clock readings only, we imagine marking one-second intervals of proper time along both worldlines and sending light pulses back and forth. That way one asks not just what the clocks finally say, but how each observer actually receives the other's signals.

2.7 Signals, Closed Spatial Loops, and Length Contraction

Now the lecture leaves the purely geometrical comparison of two final clock readings and asks what each observer actually sees. Suppose the stay-at-home twin emits one light pulse every second of proper time, and the traveling twin does the same. On the outward leg the received pulses are stretched out; on the inward leg they become bunched together. Susskind explicitly identifies this as a Doppler-type effect, though he does not need the full formalism of relativistic Doppler shift to make the point.

What matters is the rhythm of reception. Early pulses arrive late and sparse; later pulses arrive fast and compressed. No pulses go missing. Each observer eventually receives all the pulses sent by the other, but at rates that reflect the relative motion. This operational discussion is important in the lecture because it keeps proper time from becoming a purely formal invariant. The geometry is read out by actual light signals.

A question about energy is raised at this point and then postponed. Susskind refuses to answer it too early. At this stage the lecture is about coordinates, clocks, and signals, not yet about the full relativistic bookkeeping of energy for accelerated observers.

2.7.1 Question & Answer

Question. What if one spatial direction were compact, so that an observer could move uniformly around a closed loop and return without the usual turn-around? Would the asymmetry survive?

Answer. Here the lecture becomes exploratory and deliberately unfinished. Susskind first says that the asymmetry should survive, because the moving observer would discover a surprise: the same places would recur. But he then pulls back and refuses to give an unthought-out answer. The right way to preserve this beat is not to turn it into a polished theorem. It is to record the qualitative claim and the fact that the full geometry is postponed.

Only after this detour does Susskind return to the next standard effect, and he does so with a piece of methodological advice strong enough to count as doctrine: when you want to understand a special-relativity problem, draw a picture.

Take a meter stick at rest in the unprimed frame. Its left end lies at $x = 0$, its right end at $x = 1$. As time passes, the rod sweeps out a ribbon in spacetime, bounded by two vertical worldlines. The question is: what length does the moving observer assign to that ribbon?

The answer begins with the meaning of measurement. A length in the moving frame is the separation of the rod's endpoints at the same instant of *moving-frame* time. That is the local conceptual

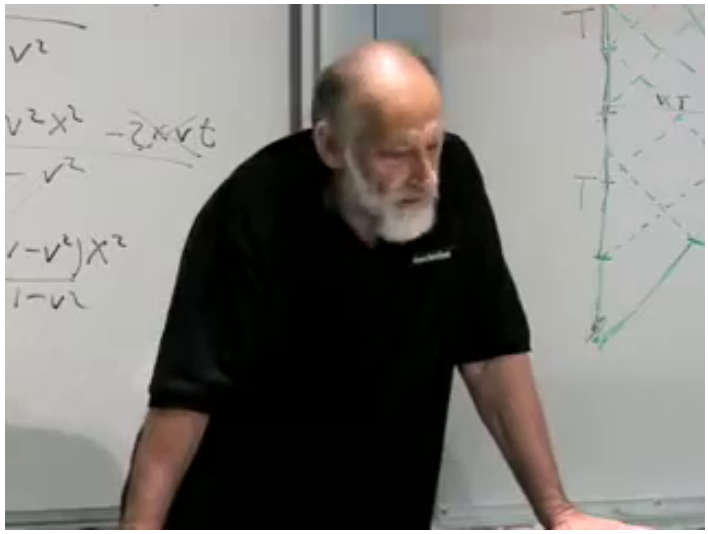


Figure 2.6: The later board during the compact-direction discussion. The left-board algebra is too fragmentary to transcribe safely, but the right-hand geometric sketch shows that the lecture had shifted into an asymmetry puzzle rather than a finished derivation.

obstacle in this part of the lecture. One must not look at arbitrary points on the ribbon; one must intersect the ribbon with a single constant-t' slice.

The clean relation is

$$t - vx = \text{const} \tag{2.44}$$

for constant t' in units with $c = 1$. In particular, the slice through the origin with $t' = 0$ satisfies

$$t = vx. \tag{2.45}$$

The lecture's spoken line wavers momentarily here, but this is the correct standard form. When the slice hits the right endpoint worldline $x = 1$, the corresponding unprimed time is therefore

$$t = v. \tag{2.46}$$

Because the separation from the origin to the selected endpoint is now space-like, it is cleaner to reverse the sign of the invariant interval and speak of the squared proper distance:

$$x'^2 - t'^2 = x^2 - t^2. \tag{2.47}$$

On the chosen slice, $t' = 0$. At the selected endpoint, $x = 1$ and $t = v$. Therefore

$$x'^2 = 1 - v^2. \tag{2.48}$$

Equivalently, in the lecture's board language one may write

$$-x'^2 = v^2 - 1, \tag{2.49}$$

which is the same statement with the signs not yet cleaned up. Either way the conclusion is

$$x' = \sqrt{1 - v^2}. \tag{2.50}$$

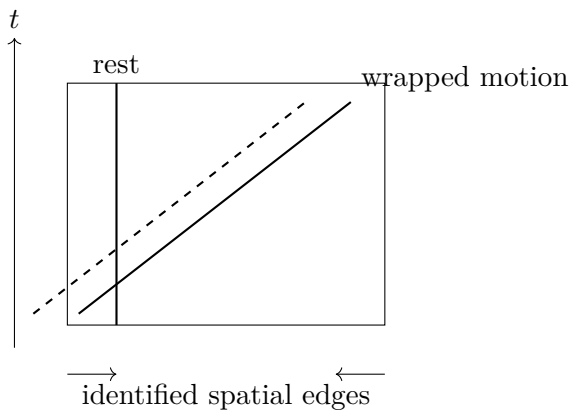


Figure 2.7: A cautious schematic reconstruction of the compact-direction question, drawn on an unwound strip. This is not a completed theorem from the lecture; it is only a clean rendering of the qualitative asymmetry being discussed.

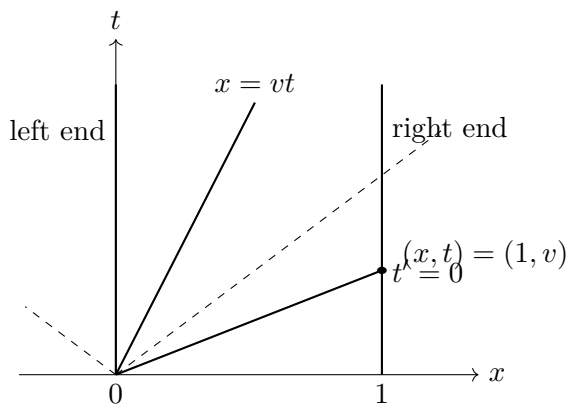


Figure 2.8: Length contraction from the spacetime picture Susskind insists on drawing. The rod is at rest in the unprimed frame, so its endpoints trace a vertical ribbon. The moving observer measures the rod by slicing that ribbon with a constant-t' line.

Thus a rod of rest length 1 is seen in the moving frame to have length

$$L' = \sqrt{1 - v^2}. \tag{2.51}$$

For a general rest length L,

$$L' = L\sqrt{1 - v^2} \tag{2.52}$$

in units with $c = 1$, or

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \tag{2.53}$$

when c is restored.

This is the companion to time dilation. Time dilation came from following a moving worldline and comparing elapsed proper time. Length contraction comes from slicing a stationary ribbon by the moving observer's simultaneity plane. Both are consequences of the same Lorentz mixing of space and time.

The lecture's final operational payoff is immediate. In the astronaut's frame the distance between distant stars is Lorentz-contracted, so a long trip may take a short proper time without anything moving faster than light. The stars are not seen to outrun light; they are simply seen at a reduced spatial separation in the moving frame.

2.8 Summary

The lecture begins with Galileo on purpose. In the Galilean picture the moving origin tilts, but time slices remain common to everyone. Einstein's correction changes exactly that point: x and t mix, the light cone is preserved, and the invariant quantity is not $x^2 + y^2$ but $t^2 - x^2 - y^2 - z^2$.

From there the lecture unfolds in a strict order. First we reinterpret the Lorentz transformation as a squeeze along null directions. Then we identify the invariant interval. Then we define proper time and recognize it as the time read by a moving clock. Time dilation follows from evaluating that invariant on the worldline $x = vt$. The reciprocity puzzle is resolved by relativity of simultaneity. The twin paradox is resolved by comparing proper times along different paths, with the ideal-clock assumption made explicit. Signal exchange gives the operational picture. A compact-direction asymmetry is raised but deliberately left provisional. Only after that does Susskind return to length contraction, again by drawing the spacetime picture first.

What remains after all the formulas is really a method. Draw the diagram. Identify the invariant. Ask which events are being compared. Then let the geometry tell you what clocks, rods, and signals will do.

Chapter 3

From Heavy-Ion Fireballs to Four-Vectors

These notes follow Leonard Susskind's lecture, with curation by LazyingArt LLC, and keep close to the order in which the lecture actually unfolds. We begin in the middle of a discussion about heavy-ion collisions and a five-dimensional black-hole picture of QCD. Only after that detour has done its motivational work does the lecture pivot back to special relativity. From there the progression is deliberate: a paradox about simultaneity, a cleanup of spacetime notation, rotations and Lorentz transformations in matrix form, the composition of boosts, and finally the relativistic replacement for ordinary velocity, together with the first appearance of proper acceleration and four-momentum.

3.1 Heavy-Ion Collisions, Horizons, and the Fifth-Dimensional Picture

The lecture does not open from a formal beginning. It opens with a question about the complexity of the black hole that is supposed to describe the aftermath of a violent hadronic collision. Susskind answers not with a formula, but with a picture.

We are asked to imagine a fifth dimension extending between a floor and a ceiling. There is a gravitational field in that fifth-dimensional direction, and it pulls excitations downward toward the floor. The excitations may be thought of schematically as little strings. When they settle near the floor, they are hadrons: protons, neutrons, mesons, the large composite objects of QCD. If they remain near the ceiling, they are more elementary excitations: gravitons, photons, and other non-hadronic degrees of freedom. In that simple geometric language, the distinction between composite hadrons and more elementary quanta is encoded in where they live along the extra dimension.

Now collide two hadrons. In the five-dimensional picture this means colliding two objects that live near the floor. They do not produce a small rigid remnant. They produce a large splash of energy, a puddle or lump deposited in the gravitational field. That lump spreads, settles, and is usefully described as a black-hole-like object. Susskind is careful about what kind of black hole this is. It is not an astronomical four-dimensional black hole, nor the Planckian object one might imagine hanging from the ceiling. It is a five-dimensional horizon-bearing lump sitting on the floor. It has a

horizon, temperature, entropy, and the familiar thermodynamic attributes of black holes.

At very high energies the incoming hadrons are Lorentz contracted into thin pancakes. Two such pancakes collide and produce a large hot region, more like a fluid lump than a rigid body. The gravitational field pulls this lump down and spreads it until it comes to something like equilibrium. If that were the whole story, the lump would simply sit there. But it radiates. In this picture the outgoing hadrons are identified with Hawking radiation emitted from the black-hole-like object. The swarm of particles seen in the detector is the field-theoretic image of horizon radiation.

The lecture then pauses over the status of this picture. One can take the fifth dimension literally, or one can regard it as a mathematically equivalent description of strongly coupled QCD, useful precisely because it turns impossible quantum-field-theoretic calculations into more tractable gravitational ones. Susskind leaves the ontological question open, but he does not leave the practical point open: the dual picture captures real features of the quark-gluon plasma, especially its unexpectedly low viscosity, in a way that was not anticipated in the older weak-coupling mindset. In that sense the picture has already earned its keep.

3.2 Collider Energetics and Why Beam Geometry Matters

The discussion then shifts, under pressure from the room, from conceptual interpretation to collider practice. Why collide two beams head-on instead of firing one beam into a stationary target? Why must accelerators be so large? Why does beam geometry matter so much? These questions are not yet the main subject of the lecture, but they motivate the return to relativity by forcing us to think about energy in the center-of-mass frame.

For two highly relativistic particles with equal and opposite momenta p, the lecture-level estimate is that the center-of-mass energy is simply

$$E_{\text{cm}}^{\text{head-on}} \approx 2p \qquad (c = 1). \tag{3.1}$$

Here we are using the relativistic fact that, for very energetic particles, energy and momentum are approximately equal in $c = 1$ units.

If instead a fast beam of momentum p strikes a stationary target of mass m, the center-of-mass energy is much smaller. The lecture gives the rough scaling

$$E_{\text{cm}}^{\text{fixed target}} \sim \sqrt{mp} \qquad (c = 1), \tag{3.2}$$

to emphasize that a tremendous amount of beam energy is wasted when one of the two participants is left at rest. The purpose of the estimate is not to provide a careful invariant-mass derivation, but to make the practical point vivid: a head-on collider is enormously more efficient at concentrating energy into the interaction region.

That is the first engineering lesson. The second concerns size. Electric and magnetic fields can only be made so strong before engineering and material constraints intervene. In a linear accelerator one gives the particle repeated pushes as it moves along. More distance means more pushes and therefore more energy. In that approximate sense the energy of the machine scales with the useful accelerating length.

There is also a second limitation, and the lecture stresses that it is very important. Charged particles moving in circles are accelerated even when their speed is constant, simply because their direction is

changing. The tighter the circle, the larger the acceleration. Large acceleration produces radiation, and radiation means energy loss. If the circular path is too tight, the machine pours energy into the beam only to have the beam radiate it away.

That is why, if the goal is to dump a large amount of energy into a small volume and see what comes out, one would like to accelerate particles as nearly in a straight line as possible. Circular machines survive because one can use the same bending magnets over and over again, repeatedly pushing the beam around the ring; but the price is synchrotron radiation.

The electron–proton contrast fits naturally into this picture. Electrons are much lighter than protons. To carry a given energy they must be pushed much closer to the speed of light, and in a circular machine that means greater acceleration and greater radiative loss. Protons, being much heavier, can circulate with much less energy loss at the same speed and therefore carry much more useful energy in a circular collider. The lecture's point is not to turn this into accelerator design theory, but to clear away the practical questions before making the announced return to relativity.

That announcement is explicit. Having answered the room, Susskind says, in effect: now let us get back to the special theory of relativity.

3.3 Relativity of Simultaneity Through a Spacetime Diagram

The re-entry point is not formalism but paradox. Special relativity, as Susskind presents it here, is best entered by confronting the conceptual mistake that repeatedly causes trouble: we carry into relativity a pre-relativistic idea of simultaneity, and then we are surprised when the answers seem contradictory.

The example is the garage-and-limousine puzzle. A short garage and a long limousine approach one another. In the garage frame the moving limousine is Lorentz contracted, so one expects that it can fit. In the limousine frame the moving garage is Lorentz contracted, so one expects that it cannot fit. The puzzle is designed to force us to ask exactly what we mean when we say that the whole limousine is inside the garage.

3.3.1 Question & Answer

Question. Can the limousine ever fit entirely inside the garage?

Answer. The lecture first makes a practical adjustment: open the back door of the garage. We are not interested in crashing the driver into the rear wall. We want only to know whether there is any instant at which the whole car is inside.

Now draw a spacetime diagram. Time is vertical, space is horizontal, and because we work in one spatial dimension here, x is the only spatial coordinate we need. In units with $c = 1$, light rays travel on lines at 45°, so the light cone gives the geometry its basic orientation. The garage frame has the ordinary t-axis and x-axis. The moving frame has a tilted t'-axis and a tilted x'-axis. Those tilted axes are the visual representation of the fact that the moving observer's notion of simultaneity is different from that of the stationary observer.

The front and back ends of the garage are at fixed positions in the garage frame, so their worldlines are vertical. The front and back ends of the limousine are moving, so their worldlines are slanted

and parallel. The question "Is the whole limousine inside the garage?" means: is there a slice of simultaneity on which both the front and rear ends of the limousine lie between the two garage worldlines?

In the garage frame the answer is yes. There are horizontal lines—constant t slices—for which the back end of the limousine has already entered and the front end has not yet emerged. On such a slice the whole Lorentz-contracted limousine lies inside the garage.

In the limousine frame the answer is no. There, simultaneity is measured along lines parallel to the x'-axis, not along horizontal lines. If we take the event at which the rear end enters and draw the moving observer's simultaneous slice through it, that slice meets the front worldline only after the front has already gone out the back. On the limousine's own slice of simultaneity, the whole car is never inside at once.

There is therefore no contradiction. The hidden word was "simultaneously." Once we distinguish the stationary observer's simultaneity from the moving observer's simultaneity, the paradox evaporates. The lecture then adds an important geometric remark: the segment cut out of the limousine by a horizontal slice is its Lorentz-contracted length in the garage frame, while the proper length of the limousine is measured on a slice simultaneous in the limousine frame. Those are not the same slice. The paradox lives in that difference.

This little puzzle is not peripheral to the chapter. It is the lecture's deliberate way back into relativity. Diagram first, moral second: simultaneity is frame-dependent, and many apparent contradictions are nothing more than a failure to say in which frame the word "simultaneous" is being used.

3.4 Four Coordinates, Covariant Notation, and the Invariant Interval

Only now does the lecture turn to the formal machinery, and even here Susskind insists that most of the mathematics is merely notation. The point of the notation is not abstraction for its own sake. The point is to package the four coordinates in a way that keeps control of the minus signs.

We begin with the four coordinates of spacetime. In the lecture these are lumped together into a single object, with the slightly idiosyncratic convention that the time coordinate is the 0-component even though it is often written last:

$$X^1 = x, \qquad X^2 = y, \qquad X^3 = z, \qquad X^0 = t. \tag{3.3}$$

Thus, schematically,

$$X^\mu = (x, y, z, t), \tag{3.4}$$

with the understanding that the final entry is really X^0.

Susskind then introduces the covariant form of the same object. Nothing new has been added physically. We are describing the same spacetime point, but in a notation that flips the sign of the spatial components:

$$X_\mu = (-x, -y, -z, t). \tag{3.5}$$

This is the lecture's sign convention. In modern language, it is the signature $+---$, but the lecture does not need that phrase. It needs only the operational rule: lower the index and the first three components change sign.

A quick example, very much in the spirit of the lecture, is useful. If a point has coordinates

$$(x, y, z, t) = (3, 2, 4, 10), \tag{3.6}$$

then

$$X^\mu = (3, 2, 4, 10), \qquad X_\mu = (-3, -2, -4, 10). \tag{3.7}$$

This is not a different point. It is the same point written in a notation that is designed to make invariant expressions come out cleanly.

The main invariant then appears immediately:

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \tag{3.8}$$

This is exactly the proper interval from the origin to the point (x, y, z, t). In differential form,

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2, \tag{3.9}$$

again in $c = 1$ units. Along a timelike worldline, $d\tau$ is the proper time measured by a clock moving with the particle.

At this point Einstein's summation convention becomes useful. Since expressions of the form "sum over one time component with a plus sign and three spatial components with minus signs" occur constantly in relativity, we adopt the rule that whenever an index appears once upstairs and once downstairs, we sum over it. Thus we no longer write the summation sign explicitly; we simply write

$$X_\mu X^\mu, \tag{3.10}$$

and the sum over $\mu = 0, 1, 2, 3$ is understood.

This is the lecture's real point about notation. Relativity generates many expressions of this kind, and without a compact device the algebra becomes cluttered with repeated sums and signs. The notation does not replace the geometry; it stabilizes it.

3.5 Rotations, Matrices, and the Structure of Lorentz Transformations

The lecture now shifts to transformation laws in general. Before mixing time and space, Susskind takes the easier symmetry first: ordinary spatial rotations. This is not a digression. It is a pedagogical ramp. One learns how a transformation acts on components in a familiar case before returning to Lorentz transformations.

Consider a rotation about the z-axis by an angle θ. Then

$$x' = x\cos\theta + y\sin\theta, \tag{3.11}$$

$$y' = -x\sin\theta + y\cos\theta, \tag{3.12}$$

$$z' = z. \tag{3.13}$$

The point of the example is not just the formula itself. It is what the formula preserves. The squared distance from the origin remains unchanged. We can verify this directly:

$$x'^2 + y'^2 = (x\cos\theta + y\sin\theta)^2 + (-x\sin\theta + y\cos\theta)^2 \tag{3.14}$$

$$= x^2(\cos^2\theta + \sin^2\theta) + y^2(\sin^2\theta + \cos^2\theta) + 2xy\sin\theta\cos\theta - 2xy\sin\theta\cos\theta \tag{3.15}$$

$$= x^2 + y^2. \tag{3.16}$$

Since $z' = z$, we conclude that

$$x'^2 + y'^2 + z'^2 = x^2 + y^2 + z^2. \tag{3.17}$$

That is the lecture's template for symmetry: the components change, but a geometrically meaningful quantity remains invariant.

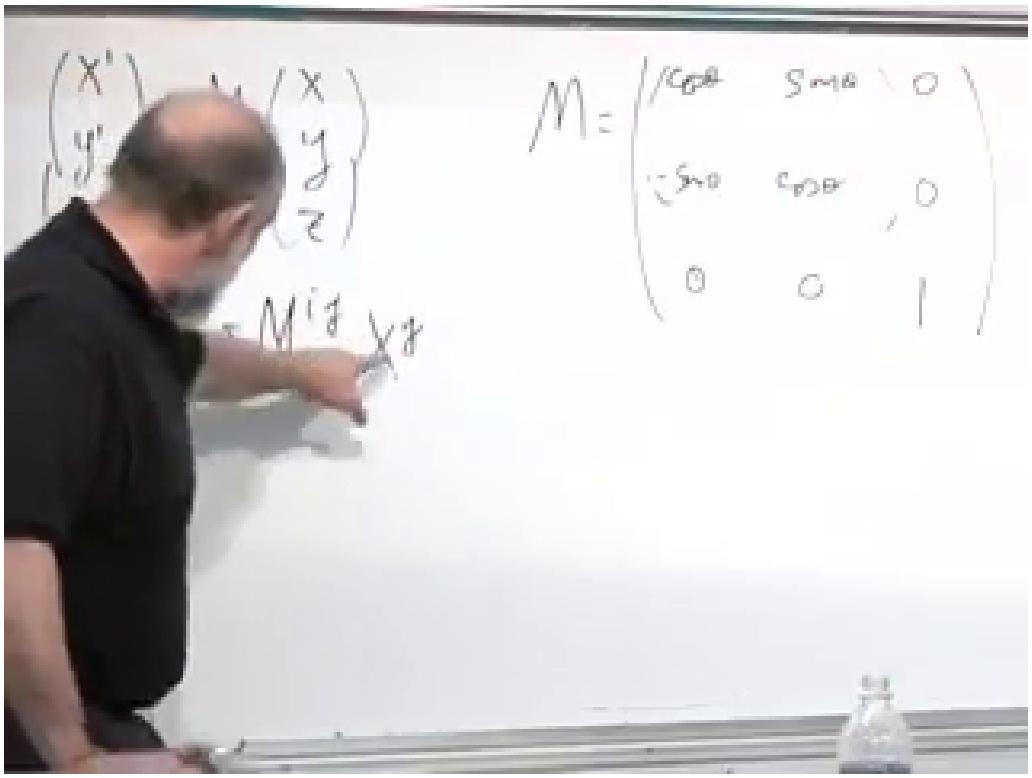


Figure 3.1: Rotation matrix acting on spatial coordinates.

The lecture then rewrites the same transformation in matrix language. That board layout is important and should be preserved. On the left is the vector relation, on the right the explicit matrix:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = M \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad M = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3.18}$$

Beneath the vector equation the board shows an indexed form, partly hidden by Susskind's arm. The natural cleaned reconstruction is

$$x'^i = M^i{}_j \, x^j, \tag{3.19}$$

with $i, j = 1, 2, 3$. We should regard this as a cautious normalization of the board line rather than a letter-for-letter transcription.

Once this is in hand, the move to Lorentz transformations becomes natural. A boost along the

x-axis mixes x and t, while leaving y and z untouched. In the lecture's $c = 1$ units,

$$x' = \frac{x - vt}{\sqrt{1 - v^2}}, \tag{3.20}$$

$$y' = y, \tag{3.21}$$

$$z' = z, \tag{3.22}$$

$$t' = \frac{t - vx}{\sqrt{1 - v^2}}. \tag{3.23}$$

It is convenient here to write

$$\gamma = \frac{1}{\sqrt{1 - v^2}} \qquad (c = 1), \tag{3.24}$$

so that the boost takes the compact form

$$x' = \gamma(x - vt), \tag{3.25}$$

$$t' = \gamma(t - vx). \tag{3.26}$$

In matrix form,

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & -\gamma v \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma v & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{3.27}$$

This is the cleaned standard form of the board calculation. The spoken algebra in the lecture is somewhat messy, but the intended object is clear: a Lorentz transformation is also a matrix acting linearly on a column of components.

At this stage the lecture makes another conceptual point. Rotations and boosts together generate all inertial frames related by uniform motion. If the relative motion is not along one of the original coordinate axes, we first rotate the spatial axes until the motion lies along the new x-direction, then perform the boost there. That is why the rotation section is not ornamental. It is part of the general structure of relativistic symmetry.

The first substantial payoff of the matrix language comes from composing boosts. Suppress the inert y and z directions and write the x-t boost as

$$B(v) = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}, \qquad \gamma_v = \frac{1}{\sqrt{1 - v^2}} \qquad (c = 1). \tag{3.28}$$

Now compose a boost of velocity v with a boost of velocity u:

$$B(u)B(v) = \gamma_u \gamma_v \begin{pmatrix} 1 & -u \\ -u & 1 \end{pmatrix} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \tag{3.29}$$

$$= \gamma_u \gamma_v \begin{pmatrix} 1 + uv & -(u + v) \\ -(u + v) & 1 + uv \end{pmatrix}. \tag{3.30}$$

Susskind's lecture does not insist on proving immediately that this composite matrix has exactly the standard boost form. Instead, he uses the easier observation: if it is a boost, then the effective velocity is read off from the ratio of the off-diagonal entry to the diagonal entry. Thus

$$w = \frac{u + v}{1 + uv} \qquad (c = 1). \tag{3.31}$$

Restoring the speed of light gives the standard relativistic addition law,

$$w = \frac{u+v}{1+uv/c^2}. \tag{3.32}$$

Newton would have written $w = u + v$. Einstein inserts the denominator.

The lecture immediately checks the moral of the formula. If u and v are both close to the speed of light, the numerator tries to become $2c$, but the denominator simultaneously approaches 2. The result does not exceed c. Even at the extreme edge, the composition of subluminal velocities remains subluminal.

3.6 Proper Velocity as a Four-Vector and as a Tangent Vector

At this point the lecture changes gears again. We have learned how coordinates transform; now we ask what kinds of physical quantities transform naturally with them. Susskind says explicitly that before speaking about velocity we should speak about four-vectors in general.

A four-vector v^μ is any object that transforms under Lorentz transformations in the same way the coordinate vector X^μ transforms. It has four components, not merely three. Its covariant form is obtained, exactly as before, by changing the signs of the spatial components:

$$v_\mu = (-v^1, -v^2, -v^3, v^0). \tag{3.33}$$

Its invariant norm is

$$v_\mu v^\mu = (v^0)^2 - (v^1)^2 - (v^2)^2 - (v^3)^2, \tag{3.34}$$

and for two four-vectors w^μ and v^μ, the invariant contraction is

$$w_\mu v^\mu = w^0 v^0 - w^1 v^1 - w^2 v^2 - w^3 v^3. \tag{3.35}$$

These are the relativistic analogs of the norm and dot product in ordinary Euclidean geometry. Their key property is invariance: the components change from frame to frame, but the contraction does not.

Only now does the lecture ask what relativistic velocity ought to be.

3.6.1 Question & Answer

Question. Why is the ordinary velocity $(dx/dt, dy/dt, dz/dt)$ not the right relativistic vector?

Answer. It is a perfectly good classical quantity, but it does not extend naturally to a four-vector. If we try to append a time component in the obvious way, we get

$$\frac{dt}{dt} = 1, \tag{3.36}$$

which is trivial. More importantly, the resulting object does not transform simply under Lorentz transformations. The lecture's point is not that ordinary velocity is meaningless, but that it is not the object that behaves most cleanly in relativistic geometry.

The cure is to divide not by coordinate time, which depends on the frame, but by proper time, which does not.

Thus the relativistic velocity is defined by

$$u^\mu = \frac{dX^\mu}{d\tau}. \tag{3.37}$$

The lecture calls this the proper velocity. The word "proper" refers to the parameter in the denominator: the derivative is taken with respect to proper time along the worldline.

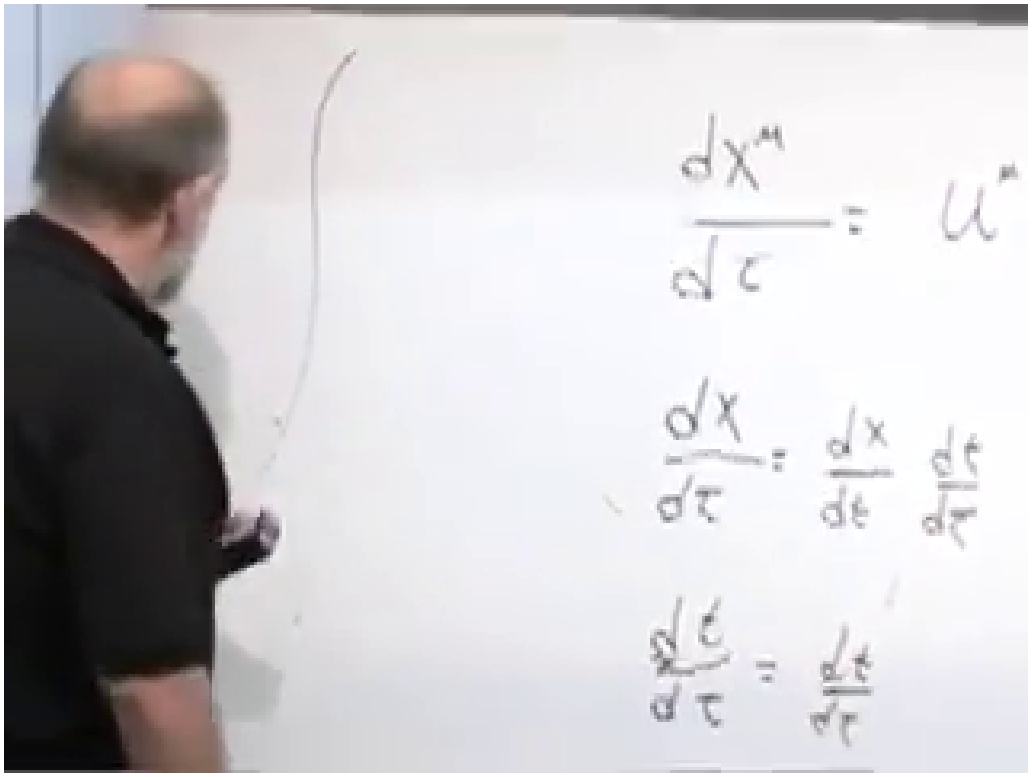


Figure 3.2: Four-velocity and the tangent-vector construction.

The board evidence is especially helpful here because it shows the defining equation placed beside a large curve on the board. The lecture is already telling us that the object is not merely algebraic. It is geometric.

The chain rule gives the first connection to the old three-velocity:

$$\frac{dx}{d\tau} = \frac{dx}{dt}\frac{dt}{d\tau}, \tag{3.38}$$

and similarly for the y- and z-components. The time component is simply

$$u^0 = \frac{dt}{d\tau}. \tag{3.39}$$

Thus everything depends on the single factor $dt/d\tau$.

To find it, start from the invariant interval:

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{3.40}$$

Now divide by dt^2:

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - \left(\frac{dx}{dt}\right)^2 - \left(\frac{dy}{dt}\right)^2 - \left(\frac{dz}{dt}\right)^2 \qquad (c = 1) \tag{3.41}$$

$$= 1 - \frac{v^2}{c^2}, \tag{3.42}$$

where in the second line we have restored c to make the units explicit. Hence

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}, \qquad \frac{dt}{d\tau} = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3.43}$$

Substituting this back into the chain-rule relation gives the cleaned component form

$$u^i = \gamma v^i, \qquad u^0 = \gamma. \tag{3.44}$$

So the spatial pieces of the proper velocity are the ordinary velocity components multiplied by γ, and the time component is the same γ. At small velocities $\gamma \approx 1$, so $u^i \approx v^i$ and $u^0 \approx 1$. At relativistic velocities the difference becomes substantial, exactly as the lecture stresses.

The lecture then gives the geometric interpretation that makes this definition feel inevitable. In ordinary Euclidean geometry, if a curve has length element

$$ds^2 = dx^2 + dy^2 + dz^2, \tag{3.45}$$

then its unit tangent vector is obtained by dividing the differential displacement by the arc length:

$$\left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}\right). \tag{3.46}$$

This is a unit vector because

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1. \tag{3.47}$$

The relativistic construction is the same idea with one crucial substitution: replace Euclidean arc length ds by proper time $d\tau$. Then

$$u^\mu = \frac{dX^\mu}{d\tau} \tag{3.48}$$

is the tangent vector to the worldline of the particle in spacetime.

This is where the lecture briefly pauses over terminology. In ordinary geometry, s is simply the distance measured along the curve. In relativity, the analogous parameter is $d\tau$, and because the interval carries the Lorentzian minus signs, one should not blur away the distinction between proper time and ordinary Euclidean distance merely by setting $c = 1$. The role is analogous; the geometry is not identical.

The next step then comes almost automatically. If u^μ is the tangent vector, its derivative with respect to proper time is the curvature data of the worldline:

$$a^\mu = \frac{du^\mu}{d\tau}. \tag{3.49}$$

This is the proper acceleration. The lecture does not yet build the full relativistic mechanics, but it clearly announces the direction: the analog of Newton's $F = ma$ must be phrased in terms of proper acceleration, not in terms of the second derivative with respect to coordinate time.

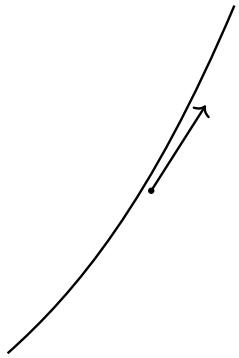


Figure 3.3: A schematic curve with its tangent vector, the Euclidean model for the relativistic construction.

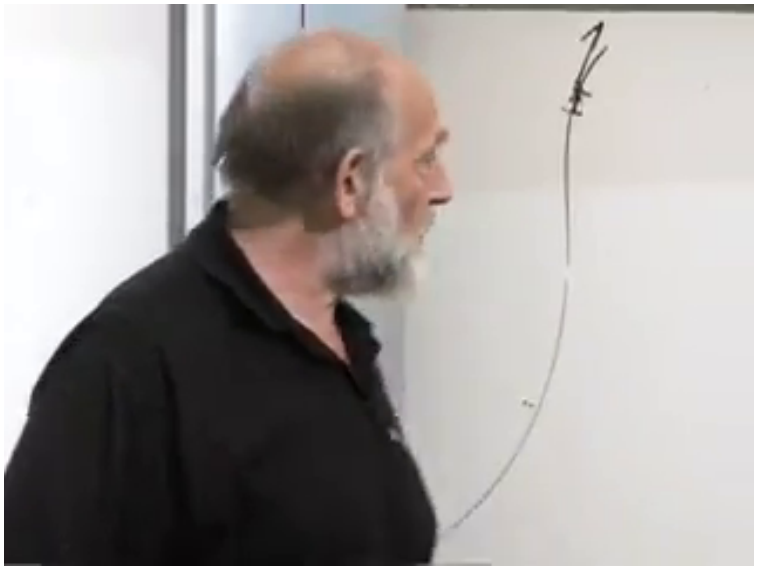


Figure 3.4: Curved trajectory for the transition from proper velocity to proper acceleration.

3.7 Momentum, Energy, and the Threshold of Mechanics

Having reached proper velocity, the lecture takes one further step into mechanics and then stops. The quantity chosen is momentum, because momentum is the basic conserved object in collision problems, and the lecture wants to hand that machinery to the next chapter rather than complete it here.

In Newtonian mechanics, one writes

$$\mathbf{p} = m\mathbf{v}. \tag{3.50}$$

Susskind pauses to stress a modern convention: mass is not treated as a velocity-dependent quantity. It is an intrinsic characteristic of the particle, what used to be called the rest mass. What changes relativistically is not the mass, but the velocity factor that must be used.

The correct relativistic definition is

$$p^\mu = mu^\mu. \tag{3.51}$$

The first three components are therefore

$$p^i = mu^i = \frac{mv^i}{\sqrt{1 - v^2/c^2}}. \tag{3.52}$$

For small velocities the denominator is essentially 1, so we recover the ordinary momentum $p^i \approx mv^i$. At larger velocities the momentum grows faster than the Newtonian formula would suggest.

The time component is the genuinely new piece:

$$p^0 = \frac{m}{\sqrt{1 - v^2/c^2}}. \tag{3.53}$$

Because the lecture is using t rather than ct as the time coordinate, the conventional energy is

$$E = c^2 p^0 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}. \tag{3.54}$$

That is the careful version of the lecture's claim that the fourth component of momentum is energy: strictly speaking, one multiplies by c^2 to get the usual energy units.

Susskind then works this out in the low-velocity limit. Let

$$\epsilon = \frac{v^2}{c^2}, \tag{3.55}$$

so that

$$E = mc^2(1 - \epsilon)^{-1/2}. \tag{3.56}$$

Using the binomial approximation

$$(1 - \epsilon)^p \approx 1 - p\epsilon \tag{3.57}$$

for small ϵ, and taking $p = -1/2$, we obtain

$$(1 - \epsilon)^{-1/2} \approx 1 + \frac{1}{2}\epsilon. \tag{3.58}$$

Therefore

$$E \approx mc^2 \left(1 + \frac{1}{2}\frac{v^2}{c^2}\right) \tag{3.59}$$

$$\approx mc^2 + \frac{1}{2}mv^2. \tag{3.60}$$

The first term is the rest energy; the second is the ordinary Newtonian kinetic energy. In that way the time component of the four-momentum reproduces familiar mechanics in the nonrelativistic limit.

This is the lecture's final conceptual ascent. Energy and momentum are not two separate conserved quantities that merely coexist. They are components of a single four-vector. Conservation of four-momentum is conservation of momentum and energy together. More than that: under Lorentz transformations the components mix. What looks like pure energy in one frame becomes a combination of energy and momentum in another. That is why the relativistic language is the right one for collisions, decays, and mechanics in general.

The lecture stops there on purpose. Proper acceleration has been named, four-momentum has been defined, and the low-velocity limit has been tied back to Newtonian mechanics. The real use of these tools is postponed to the next stage.

3.8 Summary

The lecture begins with a five-dimensional picture of heavy-ion collisions and ends at the threshold of relativistic mechanics. That is not a loose collection of topics; it is a guided progression. First the black-hole analogy is stabilized and defended as either a genuine extra-dimensional description or a mathematically equivalent avatar of QCD. Then collider practice forces us to think about effective energy and radiation loss. Only after that physical motivation is in place does the lecture return to relativity.

From there the structure is exact and deliberate. The garage paradox isolates simultaneity as the hidden source of confusion. Covariant notation is then introduced as a disciplined way to write the invariant interval. Rotations provide the easy model for transformation laws, matrices make the structure manageable, and the composition of boosts yields the relativistic addition of velocities. Finally, ordinary velocity is rejected as the wrong relativistic object, proper velocity appears as the tangent vector to the worldline, and four-momentum packages momentum and energy into a single conserved whole. The lecture closes not with completion, but with a threshold: the machinery of relativistic mechanics is now in place, and the next step is to use it.

Chapter 4

Laws of Nature, Four-Vectors, and Tensors

This lecture is not really about entanglement in the narrow sense. It is about the form that laws of nature are allowed to take. Following Leonard Susskind's order of attack, and curated here by LazyingArt LLC, we begin with a deliberately bad law, repair it by demanding invariance under rotations and then under Lorentz transformations, and only after that introduce the notation of four-vectors, scalars, and tensors. Throughout most of the chapter we set $c = 1$; when useful, we briefly restore c to make the low-velocity limit transparent.

4.1 Symmetry As The First Test Of A Law

Susskind begins in the right place: not with formalism, but with an example that is mathematically possible and physically wrong. Suppose we invent a law saying that two particles interact whenever their x-coordinates become equal. There is no contradiction in such a rule. One can write

$$x_1 - x_2 = 0. \tag{4.1}$$

If that condition holds, the particles may bounce, turn around, or fly off in some new direction. Nothing prevents us from imagining such a world.

But already one feels that the law has the wrong shape. It refers to one coordinate component as though nature cared about the way we happened to orient the blackboard. That discomfort is the point of the example. The lecture is asking what is wrong, not dynamically, but structurally.

4.1.1 Question & Answer

What symmetry is actually being violated? A student answers: homogeneity of space. Susskind's correction matters, and it should be preserved. The law does *not* violate homogeneity. It does not single out any preferred value of x; if we translate the whole configuration, the statement $x_1 - x_2 = 0$ remains the same. Translation invariance survives.

What fails is isotropy. If we rotate the axes by 90°, the same physical rule would have to say that particles interact when their y-coordinates are equal. The law has changed form under a rotation.

That is the real defect. One may use many coordinate systems, but the laws of physics should not owe their simplicity to an arbitrary choice of axis.

This is the opening moral of the lecture: a law of nature is not merely something true in one coordinate system. It should be written in a form that survives a change of coordinates.

4.2 From Spatial Coincidence To Spacetime Coincidence

The repair comes immediately. Instead of requiring equality of one component, we require equality of position itself. In two or three dimensions, the particles interact only when they arrive at the same point in space. If $\vec{r}_1$ and $\vec{r}_2$ are their position vectors, the collision rule becomes

$$\vec{r}_1 = \vec{r}_2, \qquad \text{or equivalently} \qquad \vec{r}_1 - \vec{r}_2 = \vec{0}. \tag{4.2}$$

This is now a vector equation. It says that the *whole* separation vector vanishes. And that is precisely the kind of statement that survives a rotation. A vector that is zero in one coordinate system is zero in any rotated coordinate system. Likewise, if two vectors are equal in one frame, they are equal in every rotated frame.

Susskind pauses on this point because it is the real reason we like vector equations. A single component of a vector may happen to vanish in one frame and not in another. But the statement that an entire vector vanishes is coordinate-independent. That is why physics is naturally written in terms of whole transforming objects rather than isolated pieces of them.

Only then does he bring time back in. One might try a Newtonian-looking law saying that collisions occur when the times of two particles are equal. In pre-relativistic physics that is perfectly sensible. In relativity it is not. Simultaneity is frame-dependent, so a condition on the time component alone cannot define a relativistic law.

The same logic that repaired the spatial law now repairs the spacetime law. A spacetime vector may have zero spatial components and still fail to vanish:

$$\Delta X^\mu = (0, 0, 0, \Delta t) \neq 0.$$

So it is not enough to say that the spatial parts agree. For a relativistic collision, the two events must coincide as a *whole* spacetime event. If X^μ denotes the spacetime position of an event, then the simple collision rule is

$$X_1^\mu = X_2^\mu, \qquad \text{or} \qquad X_1^\mu - X_2^\mu = 0. \tag{4.3}$$

This is the bridge from ordinary vectors to four-vectors. The lecture is still teaching one lesson, now in a larger arena: laws must be written in terms of entire objects that transform coherently.

4.3 Four-Vectors And Matrix Form Of Transformations

A four-vector is an object whose components transform exactly as the coordinates of spacetime do. In the lecture the index convention is $(1, 2, 3, 0)$, with 0 reserved for time. Thus we may write

$$X^\mu = (x^1, x^2, x^3, x^0) = (x, y, z, t).$$

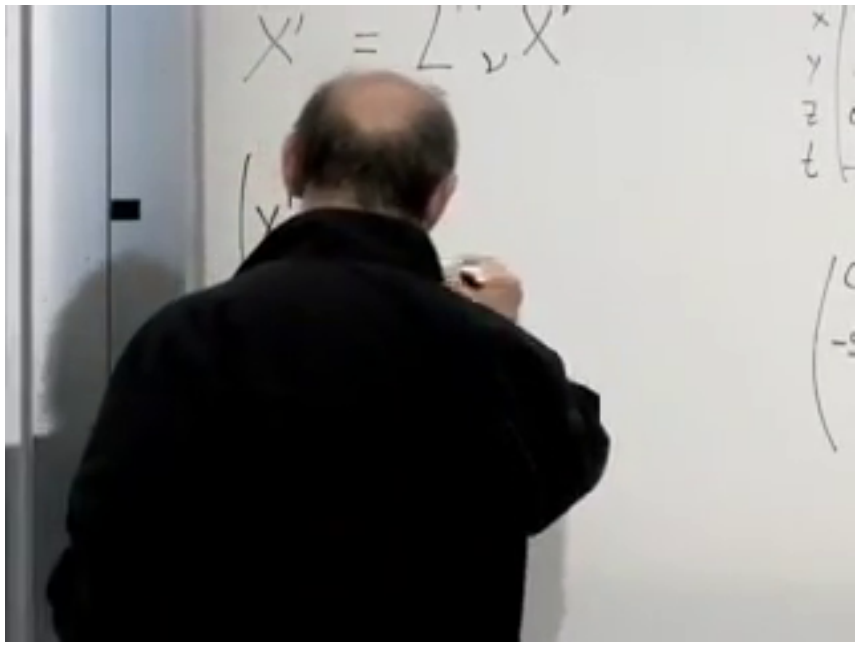


Figure 4.1: Indexed Lorentz transformation and the beginning of its column-vector form on the board.

Susskind first recalls explicit examples. For a boost in the x-direction, the cleaned standard form is

$$x' = \gamma(x - vt), \qquad y' = y, \qquad z' = z, \qquad t' = \gamma(t - vx), \tag{4.4}$$

with

$$\gamma = \frac{1}{\sqrt{1 - v^2}}. \tag{4.5}$$

The transcript around the board formulas is noisy, but the intended content is secure.

He then turns to ordinary spatial rotations. For a rotation about the z-axis, the lecture again gives a linear transformation, mixing x and y while leaving z and t alone. In cleaned form,

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{4.6}$$

Here too we are normalizing the lecture to the standard z-axis rotation matrix, since the transcript around the explicit coefficients is partially garbled.

The crucial point is not the details of any one matrix. The crucial point is that boosts and rotations are both *linear*. Their more general combinations are obtained by multiplying matrices together, and the order matters. Susskind even pauses to note that rotating and then boosting is not the same as boosting and then rotating. The transformations form a structured family, not a random collection of coordinate tricks.

The board then makes the decisive move from explicit component formulas to compact notation. The legible indexed law is

$$X'^{\mu} = L^{\mu}{}_{\nu} X^{\nu}. \tag{4.7}$$

This is the same matrix multiplication rule written with indices. The repeated index ν is understood to be summed over. That is the Einstein summation convention, and it is nothing more mysterious than the ordinary rule for multiplying a matrix by a column vector.

The lower line in the frame is only partially written, so the full column-vector form is a cautious standard reconstruction:

$$\begin{pmatrix} x' \\ y' \\ z' \\ t' \end{pmatrix} = L \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix}. \tag{4.8}$$

This pairing of the indexed law with the column-vector law is exactly the pedagogical point of the board. The lecture is showing us two ways of writing the same transformation.

From here the general definition becomes clear. A quantity A^μ is a four-vector if it transforms in the same way as spacetime position:

$$A'^\mu = L^\mu{}_\nu A^\nu. \tag{4.9}$$

And with that in hand we can write laws in two equivalent styles: either a whole four-vector is set equal to zero, or one whole four-vector is set equal to another. What is never acceptable is to write laws for disconnected pieces of a four-vector and then hope Lorentz invariance will somehow survive.

4.4 Scalars, Lowered Indices, And Invariants

At this stage Susskind deliberately slows down. Before going farther with four-vectors, he says, we should really have talked about scalars first. That is an important pedagogical beat. A scalar is a quantity that does not change under the transformation in question. Under rotations, temperature at a point is a scalar. The distance between two points in space is also a scalar.

He even gives a deliberately silly example: the distance between two points might happen, numerically, to equal the temperature in the room. Precisely because both quantities are scalars, the statement is rotationally invariant. Its truth does not depend on how we orient the axes. The example is silly, but the lesson is serious: invariant statements can be made out of invariant objects.

Vectors are different. Their components change from one frame to another. So we need a way to combine components into quantities that do *not* change.

Let A^μ be a general four-vector. Its contravariant components are written with the index upstairs. The same geometric object can also be described by covariant components, written with the index downstairs. The connection is made by the Minkowski metric:

$$A_\mu = \eta_{\mu\nu} A^\nu, \qquad \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4.10}$$

With the sign convention used throughout this lecture, lowering the index changes the signs of the three spatial components and leaves the time component unchanged:

$$A^\mu = (A^1, A^2, A^3, A^0) \qquad \Longrightarrow \qquad A_\mu = (-A^1, -A^2, -A^3, A^0).$$

The transcript briefly wobbles between 0 and 4 for the time component, and Susskind corrects an index slip on the board. In the notes we simply standardize on 0.

Now we can see why lowering the index is useful. It lets us form invariant quadratic expressions in a compact way:

$$A_\mu A^\mu = -\left(A^1\right)^2 - \left(A^2\right)^2 - \left(A^3\right)^2 + \left(A^0\right)^2. \tag{4.11}$$

For the spacetime position vector this becomes

$$X_\mu X^\mu = t^2 - x^2 - y^2 - z^2. \tag{4.12}$$

This is the invariant interval. It is the relativistic analogue of a length, with the crucial sign difference between space and time built in.

There is also a notational principle here. Whenever we sum over a repeated index, one copy should be upstairs and the other downstairs. Susskind emphasizes this before even explaining all of its consequences. In this notation, the very form of the equation is already steering us toward invariant combinations.

The next step is the bilinear product of two four-vectors. If A^μ and B^μ are four-vectors, then

$$A_\mu B^\mu \tag{4.13}$$

is also invariant. The lecture's proof is elegant and worth preserving. Since sums and differences of four-vectors are again four-vectors, the quantities

$$(A+B)_\mu (A+B)^\mu \qquad \text{and} \qquad (A-B)_\mu (A-B)^\mu$$

must both be invariant. Expanding gives

$$(A+B)_\mu (A+B)^\mu = A_\mu A^\mu + B_\mu B^\mu + 2A_\mu B^\mu, \tag{4.14}$$

$$(A-B)_\mu (A-B)^\mu = A_\mu A^\mu + B_\mu B^\mu - 2A_\mu B^\mu. \tag{4.15}$$

Subtracting one from the other isolates the cross term:

$$(A+B)_\mu (A+B)^\mu - (A-B)_\mu (A-B)^\mu = 4A_\mu B^\mu. \tag{4.16}$$

The left-hand side is a difference of invariants, so $A_\mu B^\mu$ must itself be invariant. This is the relativistic inner product. Apart from the sign structure supplied by $\eta_{\mu\nu}$, it plays the role that the dot product plays for ordinary vectors.

4.5 Proper Velocity And Four-Momentum

Only after that notational groundwork does the lecture return to a concrete physical four-vector: the infinitesimal displacement along a worldline,

$$dx^\mu = (dx, dy, dz, dt).$$

Its invariant norm defines the proper time:

$$d\tau^2 = dx_\mu dx^\mu = dt^2 - dx^2 - dy^2 - dz^2. \tag{4.17}$$

Then the proper four-velocity is defined by

$$u^\mu = \frac{dx^\mu}{d\tau}. \tag{4.18}$$

This is already a physically important shift. Ordinary velocity is $d\vec{x}/dt$. Proper velocity is the rate of change of spacetime position with respect to the particle's own proper time.

The lecture immediately points out that u^μ has four components, which seems strange at first. But they are not independent. In fact,

$$u_\mu u^\mu = \frac{dx_\mu dx^\mu}{d\tau^2} = \frac{d\tau^2}{d\tau^2} = 1. \tag{4.19}$$

So the four-velocity is a unit timelike four-vector. The fourth component is fixed by the first three, just as the components of an ordinary unit vector are constrained by its length.

Worked derivation. To connect proper velocity with ordinary velocity, we examine the spatial components:

$$u^i = \frac{dx^i}{d\tau} = \frac{dx^i}{dt}\frac{dt}{d\tau} = v^i \frac{dt}{d\tau}. \tag{4.20}$$

Now divide the defining equation for $d\tau^2$ by dt^2:

$$\left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{dx^2 + dy^2 + dz^2}{dt^2} = 1 - \frac{v^2}{c^2}. \tag{4.21}$$

Therefore

$$\frac{dt}{d\tau} = \frac{1}{\sqrt{1 - v^2/c^2}} = \gamma. \tag{4.22}$$

Hence

$$u^i = \gamma v^i, \qquad u^0 = \frac{dt}{d\tau} = \gamma. \tag{4.23}$$

The lecture carefully addresses a possible confusion here. This γ is the same mathematical function that appears in Lorentz boosts, but now its argument is the speed of a particle in a chosen frame, not the relative speed between two inertial frames. It is the same function, used in a different context.

For small velocities, $\gamma \approx 1$. Then the first three components of u^μ are close to the ordinary velocity, and the time component is close to 1 in the $c = 1$ convention. As the particle speed approaches the speed of light, all components grow large.

Momentum is introduced in exactly the same spirit. Ordinary $m\vec{v}$ is not a four-vector, because ordinary $\vec{v}$ is not a four-vector. The relativistically natural definition is

$$p^\mu = m u^\mu. \tag{4.24}$$

This is the four-momentum. In components,

$$p^\mu = (\gamma m v_x, \gamma m v_y, \gamma m v_z, \gamma m)$$

when $c = 1$. The first three components reduce to ordinary momentum at low speed. The last component is the energy component; with c restored, it contains the familiar rest energy together with the kinetic correction. The lecture does not yet dwell on that relation, but it keeps the connection with ordinary mechanics clearly in view.

4.6 The Puzzle In The Lorentz Force Law

At this point Susskind says, in effect, let us take a vacation from relativity and write the old law first. That turn in the lecture matters. The point is not to derive the relativistic force law immediately, but to see precisely why the pre-relativistic form cannot be fundamental.

He briefly notes that if one wanted a four-dimensional acceleration, the natural guess would be the proper-time derivative of four-velocity. But he deliberately does not push that construction yet. First he writes the familiar law of motion for a charged particle:

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right). \tag{4.25}$$

The electric part is velocity-independent. The magnetic part depends explicitly on the particle's velocity. This is exactly where the trouble begins.

4.6.1 Question & Answer

If the particle is at rest in one frame, where can the force come from? Suppose that in one frame there is only a magnetic field. Then the force comes entirely from the term $q\,\vec{v} \times \vec{B}$. Now move to the instantaneous rest frame of the particle. In that frame $\vec{v} = 0$, so the magnetic contribution disappears. But the particle was accelerating. A mere change of frame cannot make a genuine acceleration vanish. Therefore the new frame must contain an electric field, and that electric field must account for the force.

This is the lecture's conceptual turning point. What one observer calls a purely magnetic field cannot remain purely magnetic in every frame. Electric and magnetic fields must mix under Lorentz transformations. The decomposition into $\vec{E}$ and $\vec{B}$ is not invariant; it is frame-dependent.

So what do we need? We need an object with six components altogether, one that can mix these six quantities in a coherent Lorentz-covariant way. It is not a scalar, and it is not a four-vector. Something new has to appear.

4.7 Tensors, Symmetric Versus Antisymmetric, And The Field Tensor Preview

The need for that new object is what brings tensors into the lecture. Only after the six-component problem is on the table does Susskind introduce them. A scalar is a tensor of rank zero. A four-vector is a tensor of rank one. A rank-two tensor has two indices. The simplest examples on the board are

$$A^\mu B^\nu, \qquad A^\mu B^\nu C^\sigma. \tag{4.26}$$

The first is rank two; the second is rank three.

The frame is valuable because it preserves exactly the board logic: abstract index notation on the left, component-array thinking on the right. A rank-two tensor in four dimensions has $4 \times 4 = 16$ components, and we can display them as a 4×4 matrix-like grid.

But a tensor is not defined by being literally a product of vectors. It is defined by how it transforms. If $T^{\nu\tau}$ is a second-rank tensor, then

$$T'^{\mu\sigma} = L^\mu{}_\nu L^\sigma{}_\tau T^{\nu\tau}. \tag{4.27}$$

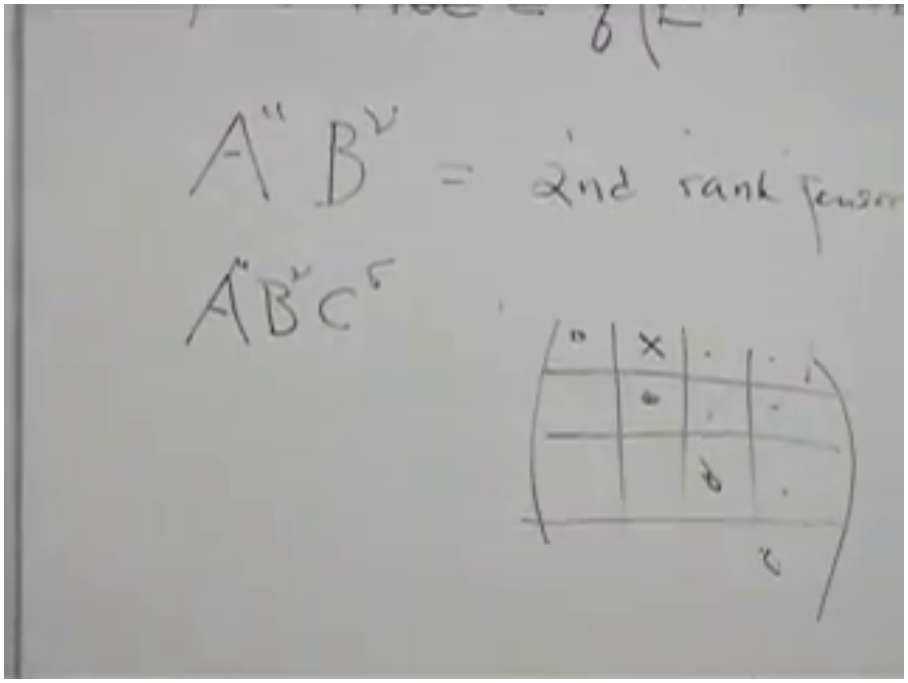


Figure 4.2: Rank-two tensor notation and the schematic component grid used on the board to discuss independent entries.

There is one transformation matrix for each index. This is the operative pattern. Products of vectors transform this way, but so do sums of such products, and more general objects as well. What matters is the transformation law.

Among rank-two tensors, two special cases are singled out. A symmetric tensor obeys

$$T^{\mu\nu} = T^{\nu\mu}, \tag{4.28}$$

while an antisymmetric tensor obeys

$$T^{\mu\nu} = -T^{\nu\mu}. \tag{4.29}$$

For a symmetric 4×4 tensor, the entries reflected across the diagonal are equal. Hence the diagonal plus one triangular half determine the whole object. That gives

$$4 + 6 = 10 \tag{4.30}$$

independent components, exactly as the lecture says.

For an antisymmetric tensor, the diagonal must vanish, because $T^{\mu\mu} = -T^{\mu\mu}$. The remaining independent entries occupy only one triangular half of the matrix, so there are

$$6 \tag{4.31}$$

independent components. That is exactly the number of components carried together by $\vec{E}$ and $\vec{B}$.

This is the answer to the six-component problem. The electromagnetic field is packaged into an antisymmetric tensor $f_{\mu\nu}$. The secure identifications in the lecture are

$$E_1 = f_{10}, \qquad E_2 = f_{20}, \qquad E_3 = f_{30}, \tag{4.32}$$

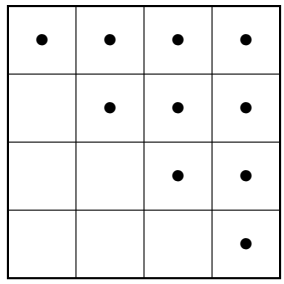


Figure 4.3: A schematic symmetric 4×4 tensor. The diagonal and one triangular half contain the ten independent components.

together with the cyclic magnetic pattern

$$f_{12} = -B_3, \qquad f_{23} = -B_1, \qquad f_{31} = -B_2. \tag{4.33}$$

Thus the electric field occupies the slots involving the time index 0, while the magnetic field occupies the purely spatial slots.

The exact sign convention in the full 4×4 matrix should be checked once more against the full board before the final typesetting pass, but the qualitative structure is clear. A cautious first-pass form consistent with the transcript is

$$f_{\mu\nu} \approx \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}. \tag{4.34}$$

What is not uncertain is the organization: $f_{\mu\nu}$ is antisymmetric, it has six independent components, the space-space block carries $\vec{B}$, and the space-time block carries $\vec{E}$.

The lecture then adds a beautiful three-dimensional remark. The magnetic field has two identities. In spacetime it is part of the two-index antisymmetric tensor $f_{\mu\nu}$. But if we ignore time and look only at the 3×3 spatial antisymmetric block, that block has three independent components and can therefore be identified with an ordinary three-dimensional vector. This is special to three dimensions. A 4×4 antisymmetric tensor cannot be identified with a four-vector in that way.

Susskind ends with a brief historical coda. Einstein did not first write special relativity in polished four-vector notation. Much of the later geometric packaging is Minkowski's. But the physical idea was already there: Maxwell's equations and the Lorentz force law ought to take a form that is the same in every reference frame. The lecture stops exactly where it should stop, at the point where the right object has been identified and the relativistic force law is ready to be built in the next step.

4.8 Summary

The lecture begins with a bad law in order to isolate a good principle. A law of nature should not depend on a preferred axis, nor on a preferred inertial frame. That is why the collision rule is rewritten first as a statement about a whole spatial vector, and then as a statement about a whole spacetime vector. The lesson is the same in both cases: whole transforming objects are acceptable; isolated components are not.

Once that principle is in place, the notation of four-vectors, covariant and contravariant components, proper time, proper velocity, and four-momentum falls into place. The old Lorentz force law then reveals its defect: electric and magnetic fields cannot be invariantly separated. The lecture therefore ends by assembling them into the antisymmetric tensor $f_{\mu\nu}$. The relativistic force law itself belongs to the continuation.

Chapter 5

Lecture 7: Charged Particles, Field Tensors, and the Relativistic Lorentz Force

These notes follow Leonard Susskind's lecture closely and are curated by LazyingArt LLC. The lecture begins with a problem already in motion: charged particles move in electric and magnetic fields, but can the Lorentz force law really be made consistent with relativity? The pressure is not merely to decorate an old equation with four-vectors. We must first understand what deserves to be called force, why the magnetic term is structurally different from the electric one, and how electric and magnetic fields can possibly remain parts of one law when different observers disagree about the particle's velocity. The discussion therefore unfolds in a deliberate order: first the left-hand side of the force law is stabilized, then the cross product is rebuilt as an antisymmetric tensor, and only after that do we move into spacetime notation.

5.1 From Newtonian Force to the Relativistic Problem

Susskind begins by recalling why the Newtonian equation

$$\mathbf{F} = m\mathbf{a} \tag{5.1}$$

ever looked invariant in the first place. In Galileo's world, or Newton's world, all inertial observers agree on the acceleration of a particle, on its mass, and therefore on the force. But once the kinematics changes, it would be surprising if the old form of the law survived untouched. Relativity forces us to rethink the concept of force at a deeper level.

There is another piece of motivation here which the lecture states very plainly. If the speed of light were infinite, there would be no magnetic force at all. Magnetic effects are already relativistic effects. That is one reason the magnetic term is the right place to look for structure. It also explains why one should not expect gravitation and electromagnetism to extend relativistically in the same way. At the nonrelativistic level, Newton's inverse-square law and Coulomb's inverse-square law look like close cousins. But relativity treats them very differently. Gravitation does not fit naturally into special relativity; electromagnetism does.

Before rewriting the force law, Susskind cleans up the left-hand side. Because the mass is time

independent, we may write

$$\mathbf{F} = m\frac{d\mathbf{v}}{dt} = \frac{d}{dt}(m\mathbf{v}) = \frac{d\mathbf{p}}{dt}. \tag{5.2}$$

This is the more durable statement. Force is the time rate of change of momentum.

Now the Lorentz force law can be written, in units where $c = 1$, as

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{5.3}$$

The lecture emphasizes that once c has been set equal to 1, the nonrelativistic limit is not $c \to \infty$; it is the limit of small velocity. The electric term already looks like an ordinary vector contribution. The magnetic term is the troublesome one: it depends explicitly on the velocity, and so it carries the most obvious trace of relativity. That is why Susskind does not go to four-vectors immediately. He pauses and asks what the cross product really is.

5.2 Re-reading the Cross Product as an Antisymmetric Tensor

From this point on it is useful to follow the lecture's index conventions. Greek indices such as μ, ν will later run over spacetime components. For the moment we work only in ordinary space, and Latin indices m, n run over $1, 2, 3$.

Take two spatial vectors $\mathbf{A}$ and $\mathbf{B}$. From them we can build the rank-two tensor

$$T_{mn} = A_m B_n. \tag{5.4}$$

In three dimensions this has nine components. It is just a matrix made from the components of the two vectors. As it stands, it has no special symmetry. But if we antisymmetrize it,

$$A_m B_n - A_n B_m, \tag{5.5}$$

then all diagonal entries vanish, and the entries below the diagonal are the negatives of those above the diagonal. So the antisymmetric tensor has only three independent components.

That counting is the whole point. The ordinary cross product also has three components, but three components by themselves do not guarantee that we are looking at a vector; in three dimensions an antisymmetric tensor also has three independent entries. The lecture therefore slows down here. The cross product is not to be thought of first as an arrow perpendicular to two other arrows. It is better to think of it as an antisymmetric tensor, which only in three dimensions may also be identified with a vector.

The board image preserves the stage on which this identification is being made. Susskind labels the upper-triangular entries by C_3, C_2, and C_1, but the upper-right entry is only a temporary placement. The lecture then cycles the component formulas carefully and corrects the sign.

The corrected component identities are

$$C_3 = A_1 B_2 - A_2 B_1, \tag{5.6}$$
$$C_1 = A_2 B_3 - A_3 B_2, \tag{5.7}$$
$$C_2 = A_3 B_1 - A_1 B_3. \tag{5.8}$$

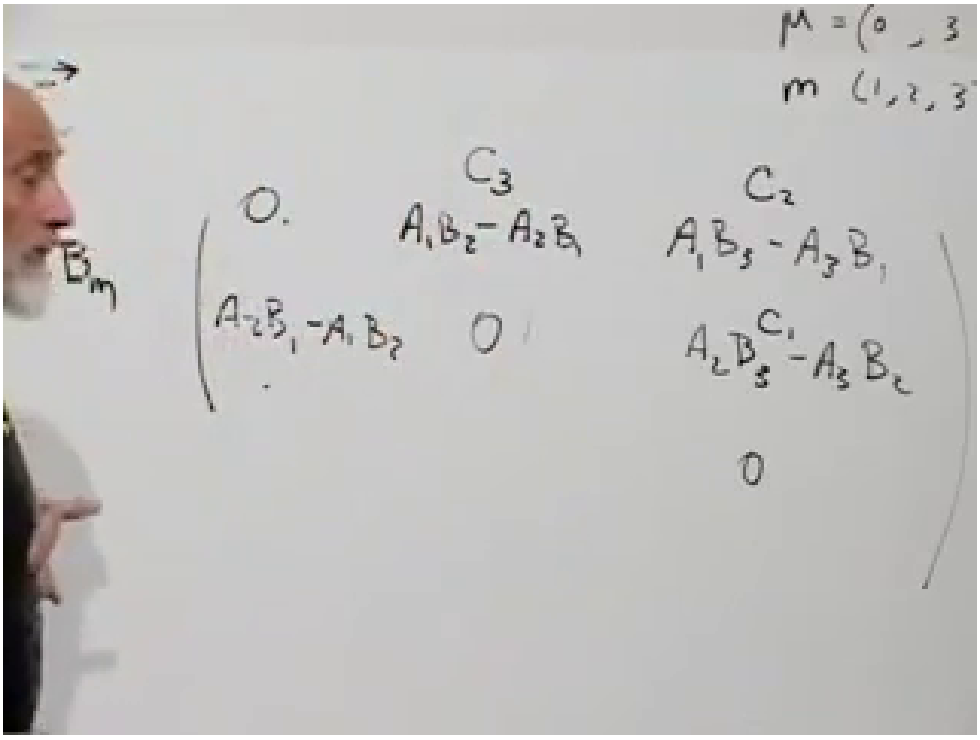


Figure 5.1: Cross product written in antisymmetric-tensor form

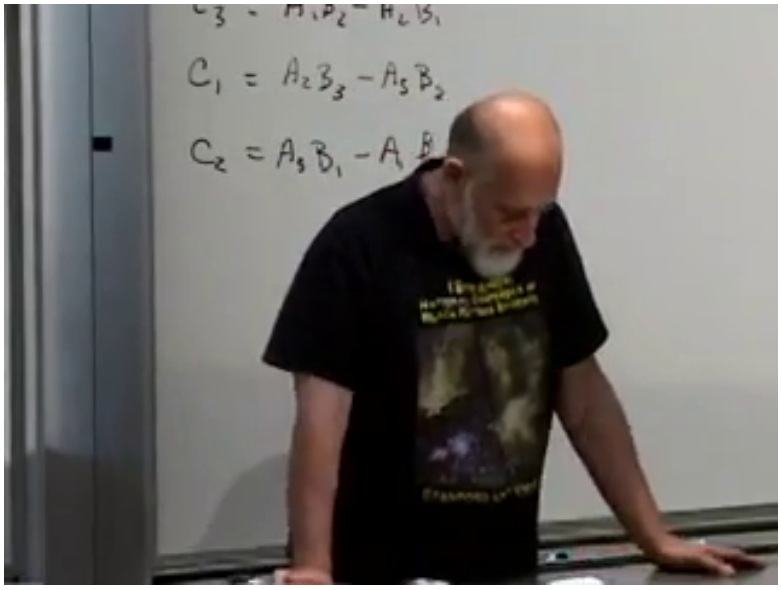


Figure 5.2: Corrected component formulas for the cross product

So the clean antisymmetric matrix associated with $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is

$$\begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix}. \tag{5.9}$$

This correction is worth keeping visible because the lecture itself does not glide over it. The board first suggests one sign, then the cyclic identities force the correction. That is part of the pedagogical rhythm.

The geometry is also important. An antisymmetric tensor has two indices, so each component is naturally associated with a plane. The xy-component is identified with the z-component of the vector, the yz-component with the x-component, and so on. In three dimensions we may describe the same object either by the arrow sticking out of a plane or by the plane to which it is perpendicular. In higher dimensions the arrow language no longer survives, but the antisymmetric tensor does. That is why this reformulation is more than a trick: it is the version that generalizes.

5.3 The Magnetic Field in Tensor Language

Only after the cross product has been re-read this way does Susskind return to the magnetic field itself. The magnetic field $\mathbf{B}$ may be represented as an ordinary vector, but it may also be represented as an antisymmetric tensor B_{mn}. In the lecture's convention there is an extra minus sign between the tensor entries and the corresponding vector components:

$$B_{12} = -B_3, \qquad B_{13} = B_2, \qquad B_{23} = -B_1. \tag{5.10}$$

So the magnetic tensor is

$$B_{mn} = \begin{pmatrix} 0 & -B_3 & B_2 \\ B_3 & 0 & -B_1 \\ -B_2 & B_1 & 0 \end{pmatrix}. \tag{5.11}$$

Remark 5.1. This sign convention is historical rather than inevitable. The invariant object is the antisymmetric tensor. The lecture is explicit that one could equally well have chosen the opposite sign and still been describing the same geometry, provided one kept the convention consistent.

Now comes the payoff. Instead of treating $\mathbf{v} \times \mathbf{B}$ as a three-dimensional cross product, we can write its components by contraction. The third component is

$$(\mathbf{v} \times \mathbf{B})_3 = v_1 B_2 - v_2 B_1. \tag{5.12}$$

Using the tensor entries, this becomes

$$(\mathbf{v} \times \mathbf{B})_3 = v_1 B_{13} + v_2 B_{23} + v_3 B_{33}. \tag{5.13}$$

Since $B_{33} = 0$, this suggests the general pattern

$$(\mathbf{v} \times \mathbf{B})_m = v_n B_{nm}. \tag{5.14}$$

Thus the magnetic part of the Lorentz force becomes

$$F_m^{(\mathrm{mag})} = q\, v_n B_{nm}. \tag{5.15}$$

This is a very small algebraic step, but it changes the style of the argument completely. The magnetic field is no longer being used as a vector that enters a special three-dimensional product. It is being used as an antisymmetric tensor acting on the velocity.

5.3.1 Question & Answer

Question. Why does a magnetic field bend the motion without changing the speed?

Answer. The lecture does not answer this by appealing to the elementary geometric fact that $\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$, even though that fact is true. Instead it answers it in the tensor language just introduced, precisely because that language will survive the later move to four dimensions.

Proposition 5.2. *In a pure magnetic field the square of the speed is constant:*

$$\frac{d}{dt}(v_m v_m) = 0. \tag{5.16}$$

Proof. Differentiate the square of the speed:

$$\frac{d}{dt}(v_m v_m) = 2\,\dot{v}_m v_m. \tag{5.17}$$

If the electric field vanishes, the acceleration is proportional to the magnetic force, so up to an overall constant,

$$\frac{d}{dt}(v_m v_m) \propto (\mathbf{v} \times \mathbf{B})_m v_m = v_n B_{nm} v_m. \tag{5.18}$$

Now B_{nm} is antisymmetric, while $v_n v_m$ is symmetric in n and m. Therefore the contraction vanishes:

$$v_n B_{nm} v_m = 0. \tag{5.19}$$

Hence

$$\frac{d}{dt}(v_m v_m) = 0. \tag{5.20}$$

□

This is exactly the sort of argument Susskind wants. One could have used a familiar theorem about three-dimensional cross products, but that would hide the mechanism. Here the mechanism is explicit: symmetry contracted with antisymmetry gives zero.

There is also a small but important nuance in the lecture. The algebra above did not actually use the assumption that the magnetic field itself is time independent. What changes in practice when $\mathbf{B}$ depends on time is that a time-dependent magnetic field produces an electric field, and the electric field can do work. So the real obstruction to constant speed is the appearance of $\mathbf{E}$, not the loss of antisymmetry.

5.4 Why Electric and Magnetic Fields Must Mix Between Frames

At this stage the lecture cashes in the tensor detour and turns back to physics. If the same laws are to hold in every inertial frame, then electric and magnetic fields cannot remain separate under a change of frame. The lecture says that this is by now more or less obvious, and then immediately gives the cleanest example.

Take a magnetic field along one axis and let a charged particle move along a second axis. The force then points along the third. It is the same three-axis picture that every student of the Lorentz force knows, but now it is used as a test of frame dependence.

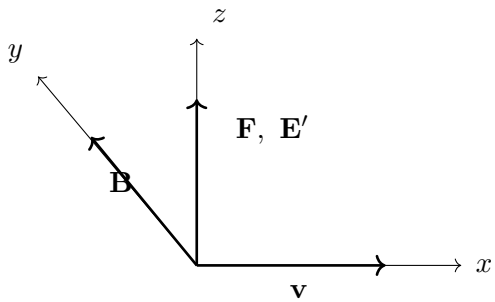


Figure 5.3: The lecture's three-axis thought experiment: the particle moves along one axis, the magnetic field points along a second, and the force appears along the third; in the co-moving frame that same force must come from an electric field

Let us call the particle's direction of motion the x-axis and the magnetic field direction the y-axis. Then $\mathbf{v} \times \mathbf{B}$ points along the z-axis, so the particle accelerates along z.

Now introduce a second observer moving with the particle.

5.4.1 Question & Answer

Question. If I move with the particle so that its velocity vanishes, where can the force come from?

Answer. In the co-moving frame the particle is instantaneously at rest, so the magnetic term $q\,\mathbf{v} \times \mathbf{B}$ cannot contribute at that instant. But in the slow-velocity Newtonian limit the acceleration is an invariant quantity, so the second observer must still see the same acceleration. Therefore the force must be blamed on something else. If the Lorentz force law is to keep its form in every frame, there is only one candidate left: an electric field. Thus a frame which began with no electric field at all must, after the change of frame, contain an electric field pointing along the direction of the force.

That is the lecture's cleanest physical motivation for unifying $\mathbf{E}$ and $\mathbf{B}$. The argument is not yet tensor algebra; it is consistency of description. One frame attributes the force to magnetism, the co-moving frame must attribute it to electricity, and so the two fields have to transform into one another.

This is also the right moment to enlarge the framework. We now have six field components altogether, and we need a single spacetime object in which they fit naturally.

5.5 The Electromagnetic Field Tensor

The lecture now asks for an object with six independent components that can hold the three electric and three magnetic field components at once. Susskind says, somewhat theatrically, that the natural such object in four dimensions is an antisymmetric tensor, and then immediately corrects the literal overstatement: one could certainly write down six scalars, or a vector and two scalars, but those would miss the point. The natural geometrical object with six independent components is the antisymmetric tensor.

To remain close to the blackboard, it is best to preserve the lecture's ordering of indices as $1, 2, 3, 0$

rather than the more common $0, 1, 2, 3$. In that ordering the electromagnetic field tensor is

$$F_{\mu\nu} = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}, \qquad \mu, \nu = (1, 2, 3, 0). \tag{5.21}$$

The 3×3 spatial block is exactly the magnetic tensor we have already written down. The entries involving one spatial index and the time index carry the electric field.

Remark 5.3. The lecture is relaxed about upstairs and downstairs positions when it is only the three spatial components that are being named. The essential content is the antisymmetric tensor itself. What matters is not to mix conventions carelessly once signs begin to enter.

This is the first genuine spacetime object in the chapter. It is also the right object conceptually: since electric and magnetic fields mix under boosts, they must be components of one and the same tensor. The question that now replaces the old Newtonian question is clear: what four-vector force law can we build from $F_{\mu\nu}$, the charge q, and the motion of the particle?

5.6 Four-Velocity, Orthogonality, and the Covariant Force Law

The lecture answers by first recalling the kinematics. The four-velocity is defined by

$$u^\mu = \frac{dx^\mu}{d\tau}, \tag{5.22}$$

where τ is proper time. In the lecture's ordering its components are

$$u^\mu = \gamma(v_x, v_y, v_z, 1), \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}. \tag{5.23}$$

The four-momentum is then

$$p^\mu = m u^\mu. \tag{5.24}$$

Because x^μ is a four-vector and τ is an invariant, u^μ is a four-vector; therefore p^μ is a four-vector; therefore

$$\frac{dp^\mu}{d\tau} \tag{5.25}$$

is the natural relativistic left-hand side. The Newtonian $d\mathbf{p}/dt$ is replaced by a proper-time derivative because we are now working in spacetime.

Before proposing any force law, Susskind proves a kinematic constraint. The four-velocity satisfies

$$u_\mu u^\mu = 1. \tag{5.26}$$

Differentiate this along the world line:

$$0 = \frac{d}{d\tau}(u_\mu u^\mu) \tag{5.27}$$

$$= \frac{du_\mu}{d\tau} u^\mu + u_\mu \frac{du^\mu}{d\tau} \tag{5.28}$$

$$= 2\, u_\mu \frac{du^\mu}{d\tau}. \tag{5.29}$$

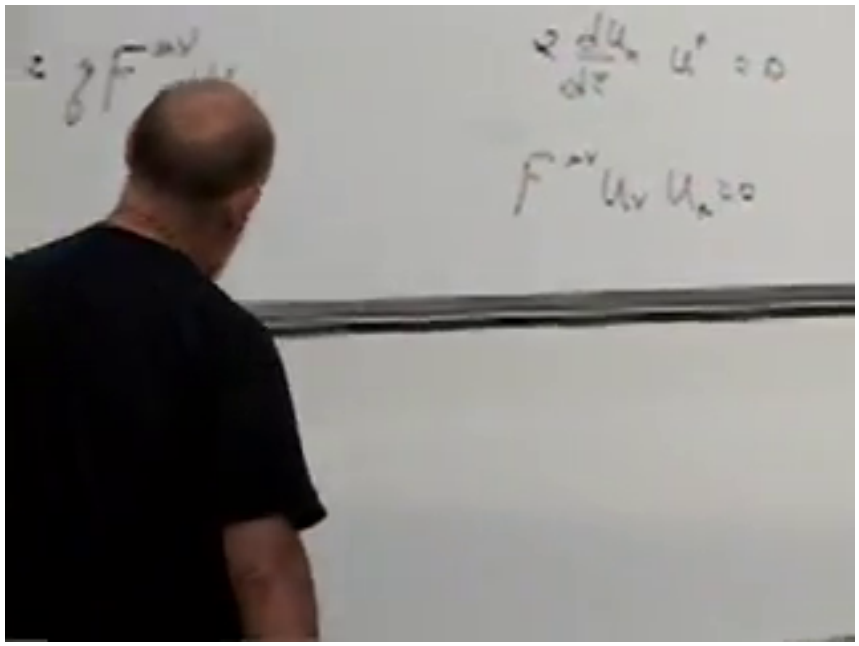


Figure 5.4: Antisymmetric-tensor force ansatz and vanishing contraction

Since $p^\mu = mu^\mu$, we may rewrite this as

$$f_\mu u^\mu = 0, \qquad f^\mu \equiv \frac{dp^\mu}{d\tau}. \tag{5.30}$$

So the four-force must be orthogonal to the four-velocity.

This is the exact four-dimensional analogue of the fact proved earlier for a magnetic field: the relevant force must contract to zero with the velocity. The lecture therefore uses exactly the same trick as before.

What is most secure in the board image is the contraction

$$F^{\mu\nu} u_\nu u_\mu = 0. \tag{5.31}$$

The left edge of the force ansatz is cropped, so the full equation is best written in the cautious standard form supported by the transcript:

$$f^\mu = qF^{\mu\nu} u_\nu. \tag{5.32}$$

Then

$$f^\mu u_\mu = qF^{\mu\nu} u_\nu u_\mu = 0, \tag{5.33}$$

because $F^{\mu\nu}$ is antisymmetric, while $u_\nu u_\mu$ is symmetric.

Now the lecture does something characteristic. It does not declare victory as soon as the formula looks covariant. It checks the low-velocity limit first. Writing out the x-component, and following the lecture's sign convention and relaxed treatment of purely spatial upper and lower indices, one finds

$$\frac{dp_x}{d\tau} = q(F_{xy}u_y + F_{xz}u_z + F_{x0}u_0) = q(E_x u_0 + B_3 u_y - B_2 u_z). \tag{5.34}$$

If the motion is slow, then $d\tau \approx dt$, $u_0 \approx 1$, and $u_i \approx v_i$. Therefore

$$\frac{dp_x}{dt} = q(E_x + v_y B_3 - v_z B_2), \tag{5.35}$$

which is exactly the x-component of

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{5.36}$$

So the covariant force law reproduces the ordinary Lorentz law in the nonrelativistic limit.

At this point Susskind tightens the argument. The equation should not merely work when $\gamma \approx 1$. The proper-time derivative and the four-velocity components themselves contain γ:

$$\frac{dp^\mu}{d\tau} = \gamma \frac{dp^\mu}{dt}, \qquad u_i = \gamma v_i, \qquad u_0 = \gamma. \tag{5.37}$$

The same factor γ therefore appears on both sides and cancels. What remains is again the ordinary-looking Lorentz force law,

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \tag{5.38}$$

but now the momentum is not $m\mathbf{v}$; it is

$$\mathbf{p} = m\gamma\mathbf{v}. \tag{5.39}$$

That is where the relativity is hiding.

The lecture then pauses over an audience question: is this mathematics or physics? The right answer is that it is both, but not in equal proportions. The mathematical part is the tensor algebra. The physical input is heavy. We assume that the same laws hold in every inertial frame, and we assume that the equations of motion are second-order rather than third- or fourth-order. Those are physical statements about how nature predicts the future, not mere formal conveniences.

5.7 Transformation Laws and the Wave-Equation Bridge

Once the covariant force law is in place, the lecture turns to the transformation law for the field tensor itself. Rather than writing down the full general tensor-transformation formula abstractly, Susskind shows how to think about it. First recall the coordinate transformation for a boost along the 1-axis:

$$x_1' = \gamma(x_1 - vx_0), \qquad x_0' = \gamma(x_0 - vx_1), \qquad x_2' = x_2, \qquad x_3' = x_3. \tag{5.40}$$

All vectors transform in this way. Therefore, if we know how a product of two vectors transforms, we know how a rank-two tensor transforms.

The first example is the component parallel to the boost. Because an antisymmetric tensor has vanishing diagonal entries and satisfies $F_{10} = -F_{01}$, the transformation collapses to

$$F_{01}' = F_{01}. \tag{5.41}$$

So the component of the electric field parallel to the direction of motion is unchanged. That is the easy case.

The more interesting case is transverse. For a boost along the 1-axis the lecture works out the transformed 02-component and finds, in the lecture's convention,

$$F'_{02} = \gamma(F_{02} - vF_{12}) = \gamma(E_2 + vB_3). \tag{5.42}$$

Thus

$$E'_2 = \gamma(E_2 + vB_3). \tag{5.43}$$

The sign here belongs to the lecture's convention and should be read that way. The physical content is simple and important: a transverse electric field mixes with a magnetic field under a boost, and similarly the magnetic field components mix with electric ones. So if a field is purely magnetic in one frame, a moving observer generally sees both electric and magnetic fields.

The later questions in the lecture sharpen this point. One cannot in general transform away a pure magnetic field merely by changing frame. If both **E** and **B** are present, transformed pieces may cancel. But if one begins with only a magnetic field, the transformed field still has a magnetic part.

The lecture then pivots away from particle mechanics and toward fields. Let $\phi(x)$ be a scalar field. Its derivative

$$\partial_\mu \phi = \frac{\partial \phi}{\partial x^\mu} \tag{5.44}$$

forms the covariant components of a four-vector. This is one of the main closing lessons: derivatives of fields carry indices, and those indices matter. In the signature used throughout the lecture, raising the index changes the sign of the spatial components but leaves the time component alone:

$$\partial^\mu \phi = \left(-\frac{\partial \phi}{\partial x^1}, -\frac{\partial \phi}{\partial x^2}, -\frac{\partial \phi}{\partial x^3}, \frac{\partial \phi}{\partial x^0} \right). \tag{5.45}$$

More generally, every differentiation adds a lower index. If we differentiate a vector, we obtain a tensor; if we differentiate a tensor, we obtain an object with one more lower index.

This is exactly the preparation needed for wave equations. The lecture wants the wave equation to be manifestly Lorentz invariant, so its left-hand side should be a scalar. The simplest possibility is therefore

$$\partial^\mu \partial_\mu \phi = 0. \tag{5.46}$$

Written out, it is

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} - \frac{\partial^2 \phi}{\partial z^2} = 0. \tag{5.47}$$

This is the lecture's bridge, not its destination. Susskind says very clearly that the goal is Maxwell's equations, but before we write them covariantly we must understand how derivatives of fields behave. The scalar wave equation is the simplest example of that structure.

5.8 Summary

The lecture begins by unsettling the Newtonian notion of force and ends by rebuilding it in a relativistic language. First we replace $m\mathbf{a}$ by the more robust definition $d\mathbf{p}/dt$. Then the magnetic term forces a detour: the cross product is re-read as an antisymmetric tensor, its three independent components are matched to a vector only after a careful correction of signs, and the same tensor language explains why a magnetic field bends motion without changing the speed. That detour is not ornamental. It prepares the central relativistic move.

Once we notice that a particle deflected by a magnetic field in one frame must be deflected by an electric field in its co-moving frame, the six field components naturally assemble into the electromagnetic tensor $F_{\mu\nu}$. Four-velocity and four-momentum then lead to the kinematic constraint $f_\mu u^\mu = 0$, and the simplest way to satisfy it is to build the force from an antisymmetric tensor:

$$f^\mu = qF^{\mu\nu}u_\nu. \tag{5.48}$$

Its low-velocity limit reproduces the ordinary Lorentz force law, while the relativistic correction is carried by the momentum $\mathbf{p} = m\gamma\mathbf{v}$. From there the lecture opens outward rather than concluding: boosts mix electric and magnetic fields, derivatives of scalar fields furnish covariant vectors, and the simplest Lorentz-invariant wave equation appears as the doorway to Maxwell's equations.

Chapter 6

Lecture 8: A First Look at Curved Spacetime and Cosmology

Susskind means to begin with cosmology, and in particular with the extension from the flat, passive spacetime of special relativity to the active spacetime of general relativity. But before that program really starts, a student interrupts with a question about the previous lecture on Maxwell's equations. That interruption is useful. It lets us begin with a reminder of what symmetry can do and what it cannot do. Only then do we move into the geometry of intervals, the meaning of the metric, the distinction between awkward coordinates and genuine curvature, and finally the cosmological metric with its scale factor, Hubble law, and null rays. These notes follow Leonard Susskind's lecture closely and are curated by LazyingArt LLC.

6.1 Why Maxwell's Equations Need More Than Relativity

The opening question is whether Maxwell's equations really followed from Lorentz invariance alone. Susskind's answer is immediate: not at all. Lorentz invariance goes a long way, but there is much more in the construction. One assumes that there are electric and magnetic fields, that they fit together into a single relativistic structure, that the equations are linear, and that they do not involve arbitrarily high derivatives. Those are physical assumptions, not consequences of relativity by itself.

6.1.1 Question & Answer

Question. Can Maxwell's equations be derived from relativity alone?

Answer. No. Relativity constrains the form of the theory, but it does not determine it uniquely. To obtain Maxwell's equations one must also assume linearity, exclude wild higher-derivative terms, and use empirical input, for example the fact that classical electromagnetic waves pass through one another freely. The general moral is the one Susskind wants on the table before cosmology begins: symmetries constrain theories, but as a rule they do not determine them.

He then pauses over the meaning of linearity. The transcript is garbled for a short stretch, but the intended point is clear from what comes immediately before and after it. If $\phi(x, t)$ is a solution of a

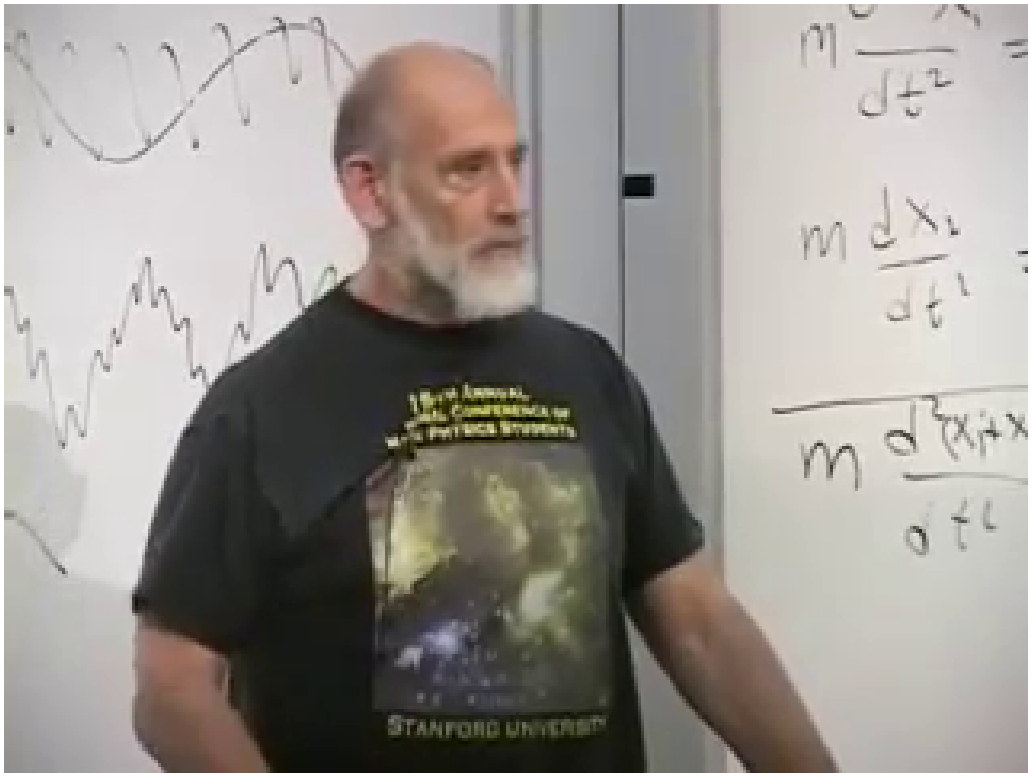


Figure 6.1: Wave superposition sketches and linear second-derivative equations. The board layout itself matters here: waveforms on the left, linear equations on the right. The exact oscillator labels on the board are only partly legible, so the displayed equations in the text use the standard reconstruction supported by the transcript.

linear equation, then multiplying it by a constant produces another solution:

$$\phi(x,t) \text{ solution} \qquad \Longrightarrow \qquad c\,\phi(x,t) \text{ solution.} \tag{6.1}$$

The more interesting statement, and the one he develops at the board, is superposition: if ϕ_1 and ϕ_2 are solutions, then $\phi_1 + \phi_2$ is also a solution.

To make this concrete he turns to the harmonic oscillator,

$$m\ddot{x} = -kx, \tag{6.2}$$

$$m\ddot{x}_1 = -kx_1, \qquad m\ddot{x}_2 = -kx_2. \tag{6.3}$$

Add the two equations, and use the linearity of differentiation:

$$m\ddot{(x_1 + x_2)} = -k(x_1 + x_2). \tag{6.4}$$

So the sum of two solutions is another solution. That is the algebra behind the superposition of waves. Two waveforms can be drawn separately, added together point by point, and the resulting waveform is still allowed by the same equation.

This is why, in classical electromagnetism, one wave coming from the left and another coming from the right simply pass through each other. In a linear theory they do not directly scatter off one another. If electromagnetic waves "talk" to one another at all, they do so indirectly, by shaking charges, and the charges then radiate new waves.

The contrast with a nonlinear equation is immediate. If we modify the oscillator by adding a quadratic term of px^2-type, then even if x_1 and x_2 solve the same nonlinear equation, their sum will not generally do so, because

$$px_1^2 + px_2^2 \neq p(x_1 + x_2)^2. \tag{6.5}$$

That single mismatch is enough to destroy superposition. So one could certainly have written nonlinear cousins of Maxwell's equations, with terms quadratic in the fields. Relativity alone would not have forbidden them. The fact that the classical electromagnetic field is linear is extra information.

This opening exchange is not a side remark. It prepares the lecture's deeper theme. Geometry, like symmetry, is powerful; but by itself it does not settle every dynamical question. With that issue cleared away, Susskind makes the pivot explicit: let us now move on to a little bit of general relativity, and then to cosmology.

6.2 From the Minkowski Interval to General Coordinate Expressions

Special relativity, in the form Susskind wants to use it, begins with a geometric statement. Take a spacetime point x^μ, with $\mu = 0, 1, 2, 3$, and a neighboring point $x^\mu + dx^\mu$. If we know the interval between every neighboring pair of spacetime points, then in principle we know the geometry of spacetime. In flat Minkowski space that interval is

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2 = dx^\mu dx_\mu = \eta_{\mu\nu}\, dx^\mu dx^\nu, \tag{6.6}$$

where

$$\eta_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1). \tag{6.7}$$

Susskind sometimes speaks here of $d\tau^2$, proper time squared, rather than ds^2. The point is the same: the invariant local spacetime separation is the fundamental geometric object.

Different observers will not agree on the individual components dt, dx, dy, dz. Rotate the axes, or perform a Lorentz transformation, and the components change. What remains unchanged is the interval itself. That is one way to formulate the content of special relativity. A second statement is that the transformations preserving the form of this interval are precisely the Lorentz transformations. A third is then the relativity principle itself: the laws of physics have the same form in frames related by those transformations.

At that point Susskind loosens the coordinates on purpose. The geometry is not the same thing as one especially nice coordinate system. Even on an ordinary flat plane, the same distance can look strange in awkward coordinates. Suppose, for example, that x is measured in centimeters and y in meters, but we insist on expressing distances in centimeters. Then the interval between neighboring points is not $dx^2 + dy^2$, but

$$ds^2 = dx^2 + \left(\frac{dy}{100}\right)^2 = dx^2 + 10^{-4} dy^2. \tag{6.8}$$

No new geometry has appeared. The odd coefficient reflects only the odd choice of units.

This is the first warning of the lecture: funny coefficients in the interval do not by themselves signal curvature. They may signal nothing deeper than a funny coordinate system.

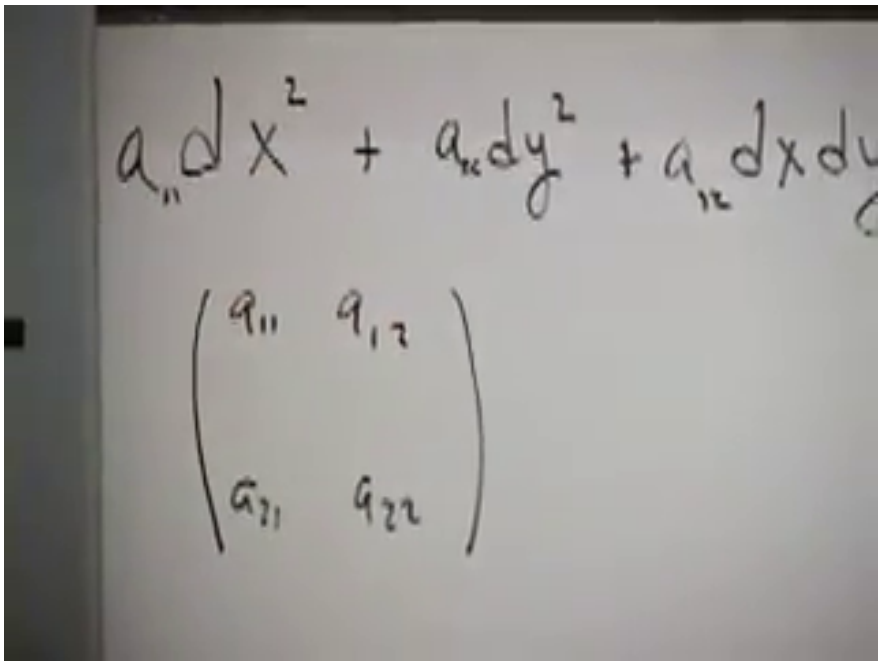


Figure 6.2: Quadratic form and metric matrix. The lecture deliberately arranges the board so that the quadratic differential form appears first and the coefficient matrix sits just beneath it.

6.3 The Metric as a Quadratic Form and as a Matrix

Susskind then makes the coordinates still more awkward. Suppose the coordinate axes are not orthogonal. Then even on a flat blackboard the interval need not take the familiar Pythagorean form. In the most general local quadratic expression for two coordinates x and y, we may write

$$ds^2 = a\,dx^2 + b\,dy^2 + c\,dx\,dy. \tag{6.9}$$

To make the symmetry manifest, he splits the mixed term into two equal pieces:

$$ds^2 = a_{11}\,dx^2 + a_{12}\,dx\,dy + a_{21}\,dy\,dx + a_{22}\,dy^2, \qquad a_{12} = a_{21}. \tag{6.10}$$

The top line on the board is then repackaged as a matrix of coefficients,

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \tag{6.11}$$

This is the board moment that the lecture wants us to stop and absorb: the matrix is called the metric.

The point is not merely notational. The metric is the object that tells us the distance between neighboring points. If we know the metric, we know the local geometry, and in principle that is enough to reconstruct the geometry globally. In two dimensions we have written this as a matrix of a_{ij}; in general one writes

$$ds^2 = a_{ij}(x)\,dx^i dx^j, \tag{6.12}$$

or in spacetime,

$$ds^2 = g_{\mu\nu}(x)\,dx^\mu dx^\nu. \tag{6.13}$$

The geometry determines which intervals are physically realized, but it does not determine one unique choice of coordinates, and therefore it does not determine one unique matrix of coefficients. The same geometry may be represented by many different metrics, related by coordinate transformations.

A second warning is now needed. Even on a flat plane, the coefficients may fail to be constant if the coordinates are chosen in a sufficiently perverse way: nonorthogonal axes, uneven spacing, curved coordinate lines, mixed units. Variable coefficients alone are not yet curvature. The real question is more subtle: can we find some coordinates in which the coefficients become constant everywhere?

6.4 Flat But Complicated Coordinates Versus Genuine Curvature

Before answering that question, Susskind first says what curvature is not. If we take a flat sheet of paper and bend it gently into a cylinder without stretching it, we have changed the way it sits in the surrounding room, but we have not changed the intrinsic distances between neighboring points on the sheet. A tiny creature living on the paper, unable to look off the surface, would not detect anything new. Mathematicians do not call that curvature. It is bending without stretching.

Intrinsic curvature appears only when a surface cannot be flattened without deformation. A hemisphere cut from a balloon is the standard example. To press it flat onto a plane one would have to stretch some parts and compress others. That failure to flatten without distortion is what curvature means in the present lecture.

The next move is important. Before turning to the sphere, Susskind gives a flat example whose metric coefficients vary with position: polar coordinates on the plane. If neighboring points differ by dr and $d\theta$, then one leg of the local right triangle has length dr, while the other has length $r\,d\theta$, not $d\theta$. Therefore

$$ds^2 = dr^2 + r^2\,d\theta^2. \tag{6.14}$$

The corresponding metric matrix is

$$g_{\text{polar}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}. \tag{6.15}$$

So here is a flat plane presented by a metric that is not constant everywhere. But the plane is still flat, because we can return to Cartesian coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta, \tag{6.16}$$

and the metric becomes $ds^2 = dx^2 + dy^2$.

This earns the central criterion. Curvature is not the same thing as having coefficients that vary from point to point. Curvature is the obstruction to finding *any* coordinate system in which the coefficients become constant everywhere.

The sphere is the decisive example. Let θ be the polar angle measured from the north pole and ϕ the azimuthal angle. On the unit sphere, the distance associated with $d\theta$ is just $d\theta$. But the distance associated with $d\phi$ depends on latitude, because the local circle of latitude has radius $\sin\theta$. Thus the transverse arc length is $\sin\theta\,d\phi$, and the local Pythagorean rule gives

$$ds^2 = d\theta^2 + \sin^2\theta\,d\phi^2. \tag{6.17}$$

For a sphere of radius R, every distance is scaled by R, so

$$ds^2 = R^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right), \tag{6.18}$$

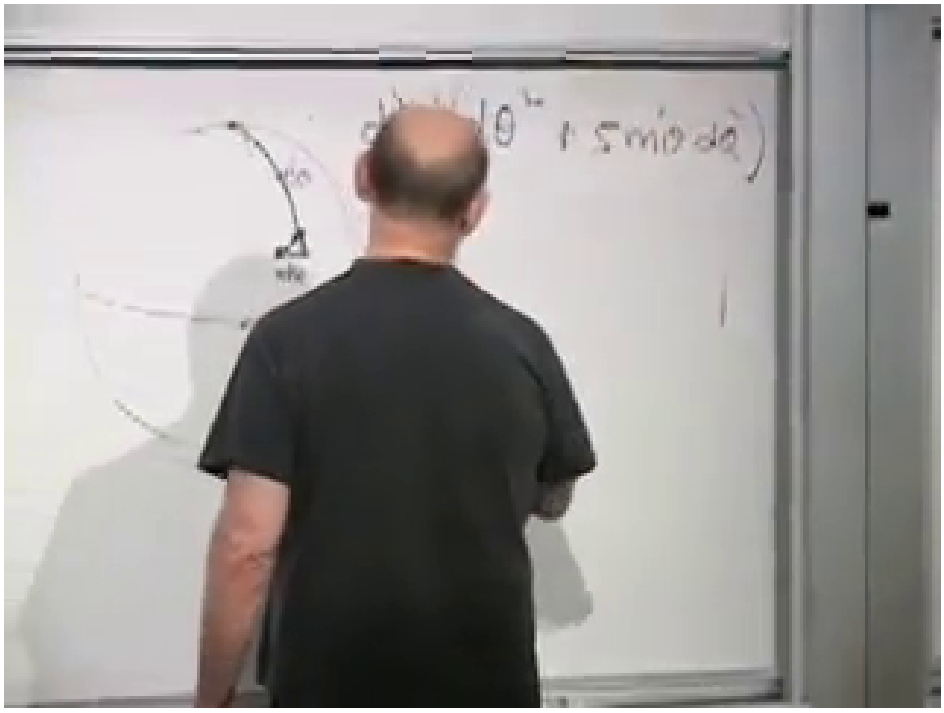


Figure 6.3: Sphere sketch and angular line element on the board. The presence of ds^2, $d\theta^2$, and the $\sin^2\theta$ factor is secure from the frame, while the clean displayed formula in the text is normalized from the transcript.

with metric matrix

$$g_{\text{sphere}} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2\theta \end{pmatrix}. \tag{6.19}$$

This metric cannot be globally flattened by any change of coordinates. That is why the sphere is intrinsically curved. A creature confined to the surface could discover that fact without ever leaving it: triangles drawn from the straightest available lines would have angle sum greater than 180°. So the lecture's criterion may now be stated in the form Susskind prefers: if one can find coordinates that make the metric coefficients constant everywhere, the space is flat; if not, it is curved.

6.5 From Curved Geometry to Cosmology

The same idea carries over to spacetime. A metric tensor $g_{\mu\nu}(x)$ may vary from point to point for trivial coordinate reasons; that in itself is not new. The new ingredient in general relativity is that spacetime itself can be intrinsically curved, so that there is no better coordinate choice that removes that variation globally. In that case the point-to-point variation of the metric is not merely a coordinate nuisance. It is geometry.

Susskind does not want to take us deeply into the general machinery of Einstein's theory. Instead he narrows immediately to the special class of geometries relevant for cosmology: spacetimes that are homogeneous in space, but time dependent. Here the lecture becomes careful about words. Homogeneous does not mean flat. A sphere is homogeneous, since every point is like every other, but it is not flat. Homogeneous also does not mean static. The universe may look the same from

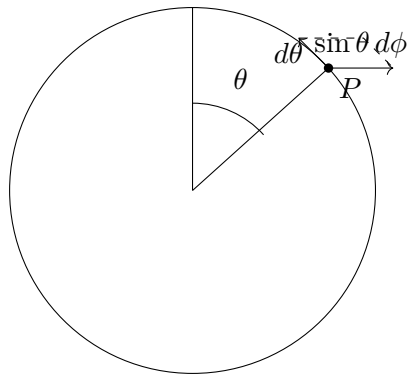


Figure 6.4: Local differential triangle on the unit sphere. The meridional leg has length $d\theta$, while the azimuthal leg has length $\sin\theta\, d\phi$.

point to point in space and still change with time.

6.5.1 Question & Answer

Question. How can the universe be homogeneous and spatially flat if we see structure everywhere, and if we cannot directly inspect distant points "all at once"?

Answer. Homogeneity is a statement about sufficiently large scales. Inside a star, inside a galaxy, and even on the scale of galaxy clusters, the universe is plainly not uniform. But on scales large compared with the intergalactic separation, the matter distribution smooths out and becomes approximately homogeneous. Nor is this an untouchable principle despite the historical name "cosmological principle"; it is an empirical claim and could in principle fail.

Answer. Likewise, spatial flatness is inferred indirectly. We cannot see three distant points at one instant and literally lay a triangle across them in real time. But we can collect signals later, correlate the data, and reconstruct what the geometry must have been. The claim that space is flat on large scales is therefore like the claim that a sufficiently small patch of the Earth looks flat: one does not prove zero curvature absolutely, one only finds that any radius of curvature must be much larger than the region explored.

This observational point matters. When astronomers say that space appears flat out to enormous distances, they mean that no deviation from Euclidean geometry is seen on those scales. If there is curvature, its radius must be much larger than the visible region. And if the universe began with appreciable curvature, the expansion may have diluted it to the point where it is now hard to detect. So the lecture keeps both possibilities alive: spatial flatness as a working approximation, and curvature as something whose radius may simply be beyond the present observational reach.

With that understood, we can ask for the simplest spacetime metric compatible with large-scale homogeneity and spatial flatness.

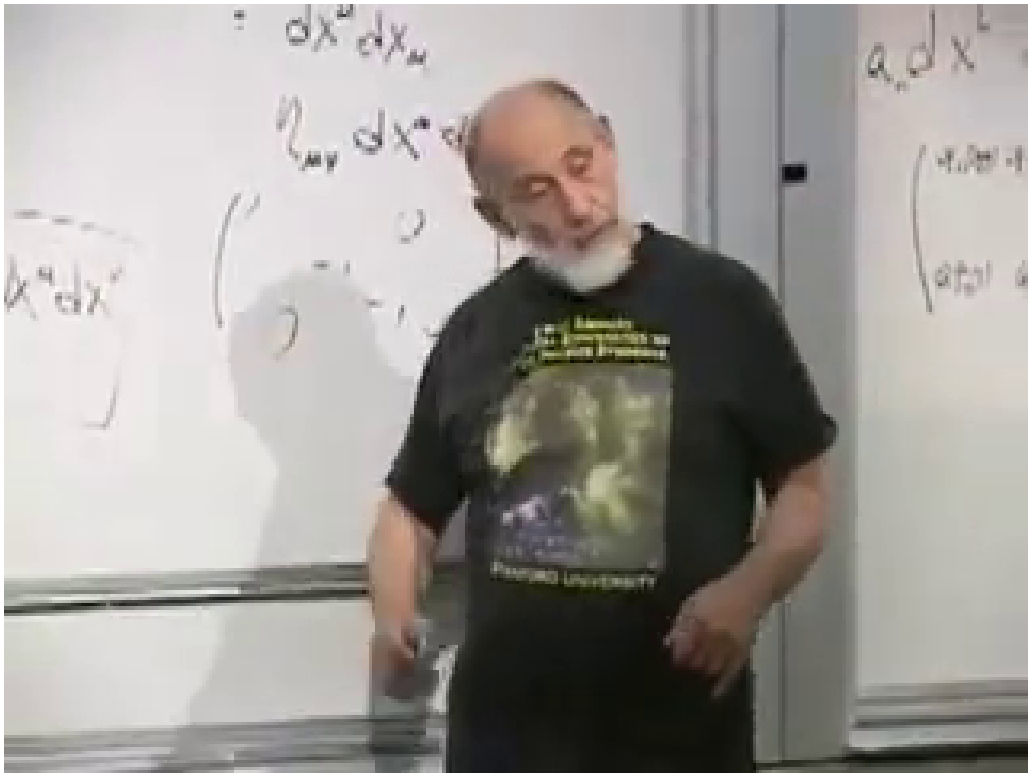


Figure 6.5: Transition from the concrete 2×2 metric example to tensor notation. The board still remembers the earlier matrix while moving to the general spacetime line element $ds^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu$.

6.6 The Expanding Metric, the Scale Factor, and the Hubble Law

At a fixed instant of time, spatial flatness means that we can choose coordinates so that the spatial metric is Cartesian. But those coordinates need not be meters or centimeters; Susskind wants us to think of them as labels attached to the cosmic flow. If one galaxy is labeled $x = 1$ and another $x = 5$, those labels can remain 1 and 5 forever, even while the actual distance between the galaxies changes. The conversion from labels to physical distance is carried by a time-dependent factor.

Because the interval is quadratic in lengths, that conversion factor enters as its square. Thus the cosmological metric is written

$$d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right). \tag{6.20}$$

This is the spatially flat, homogeneous expanding metric in comoving coordinates. The coordinates x, y, z may be taken as dimensionless labels if we like, while $a(t)$ carries the dimensions of length. One could distribute dimensions differently, but this is the convention Susskind recommends because it keeps the geometry transparent.

There is a useful check hidden in the formula. If $a(t)$ were constant, we could absorb it into a change of units for x, y, z. The metric would then be ordinary flat spacetime in disguise. So the nontrivial geometry is not in the mere presence of the symbol a; it is in the fact that a depends on time.

Now take two galaxies separated only along the x-direction. Their coordinate separation Δx is fixed,

because the coordinates expand with the flow. Their physical distance is

$$d(t) = a(t)\,\Delta x. \tag{6.21}$$

Differentiate with respect to time:

$$v(t) = \dot{d}(t) = \dot{a}(t)\,\Delta x, \tag{6.22}$$

$$= \frac{\dot{a}(t)}{a(t)}\,d(t). \tag{6.23}$$

Thus the recession speed is proportional to the distance. The proportionality factor is

$$H(t) = \frac{\dot{a}(t)}{a(t)}, \tag{6.24}$$

the Hubble parameter. It depends on when you are, not where you are. That universality is exactly what one expects from homogeneity. At a given instant of cosmic time, every sufficiently separated pair of comoving galaxies obeys the same law.

6.6.1 Question & Answer

Question. What is really expanding, and why do meter sticks, atoms, and galaxies not simply expand with the background?

Answer. The comoving coordinate mesh expands with the large-scale flow, so the physical distance between galaxies with fixed labels grows like $a(t)$. But small systems are held together by binding forces. A meter stick is held together by electromagnetic bonds; a galaxy is held together by its own gravity. On those scales the binding forces overwhelm the very weak large-scale expansion, so the object does not simply swell with the Hubble flow.

Answer. There are exceptions and near-exceptions. Andromeda, for example, is close and massive enough that its gravitational attraction overcomes the general expansion and pulls it toward us. More generally, the detailed study of how galaxies hold together led to the conclusion that there must be more mass in galaxies than is optically visible, which is one route to the inference of dark matter. So the statement that galaxies do not expand is not a slogan; it is a dynamical statement checked in considerable detail.

Answer. One should also not think of the scale factor as a force. The lecture insists on this point. A cannonball fired above escape velocity moves outward by inertia, not because a force keeps it moving outward. Gravity enters by changing the velocity, that is, by decelerating the motion, not by causing motion as such. In the same way, the cosmological expansion is not itself "a force." The dynamical question is how gravity changes $\dot{a}$, not what mysterious force keeps a going.

There is then a more philosophical objection, and Susskind takes it seriously. If the distance between galaxies divided by the length of a meter stick increases, could we not equally well say that meter sticks are shrinking? The answer is yes. Only ratios of lengths are observable. One could adopt a convention in which atomic or molecular scales vary while intergalactic distances remain fixed. The standard convention is simply to hold microscopic physics fixed and let the cosmological scale vary. It is a matter of taste and historical convenience, though a very natural one.

6.7 Light Rays, Hubble Time, and the Bridge to the Next Lecture

With the metric in hand, Susskind turns next to light. In both special and general relativity, a light ray follows a null line, so along its worldline

$$d\tau^2 = 0. \tag{6.25}$$

Restrict to motion along the x-axis and drop dy and dz. Then the cosmological metric gives

$$0 = dt^2 - a^2(t)\,dx^2. \tag{6.26}$$

So the null-ray equation is

$$dt = a(t)\,dx, \qquad \text{or equivalently} \qquad dx = \frac{dt}{a(t)}. \tag{6.27}$$

Hence the coordinate velocity of light is

$$\frac{dx}{dt} = \frac{1}{a(t)}. \tag{6.28}$$

In these comoving coordinates the light ray appears to slow down as $a(t)$ grows. But that is only because a fixed amount of coordinate distance dx corresponds to a larger and larger physical separation as the universe expands. The speed of light has not changed; the coordinate labels have become more widely spaced in physical terms.

The lecture makes one more methodological point here. Astronomers do not begin with a pile of raw observations and read cosmology directly off them. They need a geometric framework first. Very crudely, distances may be estimated from apparent luminosity and recession speeds from redshift, and that is enough to see the broad structure of Hubble's law. But once one goes to very distant galaxies, the fact that the Hubble parameter changes with time must be folded in. So the geometry comes first because it tells us what it is that observations mean.

That leads to the crude age estimate. Here Susskind makes a correction that should be preserved. One might be tempted to say, "suppose the Hubble constant were constant," but he immediately retracts that wording. The real approximation is simpler and more physical: suppose that if we run time backward, the recession velocity of a galaxy is treated as though it had been uniform. Then

$$t \sim \frac{d}{v}. \tag{6.29}$$

Using $v = Hd$, we get

$$t_{\text{age}} \sim H^{-1}. \tag{6.30}$$

This is only a rough estimate, because the recession rate changes with time. In the older decelerating picture, gravity makes the backward collision happen sooner than the uniform-velocity estimate would suggest, so H^{-1} somewhat overestimates the age. Still, it gives the right scale, and the lecture quotes a more refined age of about 13.5 billion years.

The chapter then reaches the real next question. Geometry has given us the form of the metric, but not the evolution of $a(t)$. To find that, we need dynamics.

Proposition 6.1. *For an inverse-square gravitational force and a spherically symmetric distribution of matter, the net force on a test galaxy at radius r depends only on the mass enclosed within radius r. The matter outside that radius contributes no net force, and the enclosed matter acts as if concentrated at the center.*

This is Newton's shell theorem, and it is the bridge to the next lecture. If a galaxy carries comoving label x, its physical distance from us is $a(t)x$. To determine how $a(t)$ evolves, we need only study the radial motion of that galaxy under the pull of the mass inside the sphere of radius $a(t)x$. The many-galaxy problem collapses to something very much like a cannonball moving away from the Earth under Newtonian gravity.

That is precisely where Susskind stops. He does not yet solve for $a(t)$. He only identifies the dynamical problem. The final student questions sharpen the tension without resolving it. If the universe was much smaller long ago, why can light emitted then now reach us from very large present-day distances? Why is the Big Bang not to be pictured as matter flying out in a shell from a single point? The lecture gives only the qualitative correction: the expansion is homogeneous, not shell-like. Everything expands everywhere. But the quantitative answer requires solving the null-ray equation for a definite $a(t)$, and that work is postponed.

6.8 Summary

The lecture unfolds in a carefully staged way. It begins by clearing up an issue from electrodynamics, so that the first lesson is already in place: symmetry constrains a theory but does not by itself determine it. From there Susskind reconstructs special relativity as geometry of the interval, then deliberately complicates coordinates in order to isolate the metric as the true geometric object. Only after that does he separate variable coefficients from genuine curvature, using polar coordinates on the plane and then the sphere as the decisive example.

With that language established, the lecture turns to cosmology. Large-scale homogeneity and spatial flatness lead to the expanding metric

$$d\tau^2 = dt^2 - a^2(t)\left(dx^2 + dy^2 + dz^2\right), \tag{6.31}$$

from which the Hubble law and the null-ray equation follow immediately. But the chapter ends where it should: not with a finished cosmology, but with the real outstanding problem. Geometry has told us how to describe expansion. The next step is to determine dynamically how the scale factor $a(t)$ actually evolves.

Chapter 7

Flat Expanding Space and the Dynamics of the Scale Factor

These notes follow Leonard Susskind's lecture closely, with curation by LazyingArt LLC. We begin where the lecture begins: not with dynamics, but with a recap of the kinematics already in hand. The universe looks homogeneous and isotropic on very large scales, it appears to be spatially flat to high accuracy, and the new element is a time-dependent scale factor $a(t)$. From there the argument unfolds in the same order as the lecture itself: first co-moving coordinates and Hubble's law, then the behavior of light, then the Newtonian derivation of the flat expansion law, and finally the way different forms of energy drive different histories for the universe.

7.1 Recap: Flat Expanding Space and the Scale Factor

Susskind opens by recalling the contrast with Minkowski space. In special relativity the line element is

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{7.1}$$

The transcript in this early stretch is not perfectly clean, but the intended point is unmistakable: in flat space-time the spatial piece is just the Euclidean combination $dx^2 + dy^2 + dz^2$, whereas in general relativity the coefficients of the metric become fields, capable of depending on position and time.

Cosmology then adds a crucial observational fact. On scales large compared with galaxies and clusters, the universe is astonishingly featureless. That empirical statement is what lies behind the cosmological principle: space is homogeneous and isotropic on sufficiently large scales. It is not a theorem deduced from deeper principles; it is a very successful large-scale description of what we see. A second observational fact is equally important: space appears to be very nearly flat.

That does not mean that flatness has been proved in some absolute sense. The lecture makes the point by comparison with surveying a football field. Triangles drawn on the field behave, to high precision, like Euclidean triangles, so one concludes not that the earth is exactly flat, but that its curvature is small on that scale. Cosmology proceeds the same way. Large cosmic triangles appear Euclidean, so the simplest large-scale description is flat space.

With that observational motivation in place, the flat expanding metric is

$$ds^2 = dt^2 - a^2(t)(dx^2 + dy^2 + dz^2). \tag{7.2}$$

The new object is the scale factor $a(t)$. The lecture emphasizes that one should not think of the coordinates x, y, z as fixed measuring rods. In the more natural convention here they are merely labels attached to points of space, while $a(t)$ carries the physical scale. One may picture a flat rubber sheet being stretched: the coordinate mesh rides along with the sheet, but the physical distance represented by a given coordinate separation changes with time.

That is the real opening move of the lecture. Once distances are controlled by a time-dependent scale factor, the natural question is no longer what the geometry looks like, but what equations govern $a(t)$.

7.2 Co-Moving Coordinates, Physical Distance, and Hubble's Law

Susskind now returns to a point from the earlier lecture and insists on doing it again, precisely because it is so important. The distinction between coordinate separation and true physical distance is the foundation of everything that follows.

In co-moving coordinates the large-scale matter distribution is at rest. Galaxies keep the same coordinate labels while the geometry stretches beneath them. If two galaxies are separated by Δx, then their true distance is

$$d = a(t)\,\Delta x. \tag{7.3}$$

This is the primitive statement. Hubble's law will be derived from it, not postulated independently.

Differentiate with respect to time. Since the galaxies do not move in co-moving coordinates, Δx is fixed:

$$v = \frac{dd}{dt} = \dot{a}\,\Delta x. \tag{7.4}$$

At this stage Susskind performs a small but telling algebraic maneuver: he multiplies and divides by a so that the naked coordinate separation Δx is replaced by the physical distance d. Thus

$$v = \frac{\dot{a}}{a}\,d. \tag{7.5}$$

This is Hubble's law. It tells us that recession velocity is proportional to physical distance, and it identifies the proportionality factor as

$$H(t) = \frac{\dot{a}}{a}. \tag{7.6}$$

The lecture immediately adds a useful warning. One still calls H the Hubble constant, but it need not be constant in time. If we look only modestly far away, compared with the age of the universe, we may treat it as nearly constant. If we look truly deep into the universe, we are also looking far back in time, and then we must know how $a(t)$ evolves.

7.3 Light Propagation in the Expanding Metric

At exactly the point where the scale factor first begins to matter dynamically, the lecture pauses for a student question: what does the expanding metric do to light? That detour is not ornamental. It resolves a conceptual obstacle before the lecture moves on to the dynamical equation.

For a light ray the proper time vanishes. In the language of the lecture, $d\tau = 0$, equivalently $ds^2 = 0$. Suppose the ray moves along the x-axis, so that $dy = dz = 0$. Then from the metric we obtain

$$0 = dt^2 - a^2(t)\, dx^2. \tag{7.7}$$

For motion in the positive x-direction this becomes

$$dt = a(t)\, dx. \tag{7.8}$$

Dividing by dt, we find

$$\frac{dx}{dt} = \frac{1}{a(t)}. \tag{7.9}$$

That formula is worth lingering over. In co-moving coordinates the coordinate speed of light is not constant; it decreases as $a(t)$ grows. But the lecture is careful here: this is not because light literally slows down. It is because dx is only a coordinate increment, not a physical distance. A fixed dx corresponds to a larger and larger true spatial separation as the universe expands.

7.3.1 Question & Answer

Does light really slow down in an expanding universe?

No. Only the coordinate speed changes. The physical speed of light remains the same; what changes is the amount of true distance represented by a given interval dx. As the scale factor grows, the same coordinate increment spans a larger physical length, so a light ray traced in co-moving coordinates appears to advance more slowly.

The lecture pushes the point a little further. If one insisted on describing cosmology in a perverse way, one might say that distant light appears to go slower and slower. But that language would force one into equally awkward statements about distances and measuring rods. The clean way to think is the original one: space expands, the co-moving labels are carried along with it, and the coordinate speed of light reflects that choice of coordinates.

Susskind also notes that there is strong evidence that the dimensionless constants of nature are the same everywhere. In that sense there is no sign that local atomic physics has somehow drifted from one part of the universe to another. The oddness lies in the geometry, not in a local breakdown of the speed of light.

7.4 Other Homogeneous Geometries and the Return to Flat Space

Only after the light calculation does the lecture widen the geometric backdrop. Homogeneity and isotropy do not force space to be flat. They allow three possibilities.

First there is positive curvature, represented most simply by a sphere. Here the lecture spends some time on terminology: the solid interior is a ball, the boundary is a sphere. More importantly, the geometry is intrinsic. A sphere need not be thought of as living inside some higher-dimensional ambient space in order to make sense. What matters is that triangles drawn in positively curved space have angle sums greater than 180°.

Second there is the flat case, where Euclidean geometry holds and triangle angles add to 180°.

Third there is negative curvature, for which the lecture uses the visual analogy of a saddle surface. Again the embedding picture is only a crutch. The real geometric point is that triangles in negatively curved space have angle sums less than 180°.

Thus homogeneity does not imply flatness. A sphere is homogeneous and isotropic; so is the negatively curved case. Observation is what selects flatness as the simplest description of our universe.

The lecture also takes care to correct a common misconception. If the universe were positively curved and closed, then tracing the expansion backward would not mean that matter was once concentrated at a special point and then exploded into some external void. Rather, the entire space itself would once have been small, so all points in it would have been close together.

With that clarification on the table, Susskind deliberately specializes again to the flat case. The lecture's message here is simple: curved homogeneous spaces are logically possible, but observation lets us do the main calculation in the flat approximation, and that is where the mathematics now goes.

7.5 Newtonian Cosmology and the Flat Zero-Energy Equation

The real mathematical spine of the lecture begins here. We place ourselves at the center, draw a large sphere around us, and choose a distant galaxy on the boundary. By isotropy the matter distribution is spherically symmetric about us. Then the inverse-square law gives a remarkable simplification.

Matter outside the sphere exerts no net force on the chosen galaxy. Matter inside the sphere acts as if it were all concentrated at the center. This is the shell theorem, familiar from Newtonian gravity and electrostatics, and it is the reason the cosmological problem can be turned into something that looks almost embarrassingly elementary: a projectile moving in the gravitational field of a central mass.

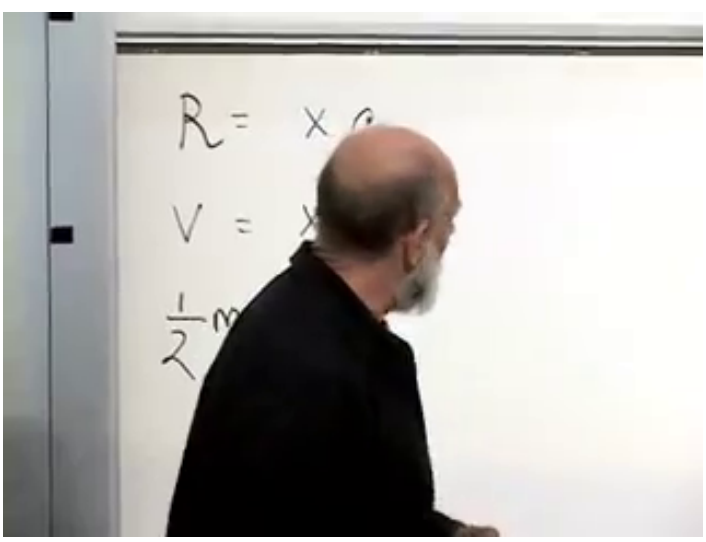


Figure 7.1: Radius, velocity, and energy setup on the board

The board is partly blocked in this frame, but its order matters and the transcript fixes the intended

equations. For a galaxy sitting at co-moving coordinate x,

$$R = x\,a(t), \qquad v = x\,\dot{a}(t). \tag{7.10}$$

Here R is the true physical distance to the galaxy and v its recession speed relative to us.

Let the chosen galaxy have mass m, and let M denote the total mass enclosed inside radius R. The Newtonian energy of the galaxy is then

$$E = \frac{1}{2}mv^2 - \frac{GMm}{R}. \tag{7.11}$$

The lecture now makes a deliberate simplification. Susskind postpones the general discussion of positive, zero, and negative total energy, and studies first the case

$$E = 0. \tag{7.12}$$

That is the escape-velocity case: kinetic energy and gravitational potential energy balance exactly. In this case

$$\frac{1}{2}mv^2 = \frac{GMm}{R}. \tag{7.13}$$

After dividing by m,

$$\frac{1}{2}v^2 = \frac{GM}{R}. \tag{7.14}$$

7.5.1 Question & Answer

Why does the enclosed mass M stay fixed while the density ρ changes?

Because the sphere is co-moving. The same galaxies remain inside it as the universe expands. The total enclosed mass therefore stays fixed. What changes is the physical volume of the sphere, and that is why the density decreases.

This is one of the best moments in the lecture's pacing. A student asks whether M varies with time, and Susskind stops to separate two ideas that are easy to blur together: conserved total mass in a co-moving region, and time-dependent mass density in the expanding volume of that region.

Now write the enclosed mass in terms of the density:

$$M = \frac{4\pi}{3}\rho R^3. \tag{7.15}$$

Substituting gives the board equation preserved in the next frame:

$$\frac{1}{2}v^2 = \frac{G}{R}\left(\frac{4\pi}{3}\rho R^3\right). \tag{7.16}$$

Now use Hubble's law in the form $v = HR$. Then

$$\frac{1}{2}H^2R^2 = \frac{G}{R}\left(\frac{4\pi}{3}\rho R^3\right), \tag{7.17}$$

$$\frac{1}{2}H^2R^2 = \frac{4\pi G}{3}\rho\,R^2. \tag{7.18}$$

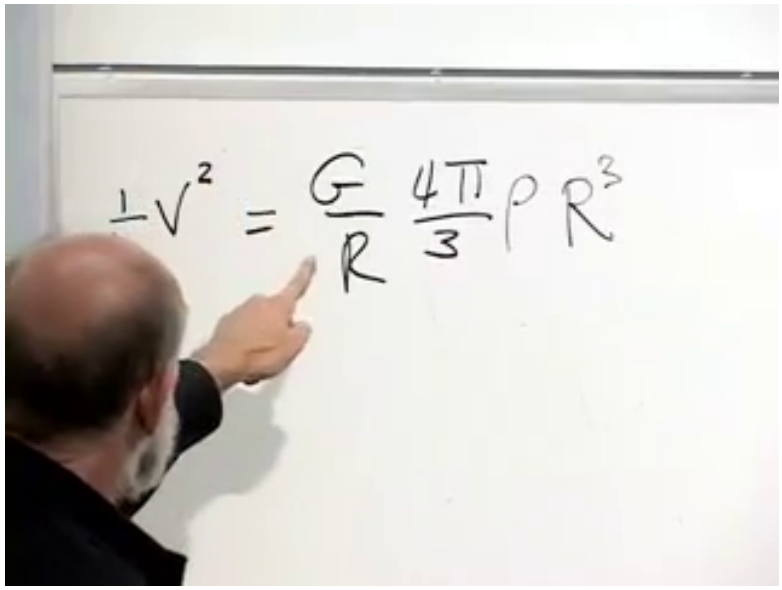


Figure 7.2: Energy equation with the enclosed mass term

The factor of R^2 cancels, and we are left with

$$\frac{1}{2}H^2 = \frac{4\pi G}{3}\rho, \tag{7.19}$$

or, in the customary form,

$$H^2 = \frac{8\pi G}{3}\rho, \qquad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho(t). \tag{7.20}$$

The lecture makes a nice observational remark at this point. The cancellation of R^2 had to occur. If the result depended on which galaxy we chose to follow, then it could not possibly define a universal Hubble parameter.

7.6 Density Dilution and the Matter-Dominated Solution

At this stage the equation for $a(t)$ is not yet closed. We have learned that $(\dot{a}/a)^2$ is proportional to $\rho(t)$, but we do not yet know how ρ changes with time. That is the next pivot of the lecture.

Susskind now shifts from the expanding sphere to an expanding co-moving box. Fix the co-moving side lengths Δx, Δy, Δz, and follow the box as the universe expands. If the box contains ordinary nonrelativistic matter, then the number of particles inside it stays fixed. The physical volume grows like a^3, and the density decreases accordingly.

The board captures the key sequence: first the raw density formula, then the normalization by present-day values, and then the time-dependent law. In the notation of the lecture,

$$\rho = \frac{M}{(\Delta x\, a)^3}. \tag{7.21}$$

So for matter we may write

$$\rho(t) = \frac{C}{a^3(t)}, \qquad \rho_{\text{today}} = \frac{C}{a^3_{\text{today}}}. \tag{7.22}$$

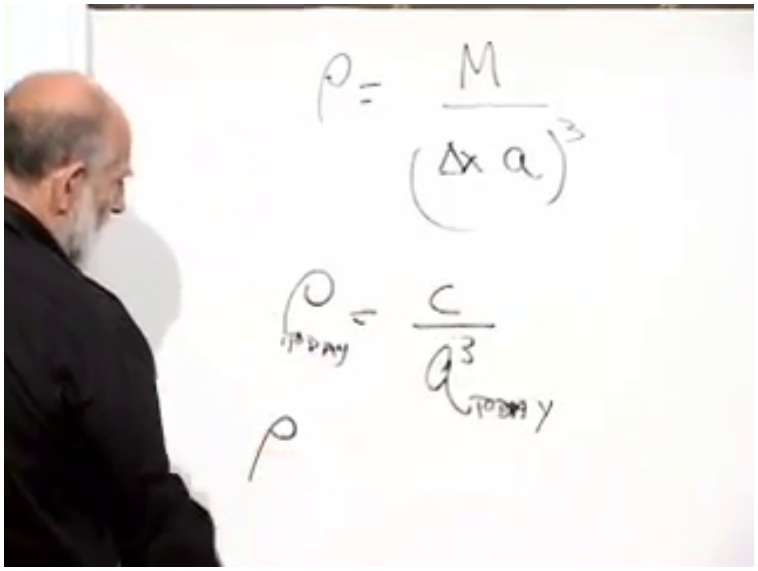


Figure 7.3: Matter density scaling as the universe expands

Dividing the two equations gives

$$\frac{\rho(t)}{\rho_{\text{today}}} = \frac{a^3_{\text{today}}}{a^3(t)}. \tag{7.23}$$

This is the lecture's matter-dilution law:

$$\rho_{\text{m}} \propto a^{-3}. \tag{7.24}$$

Now substitute this back into the flat expansion equation. If we absorb all fixed numerical factors into a new positive constant K, we obtain

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{K}{a^3}. \tag{7.25}$$

Multiply by a^2:

$$\dot{a}^2 = \frac{K}{a}. \tag{7.26}$$

This is the point where the lecture becomes especially characteristic. Susskind first tries a power-law guess, notices that something has gone wrong in the algebra, and then restarts with the cleaner route. To preserve the logic while presenting the mathematics in corrected form, we follow that cleaner route directly. Multiply by a and take the square root:

$$\sqrt{a}\,\dot{a} = \sqrt{K}. \tag{7.27}$$

Now separate variables:

$$\sqrt{a}\,da = \sqrt{K}\,dt. \tag{7.28}$$

Integrating,

$$\frac{2}{3}a^{3/2} = \sqrt{K}\,t + \text{constant}. \tag{7.29}$$

Thus

$$a^{3/2} \propto t, \qquad a(t) \propto t^{2/3}. \tag{7.30}$$

This is the matter-dominated universe.

The Hubble parameter is then

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{7.31}$$

So the universe expands forever in this model, but the expansion rate slows in the precise sense that H decreases like $1/t$. Looking backward in time, the Hubble parameter is larger; looking forward, it is smaller.

7.7 Radiation, Vacuum Energy, and the Final Energy Bookkeeping

Once the matter case is in hand, the lecture asks the right next question: is $\rho \propto a^{-3}$ really necessary? The answer is no. The expansion law depends on what kind of energy fills the universe.

Suppose the co-moving box contains photons rather than nonrelativistic matter. The number of photons in the box stays the same, but now the energy of each individual photon changes as the box expands. The lecture uses

$$E = h\nu, \tag{7.32}$$

and, in effect, $\nu \propto 1/\lambda$, so

$$E_\gamma \propto \frac{1}{\lambda}. \tag{7.33}$$

As the universe expands, the wavelength stretches with it, $\lambda \propto a$, and therefore

$$E_\gamma \propto \frac{1}{a}. \tag{7.34}$$

This contributes an extra factor of a^{-1} beyond the ordinary dilution by volume. Hence

$$\rho_{\text{rad}} \propto a^{-4}. \tag{7.35}$$

If we substitute $\rho_{\text{rad}} \propto a^{-4}$ into the same flat equation, the algebra changes only slightly. With a new positive constant K_{r},

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{K_{\text{r}}}{a^4}, \tag{7.36}$$

so

$$\dot{a}^2 = \frac{K_{\text{r}}}{a^2}. \tag{7.37}$$

Taking square roots and separating variables gives $a\,da \propto dt$, and therefore

$$a(t) \propto t^{1/2}. \tag{7.38}$$

This is the radiation-dominated universe. The lecture stresses the physical picture: if we run the expansion backward, photons gain energy as their wavelengths are squeezed, and at sufficiently early times radiation dominates over ordinary matter.

There is then one more case, and in the lecture it is the most important one for the present universe. Vacuum energy is defined here by the fact that its density does not dilute:

$$\rho_{\text{vac}} = \text{const}. \tag{7.39}$$

Substituting into the cosmological equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\text{vac}}. \tag{7.40}$$

Everything on the right-hand side is constant, so if we define

$$H = \sqrt{\frac{8\pi G}{3}\rho_{\text{vac}}}, \tag{7.41}$$

then

$$\dot{a} = Ha. \tag{7.42}$$

The solution is

$$a(t) \propto e^{Ht}. \tag{7.43}$$

Thus a vacuum-dominated universe expands exponentially.

The lecture next adds a fact that must be kept conceptually separate from the Newtonian derivation. Einstein's equations relate the sign of the total energy to the sign of the spatial curvature:

$$E > 0 \leftrightarrow \text{negative curvature}, \qquad E = 0 \leftrightarrow \text{flat space}, \qquad E < 0 \leftrightarrow \text{positive curvature}. \tag{7.44}$$

Susskind is explicit that this correspondence does not follow from Newton alone. Newton gets us astonishingly far, but not that far.

The closing figure is worth keeping because it shows the lecture's board logic: the earlier zero-energy equation remains boxed above, while the lower line rewrites the same physics in Hubble form.

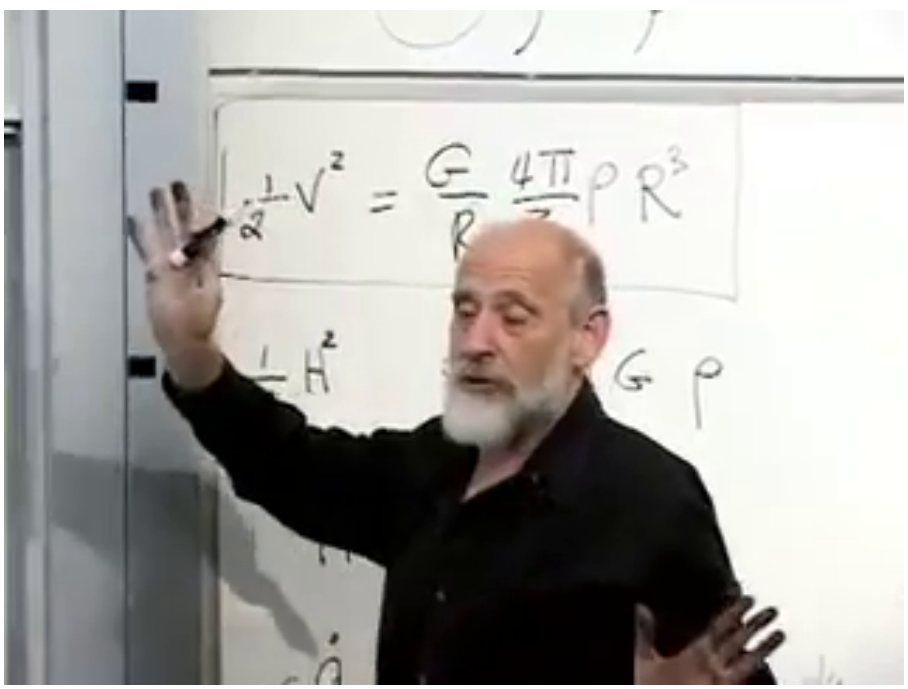


Figure 7.4: Boxed energy equation and its Hubble-form rewriting

In cleaned form, the lower board relation is

$$\frac{1}{2}H^2 = \frac{4\pi G}{3}\rho, \qquad H^2 = \frac{8\pi G}{3}\rho. \tag{7.45}$$

The frame itself does not fix the coefficient with complete reliability, but it does preserve the crucial pedagogical transition from Newtonian projectile energy to cosmological expansion.

The question period now circles back to photons and redshift. Susskind gives two complementary descriptions. In one, the wavelengths are stretched directly by the expansion of space. In the other, the photons we now see as the microwave background come from the surface of last scattering, which is receding from us and therefore Doppler-shifts the radiation it emits. The lecture insists that these are not rival explanations; they are two ways of organizing the same physics.

The surface of last scattering is also placed in a concrete cosmological picture. Looking outward means looking backward. Far enough out, the universe was once hot enough to be a plasma and therefore opaque, like the surface of the sun. That hot shell is seen today only after a large redshift, which is why radiation originally characteristic of a much hotter epoch now arrives as microwaves. If we imagine observers in the past, they would see the same last-scattering surface closer to them and moving away more slowly, and therefore they would observe less redshift. The lecture uses this changing picture to make the expansion and the redshift feel geometrically real rather than merely algebraic.

7.7.1 Question & Answer

Where does the energy of a redshifted photon go?

Susskind's answer is that one should not try to conserve the photon energy by itself. The cosmological bookkeeping involves two terms: the ordinary energy density and the energy of expansion. As radiation redshifts, the energy stored in the photons decreases, while the expansion term changes in tandem.

The lecture's analogy is a cylinder with a movable piston filled with photons. As the photons push the piston outward, their wavelengths lengthen and their individual energies decrease. That lost photon energy has not vanished; it has gone into the motion of the piston. Cosmological expansion is not literally a piston, but the analogy captures the intended transfer: the energy lost by redshifting radiation goes into the kinetic energy of expansion.

The lecture is careful not to package this as a precise theorem about globally conserved energy in arbitrary space-times. The point is narrower and more physical: in cosmology, one does not get sensible energy bookkeeping by looking at matter or radiation alone and ignoring the expansion.

A final coordinate question appears here as well. Why does the scale factor multiply the spatial part of the metric but not the time part? Susskind's answer is that a factor in front of dt^2 can be removed by a redefinition of time. If one wrote schematically

$$ds^2 = A^2(t)\,dt^2 - A^2(t)\,dx^2, \tag{7.46}$$

then by defining

$$dT = A(t)\,dt \tag{7.47}$$

one absorbs the factor in front of the time term. The spatial factor remains physically significant. So the choice to write the metric with the scale factor multiplying the spatial part is, in part, a choice of time coordinate.

The lecture ends by folding curvature and vacuum energy back into the discussion of cosmic fate. If there were no vacuum energy, then a positively curved universe would eventually turn around and recollapse. But if the vacuum energy is large enough, that recollapse can be prevented. In

the empirical picture emphasized in the lecture, the future is instead one of continuing exponential expansion, with ordinary forms of matter and radiation diluting away.

7.8 Summary

The lecture proceeds in a very deliberate order. First it fixes the kinematics: homogeneous and isotropic space, observational flatness, the metric with scale factor $a(t)$, and the distinction between co-moving labels and true distance. From that foundation it derives Hubble's law and clarifies how light moves in the expanding metric.

Only then does it turn to dynamics. The Newtonian shell-theorem construction leads, in the flat zero-energy case, to $H^2 = \frac{8\pi G}{3}\rho$. But that equation acquires content only after one specifies how ρ changes with expansion. Matter gives $\rho \propto a^{-3}$ and $a(t) \propto t^{2/3}$; radiation gives $\rho \propto a^{-4}$ and $a(t) \propto t^{1/2}$; vacuum energy gives $\rho = \text{const}$ and $a(t) \propto e^{Ht}$.

The chapter then closes where the lecture closes: with the questions that make the formulas feel physically real. How does redshift arise? What is the role of the surface of last scattering? Where does the lost photon energy go? Why is the metric written in this particular form? Those are not detachable afterthoughts. They are the final step by which the lecture turns a string of equations into a coherent picture of an expanding universe.

Chapter 8

Maxwell Equations, Covariance, and Electromagnetic Waves

We begin exactly where the previous lecture stopped. We already know how electric and magnetic fields act on charged particles; the Lorentz force law tells us that part of the story. The missing half is the reverse question: what determines the fields themselves, and how do charges and currents generate them? In this lecture Leonard Susskind answers that question in a deliberately relativistic way. Maxwell's equations are not introduced merely as a list to memorize, but as field equations to be motivated physically, written covariantly, checked component by component, and then used to derive the existence of light waves. Throughout we work in units with $c = 1$. These notes follow Leonard Susskind closely and are curated by LazyingArt LLC.

8.1 From the Lorentz Force to the Missing Field Equations

The lecture opens by looking backward before it moves forward. Electric and magnetic fields push charged particles around, and the force is proportional to the charge because an uncharged particle does not feel the electromagnetic field at all. In ordinary three-vector language the law is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{8.1}$$

This is the familiar half of electromagnetism: given the fields, we know how matter responds.

The board then pivots toward covariant notation. As written there, the second line is only schematic; it suppresses the free force index and the contraction with the four-velocity. The clean covariant statement is

$$f^\mu = q\, F^\mu{}_\nu\, u^\nu. \tag{8.2}$$

Here f^μ is the four-force, u^ν is the four-velocity, and $F^{\mu\nu}$ is the antisymmetric electromagnetic field tensor that we will reconstruct shortly. The point of the reminder is not just formal prettiness. Written this way, the Lorentz force law is manifestly the same in every inertial frame.

That is already the standard against which Maxwell's equations will be judged. The missing question is no longer how a charged particle moves in a prescribed field. The missing question is what determines the field itself: what decides whether there is an electric field here, a magnetic field

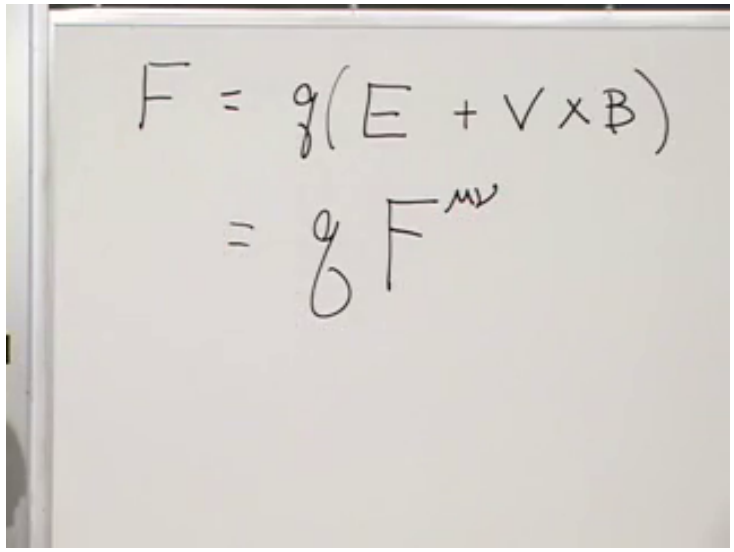


Figure 8.1: Lorentz force rewritten schematically in field-tensor notation.

there, or an electromagnetic wave moving through empty space? Those are the equations we now need, and those equations are Maxwell's equations.

As always in this course, the broader issue is relativity. If Maxwell's equations are the theory of light, and if the laws of physics are the same in all inertial frames, then light itself must come out with the same speed in all inertial frames. That is the puzzle Einstein struggled with, and the lecture keeps it in view from the start.

8.2 Why Charges Must Affect Fields

Before writing any field equations, Susskind slows down and asks why charged matter must affect the field at all. A point charge creates an electric field spreading outward from it. A current in a wire creates a magnetic field circulating around the wire. Those are familiar empirical facts, but the lecture wants the deeper logic that makes Maxwell's equations necessary.

The key point is reciprocity sharpened by conservation laws. An electromagnetic wave can hit a charged particle, change its motion, and give it energy. If that can happen, then the influence cannot run only one way. Charged matter must also be able to alter the electromagnetic field. Otherwise energy could appear in the particle without being taken from anywhere.

8.2.1 Question & Answer

Question. Why must charged matter act back on the electromagnetic field?

Answer. Because otherwise energy conservation would fail. If a field can do work on a charge, then a charge must be able to draw energy from the field and modify it. One may call this action and reaction, but the lecture's sharper point is conservation: the field cannot give energy to matter unless matter can affect the field in return.

The lecture then turns that abstract point into a causal picture. Suppose we move a point charge suddenly from one place to another. The field cannot rearrange itself instantaneously throughout all of space. A distant observer is not allowed to see the effect before a light signal could have arrived. Thus the field must rearrange by sending out a disturbance. Near the charge, the rearrangement happens first; farther away, the old field remains until the signal reaches it. If we shake the charge back and forth, the outward-moving disturbance becomes an electromagnetic wave. An antenna is precisely such a device.

The same story can be told for currents. Reverse a current in a wire, and the magnetic field around the wire must reverse as well. But that reversal cannot be known arbitrarily far away in arbitrarily short time. The change in the magnetic field must propagate outward, again as an electromagnetic disturbance. This is the basic physics Maxwell's equations must describe.

8.3 Wave Review and Vector-Calculus Review

Only now does the lecture pause for review. This is not filler. Before Maxwell's equations can be made to produce waves, we first recall what wave equations look like.

Start with a scalar field ϕ, a single number attached to each point of spacetime. The relativistic wave equation is

$$\partial_\mu \partial^\mu \phi = 0. \tag{8.3}$$

With the sign convention used throughout the lecture, this means

$$\partial_t^2 \phi - \partial_x^2 \phi - \partial_y^2 \phi - \partial_z^2 \phi = 0. \tag{8.4}$$

If the wave moves along the z-axis, then ϕ depends on z and t but not on x or y, and the equation reduces to

$$\partial_t^2 \phi - \partial_z^2 \phi = 0. \tag{8.5}$$

The lecture emphasizes the two basic traveling-wave solutions,

$$\phi = f(z - t), \qquad \phi = g(z + t), \tag{8.6}$$

representing right-moving and left-moving profiles. A particularly useful special case is the sinusoidal wave

$$\phi(z, t) = \cos\big(k(z - t)\big), \tag{8.7}$$

where k is the wave number. Large k means short wavelength; small k means long wavelength. These are exactly the shapes that will later reappear in the electric and magnetic fields themselves.

The lecture then reminds us of the vector operations Maxwell's equations use. For ordinary three-vectors,

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z, \tag{8.8}$$

$$(\mathbf{a} \times \mathbf{b})_x = a_y b_z - a_z b_y, \tag{8.9}$$

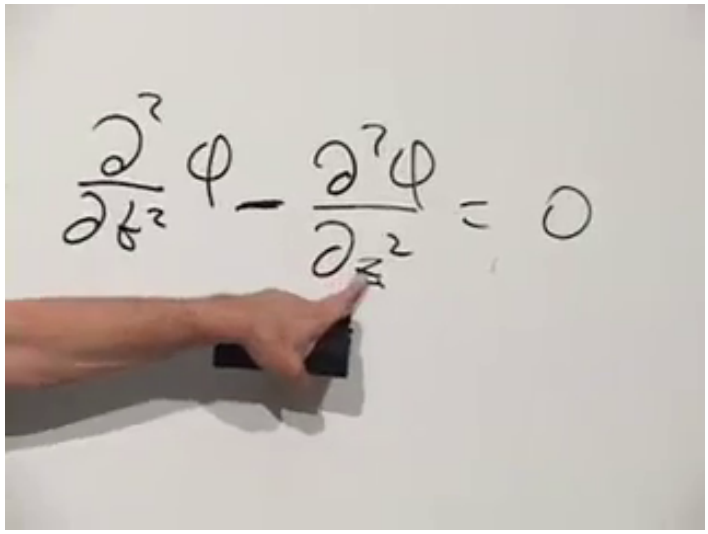


Figure 8.2: One-dimensional reduction of the wave equation.

with the remaining components obtained by cyclic permutation. For a vector field $\mathbf{A}$,

$$\nabla \cdot \mathbf{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z, \tag{8.10}$$

$$(\nabla \times \mathbf{A})_x = \partial_y A_z - \partial_z A_y, \tag{8.11}$$

again with cyclic permutation for the other components.

Susskind keeps the geometry in sight. Divergence measures the extent to which a field seems to emanate from somewhere. Curl measures the extent to which it circulates around something. A field of arrows flowing outward from a point suggests divergence; a field winding around a wire suggests curl. These are the operational meanings needed when Maxwell's equations are written down.

8.4 Vacuum Maxwell Equations and the Relativity Puzzle

With that preparation in place, the lecture writes the source-free Maxwell equations, in the order actually used on the board:

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{8.12}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{8.13}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}, \tag{8.14}$$

$$\nabla \cdot \mathbf{E} = 0. \tag{8.15}$$

These are the vacuum Maxwell equations. They describe electric and magnetic fields in the absence of charge density and current density.

Two features are immediately visible. First, the curl equations are almost symmetric under interchange of $\mathbf{E}$ and $\mathbf{B}$, but not quite: there is an essential sign difference. Second, the divergence equations are asymmetric. The electric field will later acquire a source term ρ, but the magnetic field does not acquire an analogous source term in this lecture, so $\nabla \cdot \mathbf{B} = 0$.

Susskind briefly points out where sources will eventually go. One will later write $\nabla \cdot \mathbf{E} = \rho$ and $\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}$, but for the moment those terms are set aside. This postponement is deliberate. The vacuum equations reveal the covariant structure in its cleanest form.

8.4.1 Question & Answer

Question. How can Maxwell's equations be the same in every frame if they imply the same light speed in every frame?

Answer. That is precisely the relativistic puzzle. If Maxwell's equations are the theory of light, and if those equations are the same in every inertial frame, then every inertial frame must predict the same light speed from them. The resolution is not to patch the wave speed by hand from frame to frame. The resolution is to express the laws in terms of objects with definite Lorentz transformation rules, so that the equations themselves are manifestly frame-independent.

That is why the lecture now turns to the electromagnetic field tensor. The payoff is not elegance by itself. The payoff is that once the equations are written as tensor equations, their invariance is visible, and the constancy of the speed of light follows from the same structure.

8.5 The Electromagnetic Field Tensor

The next move is the central formal pivot of the lecture. The electric and magnetic fields are assembled into a single antisymmetric 4×4 tensor $F^{\mu\nu}$. The time coordinate is placed first, so the row and column order is t, x, y, z. The board image fixes the electric entries and the antisymmetric layout.

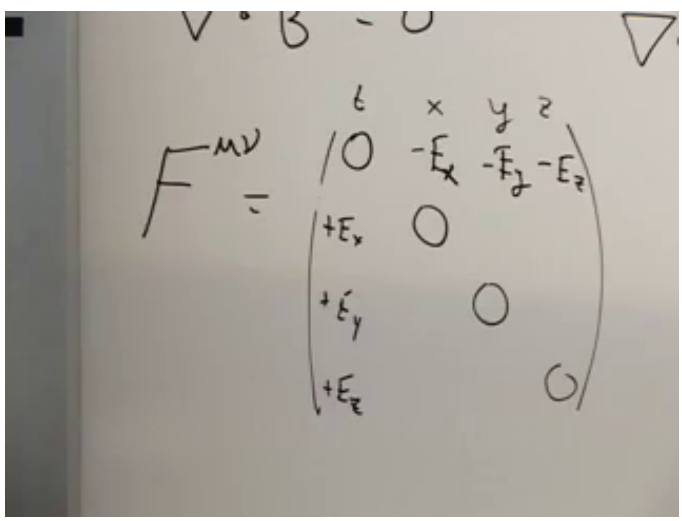


Figure 8.3: Board layout of the electromagnetic field tensor.

With the sign convention fixed by the visible board entries, we write

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \tag{8.16}$$

The diagonal entries vanish because the tensor is antisymmetric, and the entries below the diagonal are the negatives of those above it. The time-space slots contain the electric field; the purely spatial slots contain the magnetic field.

The lecture then introduces a second antisymmetric tensor, the dual tensor $\tilde{F}^{\mu\nu}$. We do not need to overdevelop it. What matters is the replacement rule, stated cautiously because Susskind corrects himself while talking through the signs:

$$\mathbf{E} \mapsto -\mathbf{B}, \qquad \mathbf{B} \mapsto \mathbf{E}. \tag{8.17}$$

With that convention, $\tilde{F}^{\mu\nu}$ packages the other half of Maxwell's equations.

At this stage the lecture's rhythm is very clear: write the tensor, inspect its slots, and then ask what covariant first-derivative equations it could possibly satisfy. Since Maxwell's equations contain only first derivatives and should leave one free index, the natural guess is

$$\partial_\mu F^{\mu\nu} = 0, \tag{8.18}$$

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{8.19}$$

There are four equations in the first line and four in the second, one for each choice of ν.

Now comes the trust-building step: we check components. For the time component,

$$\partial_\mu F^{\mu t} = \partial_t F^{tt} + \partial_x F^{xt} + \partial_y F^{yt} + \partial_z F^{zt} \tag{8.20}$$

$$= \partial_x E_x + \partial_y E_y + \partial_z E_z \tag{8.21}$$

$$= \nabla \cdot \mathbf{E}. \tag{8.22}$$

Thus $\partial_\mu F^{\mu t} = 0$ is exactly $\nabla \cdot \mathbf{E} = 0$.

To stay close to the lecture, it is also useful to check a spatial component. Taking $\nu = z$,

$$\partial_\mu F^{\mu z} = \partial_t F^{tz} + \partial_x F^{xz} + \partial_y F^{yz} + \partial_z F^{zz} \tag{8.23}$$

$$= -\partial_t E_z + \partial_x B_y - \partial_y B_x \tag{8.24}$$

$$= -\partial_t E_z + (\nabla \times \mathbf{B})_z. \tag{8.25}$$

So $\partial_\mu F^{\mu z} = 0$ is precisely the z-component of

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E}. \tag{8.26}$$

The other spatial components work the same way. Likewise, $\partial_\mu \tilde{F}^{\mu\nu} = 0$ packages $\nabla \cdot \mathbf{B} = 0$ together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

This is the formal heart of the lecture. Once Maxwell's equations are written as tensor equations, we can see immediately that they are the same in every inertial frame.

8.6 From Maxwell to Electromagnetic Waves

The lecture then pivots from invariance to content: now that Maxwell's equations are written covariantly, what do they actually say? The answer is that they allow wave-like solutions.

We begin with

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} \tag{8.27}$$

and differentiate once more with respect to time:

$$\partial_t^2 \mathbf{E} = \nabla \times \partial_t \mathbf{B}. \tag{8.28}$$

Now we use the other curl equation,

$$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}, \tag{8.29}$$

to eliminate $\mathbf{B}$. This gives

$$\partial_t^2 \mathbf{E} = -\nabla \times (\nabla \times \mathbf{E}). \tag{8.30}$$

So far the equations still involve the whole vector $\mathbf{E}$. The next step, exactly as in the lecture, is to isolate one component, say E_x.

The board algebra is long, but its strategy is simple. The x-component of the double curl can be reorganized into a clean form:

$$\partial_t^2 E_x = -\big(\nabla \times (\nabla \times \mathbf{E})\big)_x \tag{8.31}$$

$$= -\partial_x(\nabla \cdot \mathbf{E}) + \partial_x^2 E_x + \partial_y^2 E_x + \partial_z^2 E_x. \tag{8.32}$$

In vacuum, $\nabla \cdot \mathbf{E} = 0$, so the first term drops out and we obtain

$$\partial_t^2 E_x - \partial_x^2 E_x - \partial_y^2 E_x - \partial_z^2 E_x = 0. \tag{8.33}$$

Thus the x-component of the electric field satisfies the same wave equation that we earlier wrote for the scalar field ϕ. This is the exact mathematical payoff Susskind wants. The electric field itself can propagate as a wave.

A simple plane-wave example is now immediate:

$$E_x(z,t) = \cos(z-t), \qquad E_y = 0. \tag{8.34}$$

This is a right-moving wave along the z-axis in the lecture's units. It is also polarized: the electric field has been chosen to point along the x-direction. With the sign convention fixed above, a consistent accompanying magnetic field is

$$B_y(z,t) = \cos(z-t), \qquad B_x = B_z = 0. \tag{8.35}$$

This explicit B_y is not copied from the board; it is the clean consequence of the chosen sign convention together with $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

8.6.1 Question & Answer

Question. Why must the electromagnetic wave be transverse?

Answer. Because the divergence and curl equations leave no room for a propagating longitudinal component. For a wave depending only on z and t,

$$\nabla \cdot \mathbf{E} = \partial_x E_x + \partial_y E_y + \partial_z E_z = \partial_z E_z. \tag{8.36}$$

In vacuum this must vanish, so

$$\partial_z E_z = 0. \tag{8.37}$$

Therefore E_z can at most be a constant. A constant field is not part of the wave profile, so the wave part of the z-component must be absent.

Now look at the curl. If $E_x(z,t)$ is nonzero while $E_y = 0$ and the wave part of E_z vanishes, then the only nonzero component of $\nabla \times \mathbf{E}$ lies along y. But $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, so the magnetic field itself must point along y. Hence

$$\mathbf{E} \perp \hat{z}, \qquad \mathbf{B} \perp \hat{z}, \qquad \mathbf{E} \perp \mathbf{B}. \tag{8.38}$$

That is the transversality of electromagnetic radiation. The electric field, the magnetic field, and the direction of propagation are mutually perpendicular.

Remark 8.1. The next frame is kept because it preserves the board geometry of the wave, even though its subtitle metadata is mismatched. Its evidentiary value here is visual: it belongs to the wave-and-polarization discussion, not to the anti-symmetric-matrix discussion suggested by the subtitle timing.

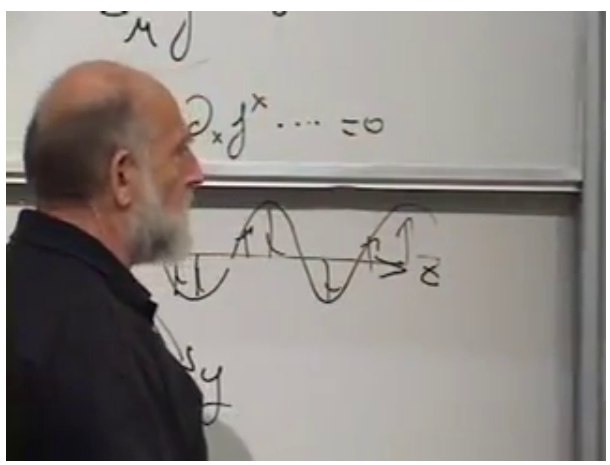


Figure 8.4: Board sketch of a transverse wave propagating along z.

This is the lecture's physical climax. Maxwell's equations imply electromagnetic waves, those waves move with speed 1 in the chosen units, and their electric and magnetic fields are transverse and polarized. Because the equations themselves are the same in every frame, these structural properties belong to every inertial observer.

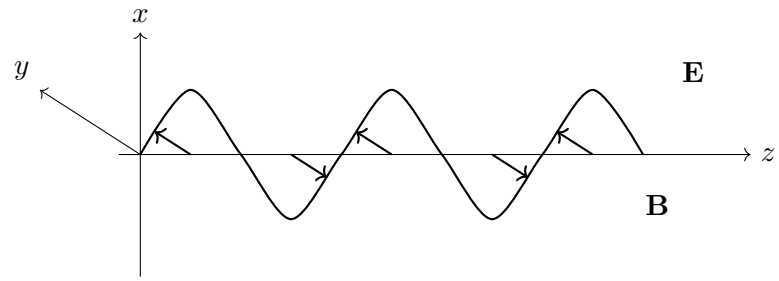


Figure 8.5: A clean reconstruction of the transverse electromagnetic-wave geometry: propagation along z, electric field along x, magnetic field along y.

8.7 Restoring Sources: Charge, Current, Continuity, and Covariance

Only after the vacuum story is clear does the lecture restore what had been postponed: charge density and current density. The Maxwell equations become

$$\nabla \cdot \mathbf{E} = \rho, \tag{8.39}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{8.40}$$

$$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B}, \tag{8.41}$$

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{8.42}$$

Here ρ is the charge density and $\mathbf{J}$ is the current density.

The charge density is defined by

$$\rho = \frac{dQ}{dV}, \tag{8.43}$$

so that the charge in a region V is

$$Q_V = \int_V \rho \, dV. \tag{8.44}$$

The lecture emphasizes the continuum approximation: in reality space may contain many individual charges, but if we coarse-grain over sufficiently small regions we may treat charge as a continuous density field.

The density ρ can depend on time as well as position, because charge can move. But it cannot change arbitrarily. The total amount of charge in the universe is conserved, and the lecture immediately strengthens that statement. Charge is not allowed simply to disappear here and reappear there with nothing in between. If charge leaves one region and appears in another, there must be a flow of charge connecting the two.

8.7.1 Question & Answer

Question. Why is charge conservation stronger than the statement that total charge is constant?

Answer. Because global constancy by itself would still allow absurd behavior: charge could vanish on Earth and reappear on Mars without anything happening in between. The lecture forbids that. Charge may move from one place to another only by flowing through the intervening space. That stronger, local form of conservation is what leads to current density and to the continuity equation.

To describe such flow, take an infinitesimal oriented surface element $d\boldsymbol{\sigma}$. The charge per unit time passing through it is

$$\frac{dQ}{dt} = \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{8.45}$$

This is why current is a vector. Its magnitude is charge per unit area per unit time, and its direction is the direction in which charge flows.

Now fix a spatial region V with outward-pointing boundary element $d\boldsymbol{\sigma}$. If net current flows outward through the boundary, the charge inside must decrease. Thus

$$\frac{dQ_V}{dt} = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{8.46}$$

Since $Q_V = \int_V \rho \, dV$, we also have

$$\frac{dQ_V}{dt} = \int_V \partial_t \rho \, dV. \tag{8.47}$$

Combining the two,

$$\int_V \partial_t \rho \, dV = -\oint_{\partial V} \mathbf{J} \cdot d\boldsymbol{\sigma}. \tag{8.48}$$

At this point the lecture invokes Gauss's theorem, also called the divergence theorem:

$$\oint_{\partial V} \mathbf{A} \cdot d\boldsymbol{\sigma} = \int_V (\nabla \cdot \mathbf{A}) \, dV. \tag{8.49}$$

Applying it to $\mathbf{J}$ gives

$$\int_V \partial_t \rho \, dV = -\int_V (\nabla \cdot \mathbf{J}) \, dV. \tag{8.50}$$

If this holds for every region V, then the integrands themselves must agree, and we obtain

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \tag{8.51}$$

This is the continuity equation, the local form of charge conservation.

Now the relativistic packaging becomes natural. The four quantities ρ, J_x, J_y, J_z form the four-current

$$J^\mu = (\rho, J_x, J_y, J_z), \tag{8.52}$$

and the continuity equation becomes

$$\partial_\mu J^\mu = 0. \tag{8.53}$$

The sourced Maxwell equations themselves then collapse into a single covariant equation:

$$\partial_\mu F^{\mu\nu} = J^\nu. \tag{8.54}$$

For $\nu = t$, the right-hand side is $J^t = \rho$, so the equation becomes

$$\nabla \cdot \mathbf{E} = \rho. \tag{8.55}$$

For spatial ν, it becomes

$$\nabla \times \mathbf{B} = \partial_t \mathbf{E} + \mathbf{J}. \tag{8.56}$$

The other half of Maxwell's equations remains

$$\partial_\mu \tilde{F}^{\mu\nu} = 0. \tag{8.57}$$

So the return of sources does not spoil covariance. On the contrary, it completes it. Charges and currents enter as a four-vector, and the field equations remain tensor equations. The way fields act on matter is Lorentz invariant, and the way matter generates fields is Lorentz invariant as well.

8.8 Summary

The lecture has a very definite arc. We begin with the Lorentz force law and identify the missing half of the subject: not how fields act on matter, but how matter determines fields. Maxwell's equations answer that question. In vacuum they can be written as the covariant equations $\partial_\mu F^{\mu\nu} = 0$ and $\partial_\mu \tilde{F}^{\mu\nu} = 0$, and from them we derive wave equations for the field components themselves. A plane-wave solution then shows that electromagnetic radiation is transverse and polarized, with $\mathbf{E}$, $\mathbf{B}$, and the direction of propagation mutually perpendicular.

Finally, when charges and currents are restored, the continuity equation and the four-current J^μ lead to the sourced covariant equation $\partial_\mu F^{\mu\nu} = J^\nu$. That is the closing point Susskind wants to leave with us: electromagnetism can be written entirely in Lorentz-covariant form, and because those same laws imply waves moving with speed c, light travels with the same speed in every inertial frame.